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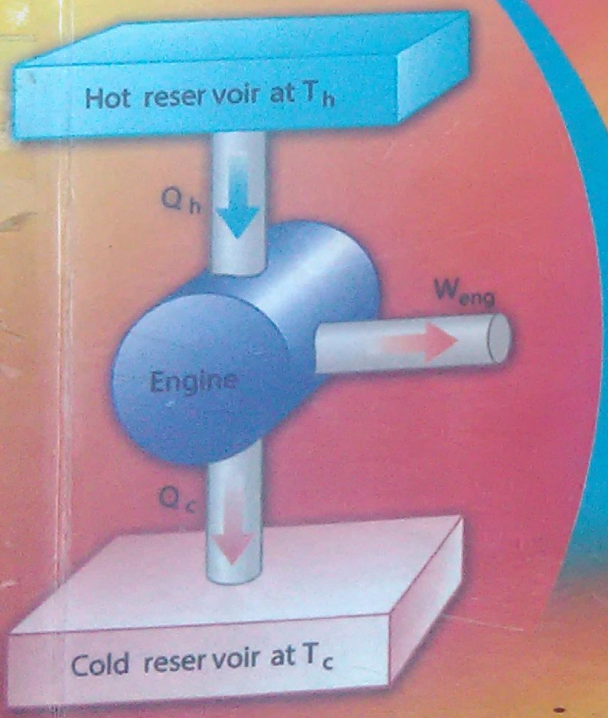
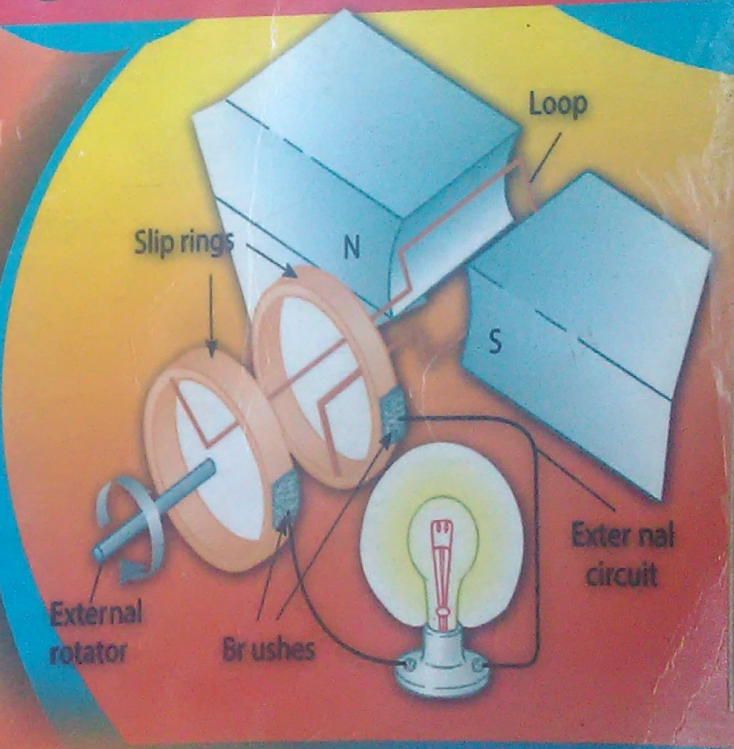
PHYSICS

GRADE 11





Based on Common Currently Used National and International Curriculum

2010 Edition

Abdu Yesuf
Arebu Abdella

Extreme

PHYSICS

Grade 11

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Unit 1

Measurement and Practical Work

Physics is the study of the basic laws of nature. Its beauty lies in the discovery and application of these laws of nature to make life simple and create a better world. For example, from rockets sent to space to integrated chips in computers, to lasers used to perform eye surgery. From atoms and subatomic particles to solar system and galaxies all obey the laws of physics.

Rocketry and space travel have their roots firmly planted in the physical laws. The transportation industry relies heavily on physics in the development of engines and the design of aerodynamic vehicles. Entire electronics and computer industries owe their existence to the invention of transistors, that come from the electrical nature of solids. The telecommunication industry depends on electromagnetic waves from theory of electricity and magnetism. x – ray, ultrasonic and magnetic resonance concepts of physics are also used for observing images of the interior of the human body. The laser used in space exploration and in Medicine is a direct application of the principles of atomic physics … etc.

1.1 Science of Measurement

Physics is all about observing natural phenomena and then experimenting. Hence, it is a science of measurement.

The Scientific Method

Science is an organized body of knowledge that is based on a method of looking at the world. The same problem may be analyzed differently by different scientists. However, there are common guidelines for the practice of science that most scientists pass through. These general approaches to problems are called *scientific methods*.

Scientific Method: Is a systematic procedure scientists use to acquire knowledge in science.

Scientific method ensures a *rigorous, evidence based structure* where only ideas that have been carefully tested are accepted as scientific theory.

- The common steps or processes in scientific method are:
  - **Ask a Question**
    This may be based on an observation. Such as why is the sky blue? Why does the sun shine?
  - **Do Background Research or Use Existing Knowledge**
    Your question might be already answered by other. If so, search what has been said about it.
  - **Form a Hypothesis**
    Make a temporary explanation or guess possible answer for your observation or question
  - **Make predications from your Hypothesis**
  - **Do Experiment**
    Make practical work under known conditions to test the truth of your hypothesis.
  - **Analyze your Experimental Data**
    Examine in detail using mathematical relations or qualitatively the meaning of your set of data
  - **Draw Conclusion**
    Generate the outcome of the overall result. See if your hypothesis was correct. If not, construct a new hypothesis and repeat.
- If your hypothesis is correct, it will be cross checked and tested by others before it becomes an accepted scientific theory. This process of peer review is important to get evidence based and not made up scientific theory.

1.2 Significant Figures

Significant figures are those digits that correctly indicate the presision of a measurement. They show the limits of accuracy and the beginning of uncertainty.

Significant Figures: Are reliably known or meaningful digits in a measured or calculated quantity.

Rules For Determining The Number of Significant Figures

1. All non- zero digits are significant
   For example, 132.73 contains 5 significant figures
2. All zeros between two non zero digits are significant
   For example, 207.009 contains 6 significant figures
3. A) All zeros to the left of understood decimal point but to the right of a non- zero digit are not significant
   For example, 307000 contains 3 significant figures

B) All zeros to the left of understood decimal point but to the right of a non- zero digit are significant if they come from a measurement.

For example, 400m has 3 significant figures

4. All zeros to the left of an expressed decimal point and to the right of a non- zero digit are significant

For example, 307000. Contains 6 significant figures

5. All zeros to the right of a decimal point but to the left of non-zero digit are not significant provided there is only a zero to the left of the decimal point.

For example, 0.000345m contains 3 significant figures

6. All zeros to the right of a decimal point and to the right of a non-zero digit are significant

For example, 0003040 and 40.00m each has 4 significant figures

Illustrative Example

1. State the number of significant figures for the following

i. 600900
ii. 25300
iii. 648700.
iv. 400m
v. 7785
vi. 5212.0
vii. 978.850
viii. 984.06
ix. 426.071
xi. 6.032
xii. 0.0631
xiii. 0.007
xiv. 0.0050
xv. 6.320
xvi. 9.1×10^{-31}
xvii. 2.64×10^{24}
xviii. 0.2370
xix. 0.045
xx. 0.0006032

Solution

i. 4
ii. 3
iii. 6
iv. 3
v. 4
vi. 5
vii. 6
viii. 5
ix. 6
xi. 4
xii. 3
xiii. 1
xv. 4
xvi. 2
xvii. 3
xviii. 4
xix. 2
xx. 4

Arithmetic operations with significant figure

1. Addition and Subtraction

> In addition or subtraction, the number of decimal places which are significant in the result is the same as the smallest number of significant decimal places in any number used in the sum or difference.

☛ Use the least number of decimal places

2. Multiplication and Division

> In multiplication or division, the number of significant figures in the product or the quotient is the same as the smallest number of significant figures in any of the factors.

☛ Use the least number of significant figures

Illustrative Example

2. a. Add 17.35g to 25.6g

   b. Add 6.75×10^3 cm and 4.52×10^2 cm

   c. Subtract 36.8 km from 97 km

   Solution: a. 17.35 g + 25.6g = 42. 95 g

   - The answer is 43.0g

   b. First convert to the same power of ten

   4.52×10^2 cm = 0.452×10^3 cm = 0.45×10^3 cm

   6.75×10^3cm + 0.45×10^3 cm = 7.20×10^3 cm

   c. 97 km – 36.8 km = 60.2 km

   - The answer is 60 km

3. a. Multiply 3.8 and 0.125 with due regard to significant figure

   b. Divide 8.2 by 3.885

   Solution: a. $3.8 \times 0.125 = 0.475$

   - The answer is 0.48

   b. $\frac{8.2}{3.885} = 2.11068$

   - The answer is 2.1

1.3 Uncertainty in Measurements

Numbers obtained by measurement are always inexact. There are many causes of uncertainty, but the most common are:

- The person taking the measurement
- The measuring device

- The environment where the measurement is being made.
- The nature of the object being measured.

Uncertainty: Is the amount of doubt in a measurement

For example, if the width of this book is reported as:

$$10\ \text{cm} \pm 0.25\ \text{cm}$$

It means that, the true value is unlikely to be less than 9.75 cm or greater than 10.25 cm. hence, we can observe that, 0.25cm is the uncertainty

Percentage Uncertainty

It is a way of expressing the uncertainty in terms of percentage.

$$\text{Percentage uncertainty} = \frac{\text{Uncertainity}}{\text{measured value}} \times 100$$

For example, the percentage uncertainty of 10 ± 0.25 is,

$$\%\ \text{uncertainty} = \frac{0.25}{10} = 2.5\%$$

So we can write as, $10\text{cm} \pm 2.5\%$

☛ **Keep in Mind That:**

- Percentage uncertainty tells us the amount in a measurement we are not sure of in percent.
  For instance "$47\Omega \pm 10\%$" means, the true value differs from 47Ω by 10%. So it is between 42 and 52 Ω.
- As a rule, percentage uncertainty should be expressed in two significant figures.
- In multiplication and division, the percentage uncertainty of the given values should be added.

Illustrative Example

4. If $I = 2.0\ A \pm 0.05A$. Find the percentage uncertainty

   Solution: Percentage uncertainty $= \frac{0.05}{2.0} \times 100 = 2.5\%$

   So we can write $I = 2.0A \pm 2.5\%$

5. If $v = 6.25\ v \pm 0.8\%$ and $R = 4.5\Omega \pm 2\%$, find I

   Solution: $I = \frac{V}{R} = \left(\frac{6.25}{4.5}\right) \pm (0.8 + 2)\%$

   $\Rightarrow I = 1.4A \pm 2.8\% = 1.4A \pm 0.04A$

6. If $\overline{V_{av}} = 3.86\ \text{m/s} \pm 3.5\%$ and $t = 2.4\ \text{sec} \pm 4.7\%$

Find $\vec{S}$

Solution: $\vec{S} = \left(\overrightarrow{V_{av}}\right)(t) = (3.86)(2.4) \pm (3.5 + 4.7)\%$

$\Rightarrow \vec{s} = 9.3\text{m} \pm 8.2\%$

7. If $r = 2.6\text{ cm} \pm 2.34\%$, Find area of the circle

Solution: $A = \pi r^2 = \pi(2.6)\,(2.6) \pm (2.34 + 2.34)\%$

$6.8\pi\text{ cm}^2 \pm 4.7\%$

8. Given $I = 6.2\text{A} \pm 1.6\%$, $V = 24.7\text{v} \pm 0.2\text{v}$.
Find R

Solution: $R = \dfrac{V}{I} = \left(\dfrac{24.7}{6.2}\right) \pm (0.8 + 1.6)\%$

$R = 4.0\,\Omega \pm 2.4\%$

Note that, the percentage uncertainty of 24. $7\text{v} \pm 0.2\text{v}$ is

$$\left(\frac{0.2}{24.7}\right) \times 100 = 0.8\%$$

1.4 Errors in Measurement

The devation of a measured value from the standard or accepted value is called *error*.

Error: Is a value which in not actual or true.

☛ Accepted or true value is the actual value of the measured object without any experimental error.

The most common types of errors in measurement are:

1. **Random error:** Is unpredictable error that has no pattern or bias and it makes the reading above or below the true value.
   → Can be reduced by taking several readings and average them
2. **Parallex error:** Is caused by improper position of the eye while taking record of a measured value
   → Can be reduced by taking reading of the measured value perpendicularly
3. **Systematic error:** It shows bias or trend and makes your reading too high or too low
   → Can be reduced by redesigning the measuring instrument or by taking measurement properly
4. **Zero error:** Is caused by an instrument that does not start from the zero mark
   → Can be reduced by making the appropriate instrument adjustment.

☛ In a more general sense, errors are of three types

- Personal error: Example, parallax error, systematic error
- Instrumental error: Example, systematic error, zero error
- Environmental error: Example, random error.

1.5 Precision and Accuracy

Precision and accuracy are terms which are used to express uncertainty in measurement.

Precision: Is the closeness or exactness of measured values to each other

The precision is good or high if each of the measurements is close to the average value

☛ The more precise a reading is, the smaller the uncertainty. 1.0m ± 0.1m, 1.00m ± 0.01m, 1.000m ± 0.001m are in order of increasing precision.

Accuracy: Is the closeness of a measured value to its true or correct value.

For example, if the true or accepted value of $g = 9.8066\ m/s^2$, and if you make three measurements of $g_1 = 9.76\ m/s^2$, $g_2 = 9.86\ m/s^2$, and $g_3 = 9.82\ m/s^2$ the most accurate value is $g_3 = 9.82\ m/s^2$ since it is closest to $g = 9.8066\ m/s^2$.

☛ Measurements of high precision are more likely to be accurate than are those of poor precision, but even highly precise measurements are sometimes inaccurate.

1.6 Writing a Laboratory Report

A Laboratory report is a written composition of the result of an experiment, written precisely and clearly using good grammar and punctuation and includes;

Title: Tells name of the experiment
Objective: States the purpose of experiment, i.e what to achive
Theory: Describes the equations, concepts and principles to be used
Material: Describes substances, equipments or instruments to be used
Procedure: Tells steps used in doing the experiment
Data: Represents the measured values and observed information
Data Analysis: Describes the data in meaningful manner using graphs, tables, software results.
Result and Discussion: Interpretation of the data analyzed with the possible sources of error.
Conclusion: Summary of the implication of experiment result

Evaluation: Assessment of the reliability and validity of the experiment result and designing a way of improving the experiment to make it realistic.

Illustrative Example

9. Which one of the following experimental errors can be reduced by taking repeated measurements (UEE: 2007)
   A. parallel errors C. Random errors
   B. Systematic errors D. Zeros errors **Answer: C**

Unit 1: Additional Examples

10. The number of significant figures in 0.00040 m is
    A. 1 B. 2 C. 3 D. 4
    Explanation: All zeros to the right of a decimal point and a non-zero digit are significant. **Answer: B**
11. The number of significant figures in 600m is
    A. 1 B. 2 C. 3 D. 4
    Explanation: Zeros to the right of non-zero digit are significant when they come from a measurement.
    Answer: C
12. The number of significant figures in 3.80 m is
    A. 1 B. 2 C. 3 D. 4 **Answer: C**
13. Which of the following is false regarding significant figures
    A. All non – zero digits are significant
    B. The zeros appearing in the middle of a number are significant while those at the end of a number without a decimal point are ambiguous
    C. The power of 10 are counted while counting the number of significant figures.
    D. Greater the number of significant figure in a measurement, smaller is the percentage error. **Answer: C**
14. The number of significant figures in 0.01020
    A. 3 B. 4 C. 5 D. 6 **Answer: B**
15. The number of significant figures in 0.900 is
    A. 1 B. 2 C. 3 D. 4
    Answer: C
16. The number of significant figures in 0.007 is
    A. 1 B. 2 C. 3 D. 4 **Answer: A**
17. The number of significant figures in 0.0060×10^{-18}m is
    A. 2 B. 3 C. 4 D. 5 **Answer: A**

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18. The decimal equivalent of $\frac{1}{20}$ up to three significant figures is

A. 0.05 B. 0.050 C. 0.500 D. 0.0500 **Answer: D**

19. Which of the following measurement is most accurate?

A. 9×10^{-2} m B. 90×10^{-3}

C. 900×10^{-4}m D. 0.090 m

Explanation: In (a), there is an error of 1 in 9. In (b), there is an error of 1 in 90. In (c), there is an error of 1 in 900. In (d), there is an error of 0.01 in 0.09.

Answer: C

20. When 999mm is added to 100m, we get

A. 101 m B. 100.999m

C. 100.99m D. 100.9m

Explanation: The result must be three significant figures

100m + 0.999m = 101m **Answer: A**

21. With due regard for significant figures, $(12.5)^2$=

A. 156.250 B. 156.25

C. 156.2 D. 156 **Answer: D**

22. Which of the following measurement is most precise?

A. 5648 sec B. 56.78 sec

C. 567.8 sec D. 5.678 sec

Explanation: The one with maximum number of decimal places is most precise. **Answer: D**

23. Which of the following measurement is most significant?

A. 0.003 mm B. 3.00m

C. 30.00 mm D. 3.0 mm

Explanation: 30.00mm has four significant figure. **Answer: C**

24. The length of a pencil is 10.4 cm. out of it, 2.01 cm is cut out. The remaining length of the pencil is

A. 8 cm B. 8.4 cm

C. 8.3 cm D. 8.39 cm

Explanation: The number of decimal places in the result is the same as the smallest number of significant decimal places in any number used in the sum or difference. **Answer: B**

25. The heat generated in a circuit is given by $Q = I^2 Rt$. If the percentage errors in measuring I, R and t are 2%, 1% and 1% respectively, then the maximum error in measuring heat will be.

A. 2% B. 3% C. 4% D. 6%

Solution: Total error = 2 × 2% + 1% + 1% = 6% **Answer: D**

26. The sides of a rectangle are 6.01m and 12m. Taking the significant figures into account, the area of the rectangle is.
A. $72\ m^2$ B. $72.1\ m^2$
C. $72.00\ m^2$ D. $72.12\ m^2$
Hint: the result should have two significant figures **Answer: A**

27. Error in the measurement of radius of sphere is 2%.
The error in the measurement of the sphere volume is
A. 1% B. 5% C. 3% D. 6%
Explanation: $v \sim r^3 \Rightarrow$ Required error $= 3 \times 2\% = 6\%$ **Answer: D**

28. The percentage errors in the measurement of mass and speed are 2% and 3% respectively. What is the error in measuring kinetic energy then?
A. 11% B. 8% C. 5% D. 1%
Solution: $K_E = \frac{1}{2}mu^2 \Rightarrow$ Required error $= 2\% + 2 \times 3\% = 8\%$
Answer: B

29. The pressure on a square plate is given by $P = \frac{F}{\ell^2}$. If the maximum errors in measuring force and length are 4% and 2% respectively, then the maximum error in the pressure is.
A. 1% B. 2% C. 8% D. 10%
Solution: Maximum percentage error in pressure is
$4\% + 2 \times 2\% = 8\%$ **Answer: C**

30. In a simple pendulum experiment, time period is measured with an accuracy of 0.2% while length is measured with an accuracy of 0.5%. The percentage accuracy in the value of acceleration due to gravity is
A. 0.25% B. 0.7% C. 0.9% D. 1.0%
Solution: The word 'accuracy' is used in the sense of percentage error. Now, $T = 2\pi\sqrt{\frac{\ell}{g}} \Rightarrow g \sim \frac{\ell}{T^2}$
The required error $= 0.5\% + 2 \times 0.2\% = 0.9\%$ **Answer: C**

31. The dimensions of a rectangular block measured with least count of 0.01cm are 5mm × 10mm × 5mm. The maximum percentage error in the volume of block is
A. 5% B. 10% C. 15% D. 20%

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Solution: Percentage error in volume

$$=\frac{0.01}{0.5}+\frac{0.01}{0.1}+\frac{0.01}{0.5}=(0.05)\times 100\%=5\%$$ **Answer: A**

32. A current of 2.34 A flows in a resistance of 11.11 11 11Ω. The potential difference across the resistance with due regard for significant figures is

A. 26.000v B. 26.00v C. 26.0v D. 26v

Solution: V = TR = 26.0v **Answer: C**

33. A resistance of 10k Ω having tolerance of 10% is connected in series with another resistance of 20k Ω having a tolerance of 20%. The tolerance of the combination is

A. 10% B. 13% C. 17% D. 20%

Solution: $\Delta R=\Delta R_1+\Delta R_2=\frac{10}{100}\times 10+\frac{20}{100}\times 20=5k\Omega$

$$R_T=R_1+R_2=10+20=30\ k\Omega$$

Percentage error $=\frac{\Delta R}{R_T}\times 100\%=\frac{5}{30}\times 100\simeq 17\%$ **Answer: C**

34. A resistor of 4kΩ with tolerance 10% is connected in parallel with a resistor of 6kΩ with tolerance 10%. The tolerance of the parallel combination is

A. 10% B. 20% C. 30% D. 40%

Solution: Percentage error

$$\frac{\Delta R_1}{R_1}\times 100+\frac{\Delta R_2}{R_2}\times 100+\frac{\Delta R_1+\Delta R_2}{R_1+R_2}\times 100$$

$$\Delta R_1=\frac{10}{100}\times 4=0.4$$

$$\Delta R_2=\frac{10}{100}\times 6=0.6$$

$$\Rightarrow\frac{0.4}{4}\times 100+\frac{0.6}{6}\times 100+\frac{0.4+0.6}{4+6}\times 100=30\%$$ **Answer: C**

35. Two quantities A and B have different dimension. Which mathematical operation given below is physically meaning full?

A. $\frac{A}{B}$ B. $A+B$ C. $A-B$ D. $A=B$

Answer: A

36. A cube has a side of length 1.2×10^{-2}m. calculate its volume.
A. 1.7×10^{-6} m^3 B. 1.73×10^{-6} m^3
C. 1.70×10^{-6} m^3 D. 1.732×10^{-6} m^3

Answer: A

37. The volume of a sphere is 1.76 cm^3. The volume of 25 such spheres is
A. 44.00 cm^3 B. 44 cm^3
C. 44.0cm^3 D. 0.44×10^2 m^3 **Answer: C**

38. The measured mass and volume of a body are 22.42g and 4.7 cm^3 respectively with possible errors of 0.01g and 0.1 cm^3. The maximum error in density is
A. 0.2% B. 2% C. 5% D. 10%

Solution: percentage error of mass $\frac{0.01}{22.42} \times 100 = 0.04\%$

Percentage error in volume $= \frac{0.1}{4.7} \times 100 = 2\%$

Percentage error in density = 2% + 0.04% ≃ 2% **Answer: B**

39. What is the difference between 11.5 ± 0.1cm and 7.8 ± 0.1 cm
A. 3.7 ± 0 cm B. 3.7 ± 0.2 cm
C. 3.7 ± 0.1 cm D. 3.7 cm

Solution: (11.5 ± 0.1 cm) – (7.8 ± 0.1 cm) = 3.7 ± 0.2 cm

Answer: B

Unit 1: Problems

1. Compute the following using the correct number of significant figures
A. 652.397 m + 48. 542 m C. 5.67 km × π km
B. 79. 54 Ω – 56. 24Ω – 56.24Ω1R D. 3.2 cm ÷ 1.4577 cm

2. Determine the unknown quantity with the correct number of significant figures.
A. R = 2.3Ω ± 0.05 Ω
V = 9.47v ± 0.8 %
I = ?
B. $\vec{v} = 8.75\,m/s \pm 0.12\,m/s$
t = 2.4 sec ± 0.3 sec
$\vec{s}$ = ?
C. v = 10.4 m/s ± 0.25 m/s
s = 46.7 m ± 5.3%
$\vec{s}$ = ?
D. r = 5.3cm ± 0.6cm
Area of circle = ?

3. A cube box of mass 12 kg has sides of 0.58 m. what is the density of the box with the correct number of significant figures?

4. The density of a rectangular prism is $98.4\ g/cm^3 \pm 0.45\ g/cm^3$. Calculate the mass of the prism if its height is $7.2\ cm \pm 0.9\%$ and base area is $16.5\ cm^2 \pm 0.35\ cm^2$

Unit 1: Solution to Problems

1. a. 700.94 m b. 23.30Ω C. $17.8\ km^2$ D. 2.2

2. a. $I = \frac{V}{R} = \frac{9.47}{2.3} \pm (0.8 + 2.2)\% = 4.1A \pm 3.0\%$

   b. Fist change to percentage uncertainty, $21m \pm 14\%$

   c. $4.49 \pm 7.7\%$ d. $28\pi\ cm^2 \pm 22\%$

3. $\rho = \frac{m}{v} = \frac{12.5}{(0.58)^3} = 64kg/m^3$

4. $m = \rho v$, but $v = Ah = (7.2)(16.5) \pm (0.9 + 2.1)\%$

   $v = 120\ cm^3 \pm 3.0\%$

   $m = (98.4)(120) \pm (0.46 + 3)\%$

   $m = 1.2 \times 10^4\ g \pm 3.5\%$

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Unit 2

Vectors

The pilot, the captain, the car drive, the motorist and all of us use the concept of vectors in our day to day activities specially, when we are thinking of the start and finish of our journey. Moreover, Engineers do also use the concept of vectors while designing a stable structure such as machines, buildings, bridges, dams, railways and highways, . . . etc

2.1 Scalars and Vectors

Based on the International system of units called the *SI* units system, physical quantities are classified as fundamental or derived.

Fundamental or Basic quantities: Are quantities which do have units that are internationally agreed definitions.

☛ A fundamental quantity cannot be expressed in terms of any other physical quantity. The units in which these quantities are measured are called *fundamental* or *basic* units. Examples of fundamental quantities along with their units are given in the following table.

Fundamental Quantity	Fundamental Unit	Unit Symbol
Length	Meter	m
Mass	Kilogram	Kg
Time	Second	S
Current	Ampere	A
Amount of substance	Mole	Mol
Luminous intensity	Candela	Cd
Temperature	Kelvin	K

Derived Quantities: Are quantities whose units can be obtained by combining the units of the fundamental quantities.

☛ Derived quantities can be obtained by combining that is multiplying or dividing the fundamental quantities. Some examples of derived physical quantities along with their units are:

Derived Quantity	Derived Unit
Speed	m/s
Acceleration	m/s^2
Force	kgm/s^2
Momentum	kg m/s
Work	$kg\ m^2/s^2$
Volume	m^3

Based on direction as a criteria, physical quantities again are classified as scalars and vectors.

Scalars: Are quantities that can be completely specified or described by their magnitude only.

Example of scalar quantities include, time, mass, speed, length, work, energy, power, temperature, . . . etc

☛ **Important points to remember**

- Scalars do not need direction. That is, we can say the time is 3 PM or the mass is 5 kg. But, it does not make sense if we say 3PM towards west or 5 kg due North. Okey?
- Scalars can be added or subtracted like any number. For example, 2 kg + 5 kg = 7 kg
- Scalars can be combined if and only if they represent the same quantity.
  For example, 2 kg + 5 kg = 7 kg, possible
  2 kg + 5m = ?, impossible to add
- Scalars can be positive or negative number.
  For example, −25°C or 25°C are both scalars

Vectors: Are quantities that need both magnitude and direction for their complete description.

Velocity, displacement, acceleration, momentum, torque, electric field strength are some examples of vector quantities.

☛ **Important points to remember**

- Vectors need direction. That is, we can say, the car is moving at a velocity of 10 m/s towards East or the man is pushing the cart by a force of 100N towards South. But the car is moving at a velocity of 10 m/s or the man is pushing the cart by a force of 100N, both do have incomplete meaning since they require "which direction."
- Vectors cannot be combined by the rule of ordinary arithmetic, rather, they follow vector rules of addition.

- Physical quantities such as density, stress, strain, refractive index, dielectric constant, electrical conductivity etc are scalars in isotropic media. But in anisotropic media, these quantities assume different values in different directions, and they are called tensors.

Conceptual Example

1. A vector quantity possesses both magnitude and direction. If so, can you conclude that any quantity which has magnitude and direction is necessarily a vector? Explain

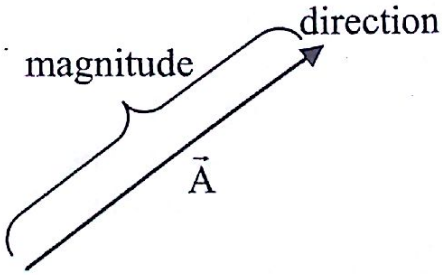
   Explanation: Not necessarily, For example, electric current and rotation of a rigid body about an axis both have magnitude and direction. However, both quantities that is current and rotation of rigid body are not vectors since they do not obey the parallelogram law of addition.

☛ For a quantity to be a vector, it should not only have magnitude and direction, but it should also satisfy the parallelogram law of vector addition.

2.2 Representation of Vectors

Analytically, a vector is represented by bold face letters as **A** in print or books. Or a letter with an arrow over its head as $\vec{A}$. The magnitude or modulus of vector A is represented by |A| or simply by a light face letter A.

Graphically, a vector is represented by a ray. The length of the ray gives the magnitude of the vector and the arrow of the ray indicates the sense of direction of the vector.



Vector A graphically

Equal vectors: Two vectors are said to be equal if and only if they have the same magnitude, the same direction and represent the same type of physical quantity.

For example, if we have the vectors;

$\vec{A}$ = 10N south, $\vec{B}$ = 10 m/s south, $\vec{C}$ = 10N East and $\vec{D}$ = 10N south

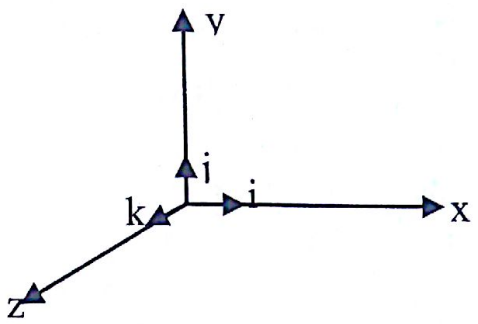
Then, $\vec{A} = \vec{D}$ is true.

2.3 Types of Vectors

Vectors can be of the following types;

Unit vector: Is a vector with magnitude of one and is used to tell or indicate the direction of a given vector.

In general, the unit vectors that point in the positive x, y and z axes are denoted by i, j and k respectively as shown in the figure.



☛ **Note that;** The three unit vectors i, j and k are mutually perpendicular. That is $i \perp j$, $j \perp k$ and $i \perp K$.

A vector in space can be expressed using the unit vector notation. That is, let vector $\vec{A}$ be a vector in space. Then, it can be written using the unit vector notation as.

$$\boxed{\vec{A} = A_x i + A_y j + A_z k}$$

The unit vector in the direction of vector $\vec{A}$ is defined as:

$$\boxed{\vec{A} = \frac{\vec{A}}{|A|} = \frac{A_x i + A_y j + A_z k}{\sqrt{(A_x)^2 + (A_y)^2 + (A_z)^2}}}$$

Where: $\vec{A}$ is a unit vector and is read as A – cap or A – hat.

$|A|$ is the magnitude or modulus of vector $\vec{A}$

Illustrative Examples

2. Vector $\vec{A}$ has components $A_x = 4$, $A_y = 3$ and $A_z = -5$.
   a) Express $\vec{A}$ in terms of unit vectors.
   b) Find the magnitude of $\vec{A}$.

   Solution: a) $\vec{A} = A_x i + A_y j + A_z k = 4i + 3j - 5k$

   b) $|A| = \sqrt{(A_x)^2 + (A_y)^2 + (A_z)^2} = \sqrt{(4)^2 + (3)^2 + (-5)^2} = 5\sqrt{2}$

3. What is the unit vector in the direction of vector, $\vec{P} = 2i + 3j - 5k$?

   Solution: $P = \frac{\vec{P}}{|P|} = \frac{2i + 3j - 5k}{\sqrt{(2)^2 + (3)^2 + (-5)^2}} = \frac{1}{\sqrt{38}}(2i + 3j - 5k)$

4. The head of vector $\vec{A}$ is at the coordinate (3, 4, 9) and its tail is at (-2, 3, 5). Write the vector.

   Solution: $\vec{A} = A_x i + A_y j + A_z k = (3 - (-2))i + (4 - 3)j + (9 - 5)k$

   $\vec{A} = 5i + j + 5k$

Null or zero vectors: Is a vector whose magnitude is zero and whose direction is indeterminate or arbitrary.

Important points to remember

- A zero vector has head and tail at the same point.
- Zero vector is identity vector of vector addition. i.e., $\vec{A} + 0 = \vec{A}$.

Conceptual Example

5. Is it possible to talk of the direction of a zero vector? Explain

Explanation: No. A zero vector has arbitrary direction, that is, its direction is indeterminate.

Illustrative Example

6. If $\vec{A} = 2i + 9j + k$, find $\vec{A} - \vec{A}$.

Solution: $\vec{A} - \vec{A} = (2i + 9j + k) - (2i + 9j + k)$

$= (2 - 2)i + (9 - 9)j + (1 - 1)k$

$= 0i + 0j + 0k = 0$ ⟵ null or zero vector

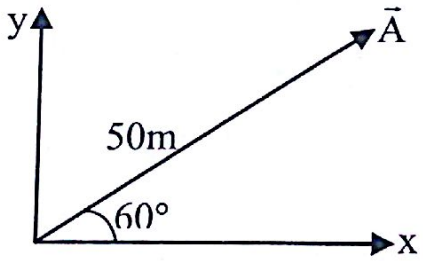
Position Vector: Is a vector that tells the position of a body with respect to another point.

For example, the position of vector $\vec{A}$ can be reported in one of the following ways.

→ $\vec{A}$ is 50m on a bearing of 30°

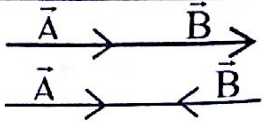
→ $\vec{A}$ is 50m at 60° North of East

→ $\vec{A}$ is 43.3m East and 25m North from the origin.



Collinear Vectors: Are vectors acting on the same line. These vectors can be parallel or anti parallel to each other with the same line of action.

For example, vector $\vec{A}$ and $\vec{B}$ are collinear in the two cases shown.



Coplanar vectors: Are vectors that act on the same plane.

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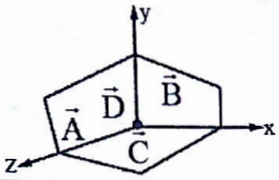
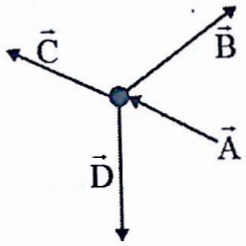
For example, vector $\vec{A}$ and vector $\vec{D}$ are coplanar in the given sketch.



Concurrent vectors: Are vectors acting on the same point.

The given vectors are concurrent.



Conceptual Example

7. Are all collinear vectors coplanar and vice versa? Explain

Explanation: Collinear vectors are coplanar but coplanar vectors may not be collinear.

2.4 Resolution of Vectors

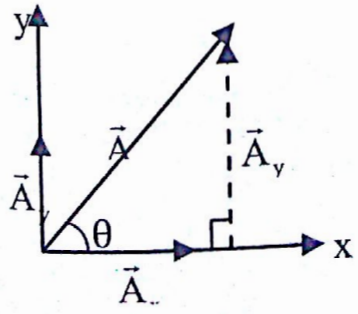
Any non zero vector can be decomposed into x, y, z parts.

Resolution: Is a process of spitting a vector into components.

splitting

For example, a vector in xy plane can be resolved into two components. To understand this, consider vector $\vec{A}$ lying in the xy plane making an angle θ with the positive x axis. Vector $\vec{A}$ can be expressed as the sum of two vectors $\vec{A}_x$ and $\vec{A}_y$ in the x and y direction respectively. Thus, $\vec{A} = \vec{A}_x + \vec{A}_y$. The vectors $\vec{A}_x$ and $\vec{A}_y$ are called the x and y or the *horizontal* and *vertical components* of vector $\vec{A}$ respectively.

$\vec{A} = A_x i + A_y j$ from the trigonometry relations.



$$\cos\theta = \frac{A_x}{A}$$

$$\Rightarrow \boxed{A_x = A\cos\theta}$$

$$\sin\theta = \frac{A_y}{A}$$

$$\Rightarrow \boxed{A_y = A\sin\theta}$$

The magnitude and direction of vector $\vec{A}$ can be obtained from the components using Pythagorean theorem and the concept of trigonometry.

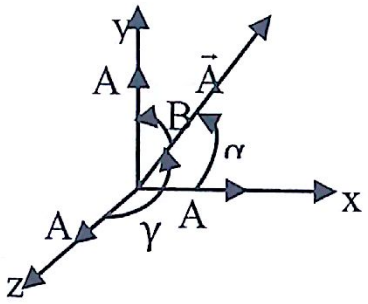
- The magnitude of vector $\vec{A}$ is related to its components by

$$\boxed{A = \sqrt{A_x^2 + A_y^2}}$$

- The angle that vector $\vec{A}$ makes with the positive x-axis.

$$\boxed{\tan\theta = \frac{A_y}{A_x}}$$

☛ A vector in space can be resolved into three components and we can have the following.



$$\boxed{A = A_x i + A_y j + A_z k}$$

$$\boxed{A = \sqrt{A_x^2 + A_y^2 + A_z^2}}$$

If α, β and γ are angles which vector $\vec{A}$ makes with positive x, y and z axis respectively, then

$$\left.\begin{aligned} \cos\alpha &= \frac{A_x}{A} = \frac{A_x}{\sqrt{A_x^2 + A_y^2 + A_z^2}} \\ \cos\beta &= \frac{A_y}{A} = \frac{A_y}{\sqrt{A_x^2 + A_y^2 + A_z^2}} \\ \cos\gamma &= \frac{A_z}{A} = \frac{A_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}} \end{aligned}\right\} \text{Direction Cosines}$$

These values are called *direction cosines* and have the property.

$$\boxed{\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1}$$

Conceptual Examples

8. Can you find a vector quantity that has a magnitude of zero but components that are different from zero? Explain

Explanation: No. Because, if $A = 0 \Rightarrow A = \sqrt{A_x^2 + A_y^2 + A_z^2} = 0$.

This is possible only when $A_x = A_y = A_z = 0$.

9. Can the magnitude of a vector be less than the magnitude of any one of its components? Explain

Explanation: NO. Because, $A = \sqrt{A_x^2 + A_y^2 + A_z^2} \Rightarrow A \geq A_x, A_y, A_z$.

10. If $\vec{A} = 0$, does it follow that $A_x = -A_y$? Explain

Explanation: $\vec{A} = A_{xi} + A_{yj} = 0$. If $\vec{A}$ is a null vector then, $A_x = A_y = 0$. But it does not mean $A_x = -A_y$.

11. Two vectors $\vec{P}$ and $\vec{Q}$ both lie in the xy plane.

A. Is it possible for $\vec{P}$ to have the same magnitude as $\vec{Q}$ but different components?

B. Is it possible for $\vec{P}$ to have the same components as $\vec{Q}$ but different magnitude?

Explanation:

A. Yes, consider $\vec{P} = 3i + 4j$, $\vec{Q} = i + \sqrt{24}j$, but $P = Q = 5$

B. No. This is impossible

12. If one component of a vector is not zero, can its magnitude be zero? Explain

Explanation: No. Because, $A = \sqrt{A_x^2 + A_y^2 + A_z^2}$

$A = 0$, iff $A_x = A_y = A_z = 0$, get it?

13. If the component of vector $\vec{A}$ along the direction of vector $\vec{Q}$ is zero, what can you conclude about the two vectors?

Explanation: The two vectors must be mutually perpendicular.

14. Can a unit vector have any component with magnitude greater than one? Can it have any negative component?

Explanation: No, components cannot be greater than the original vector. But, unit vectors can have negative component when they point the negative axes.

15. If $\vec{A} = \alpha(3i + 4j)$, determine the constant α that makes $\vec{A}$ a unit vector.

Explanation: If $\vec{A}$ is a unit vector, then its magnitude is 1

$$A = \sqrt{(3\alpha)^2 + (4\alpha)^2} = 1 \Rightarrow \alpha = \pm 0.2$$

Illustrative Examples

16. The x and y components of vector A are 30m and 40m respectively. What is

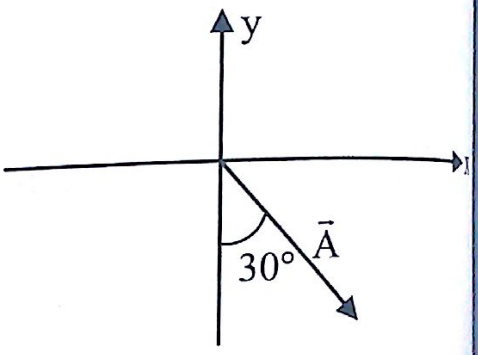
A. The magnitude of $\vec{A}$.

B. The direction of $\vec{A}$.

Solution: A. $A = \sqrt{A_x^2 + A_y^2} = \sqrt{(30)^2 + (40)^2} = 50\text{m}$

B. $\tan\theta = \frac{A_y}{A_x} = \frac{40}{30} \Rightarrow \theta = 53°$

17. What are the x and y components of vector $\vec{A}$ if its magnitude is 3m?



Solution: $A_x = A \sin\theta = (3)$ $\sin 30° = 1.5\text{m}$

$A_y = -A\cos\theta = -(3)$ $\cos 30° = -2.6\text{m}$

18. Let $\vec{A} = i + 2j + 3k$. Find

A. The magnitude of $\vec{A}$.

B. The direction cosine of $\vec{A}$.

C. $\cos^2\alpha + \cos^2\beta + \cos^2\gamma$

Solution:

A. $A = \sqrt{A_x^2 + A_y^2 + A_z^2} = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}$

B. $\cos\alpha = \frac{A_x}{A} = \frac{1}{\sqrt{4}}, \cos\beta = \frac{A_y}{A} = \frac{2}{\sqrt{14}}, \cos\gamma = \frac{A_z}{A} = \frac{3}{\sqrt{14}}$

C. $\left(\frac{1}{\sqrt{14}}\right)^2 + \left(\frac{2}{\sqrt{14}}\right)^2 + \left(\frac{3}{\sqrt{14}}\right)^2 = \frac{1}{14} + \frac{4}{14} + \frac{9}{14} = 1$

2.5 Vector Addition and Subtraction

If we have 2 kg of sugar and add 3 kg of sugar to it, we will have a total of 5 kg of sugar. This is always the case when scalars add, but vectors do not add like this because their direction may be different. For instance, the sum of a force of 2N and 3N may be in the range of 1N and 5N based on their orientation.

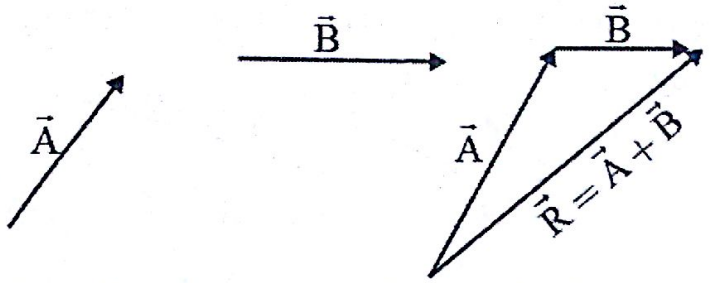
Vectors can be added graphically by drawing a scaled diagram of the magnitude and with proper orientation of the vectors or analytically by using algebraic and geometrical equations as discussed below.

A. Graphical Method of Vector Addition

Graphically, vectors can be added using the following methods.

Triangle method: In this method, two vectors are joined head to tail. The resultant vector is drawn from the tail of the first vector to the head of the second vector.

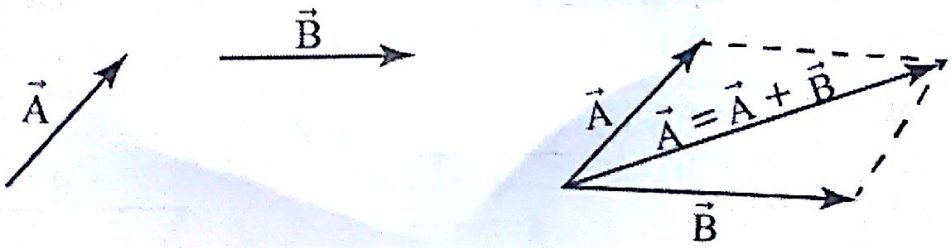
For example, if we have two vectors $\vec{A}$ and $\vec{B}$, triangle method gives us the resultant, $\vec{R} = \vec{A} + \vec{B}$.



☛ **Remember,** Resultant vector is that single vector which would have the same effect as all the original vectors taken together.

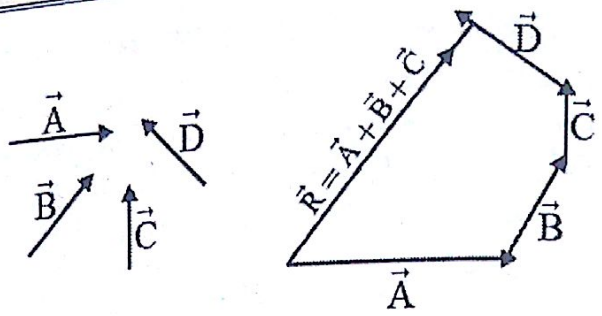
Parallelogram Method: Here, two vectors are joined tail to tail. The resultant vector is then the diagonal of the parallelogram, formed with the two vectors as its sides.

For example, consider the two vectors $\vec{A}$ and $\vec{B}$, the parallelogram method gives us the resultant, $\vec{R} = \vec{A} + \vec{B}$, as

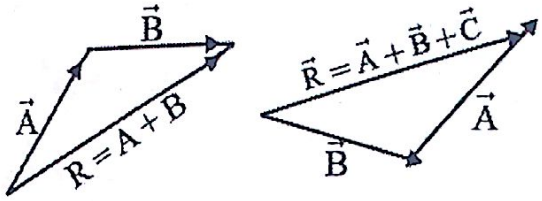


Polygon Method: In this case, more than two vectors are added by joining the given vectors head to tail. The resultant is the vectors that joins the tails of the first vectors to the head of the last vector.

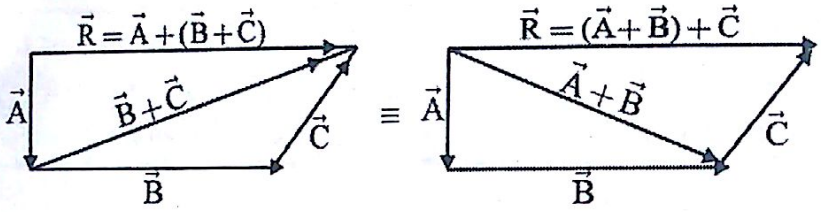
For example, if we have the vectors, $\vec{A}, \vec{B}, \vec{C}$ and $\vec{D}$, the polygon method gives us the resultant $\vec{R} = \vec{A} + \vec{B} + \vec{C} + \vec{D}$, as



- Vectors addition obeys the commutative as well as distribution laws. That is, order of addition doesn't matter. Give the same result. That means, $\vec{R} = \vec{A} + \vec{B} = \vec{B} + \vec{A}$,

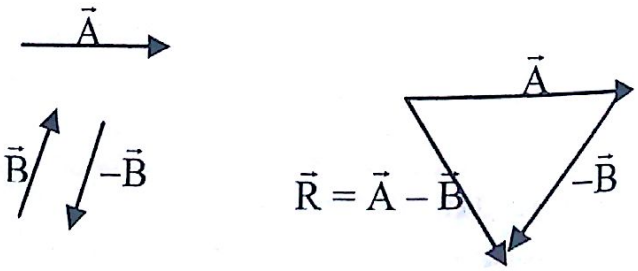


And $\vec{R} = \vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C}$, thus



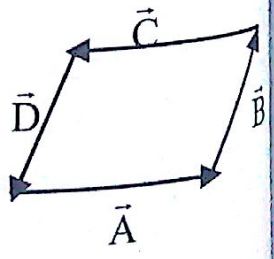
Subtraction of vectors: It means adding a vector to a negative vector. Thus, $\vec{A} - \vec{B}$ implies $\vec{A} + (-\vec{B})$.

For example, if we have two vectors $\vec{A}$ and $\vec{B}$, then their difference, $\vec{R} = \vec{A} - \vec{B} = \vec{A} + (-\vec{B})$.



Conceptual Examples

19. Vector $\vec{A}$ is 6m along the horizontal, and vector $\vec{B}$ is 9m and perpendicular to vector $\vec{A}$. If you take vector addition of the two, which one is greater? $\vec{A} + \vec{B}$ or $\vec{B} + \vec{A}$? Explain

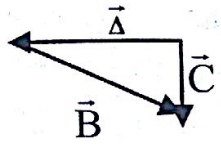


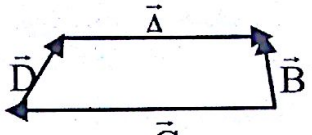
Explanation: The order of addition doesn't make any difference to the result.
Therefore $\vec{A} + \vec{B} = \vec{B} + \vec{A}$.

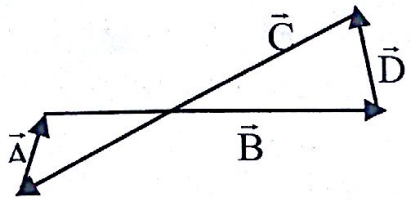
20. From the given figure identify the resultant vector and justify you answer.

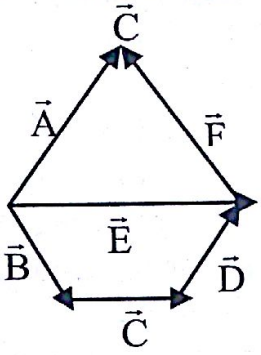
Explanation: The resultant is zero. Because, the head to tail connection has formed a closed loop by itself. Which means, end up where we started?

21. Identify the resultant from the given diagrams.

a) 

b) 

c) 

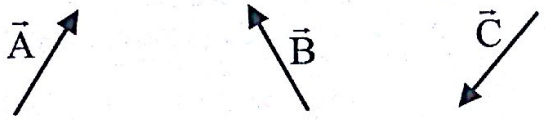
d) 

Explanation: Graphically, the resultant is drawn from the tail of the first vector to the head of the last vector. Thus:

a) $\vec{C}$ b) $\vec{B}$ c) zero d) $\vec{A}$, are resultants

Illustrative Examples

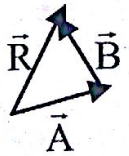
22. Perform the following vector additions and subtractions graphically if you are given vector $\vec{A}$, $\vec{B}$ and $\vec{C}$ as shown:

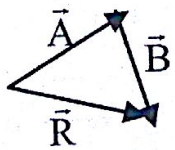


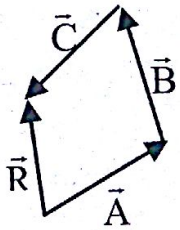
a) $\vec{A}+\vec{B}$ b) $\vec{A}-\vec{B}$ c) $\vec{A}+\vec{B}+\vec{C}$ d) $\vec{A}+\vec{B}-\vec{C}$

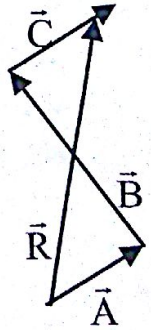
Solution:

a) $\vec{A}+\vec{B}$ b) $\vec{A}-\vec{B}$ c) $\vec{A}+\vec{B}+\vec{C}$ d) $\vec{A}+\vec{B}-\vec{C}$

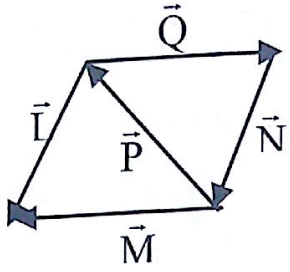








23. Use the given figure and express vectors $\vec{N},\vec{M}-\vec{L}$ $\vec{N}+\vec{M}-\vec{L}$ in terms of vectors $\vec{P}$ and $\vec{Q}$.



Solution:

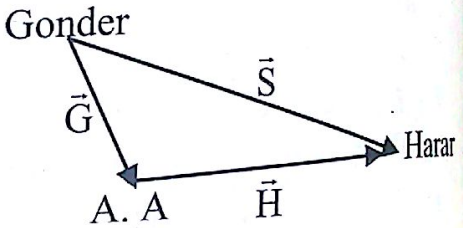
i. $\vec{P}+\vec{Q}+\vec{N}=0 \Rightarrow \vec{N}=-(\vec{P}+\vec{Q})$

ii. $\vec{M}-\vec{L}=\vec{P}$

iii. $\vec{N}+\vec{M}-\vec{L}+\vec{Q}=0 \Rightarrow \vec{N}+\vec{M}-\vec{L}=-\vec{Q}$

24. A pilot wants to know the real displacement from Gonder Harar as shown. He can determine the displacement by:

A. $H+G$
B. $H-G$
C. $G-H$
D. $H+S$



Answer: A

B. Analytical Method of Vector Addition

Analytically, vectors can be added using the following methods.

Component Method: Here vectors in the same direction are added.

For example, if we have two vectors $\vec{A}$ and $\vec{B}$ in space in component notation as:

$$\vec{A}=A_x i+A_y j+A_z k \text{ and } \vec{B}=B_x i+B_y j+B_z k$$

then, their sum or resultant is obtained by

$$\vec{R}=\vec{A}+\vec{B}=(A_x i+A_y j+A_z k)+(B_x i+B_y j+B_z k)$$

$$\boxed{\vec{R}=(A_x+B_x)i+(A_y+B_y)j+(A_z+B_z)k}$$

Or $\vec{R}=R_x i+R_y j+R_y j$ where,

$$R_x=A_x+B_x$$
$$R_y=A_y+B_y$$
$$R_z=A_z+B_z$$

And, $\boxed{R = \sqrt{R_x^2 + R_y^2 + R_z^2}}$

Conceptual Examples

25. Can vectors $\vec{A}$ and $\vec{B}$ having different magnitude be combined to give a zero sum? Explain

Explanation: No. Two vectors must be equal in magnitude and opposite in direction to give a zero sum.

26. Can the sum of the magnitude of two vectors ever be equal to the magnitude of their sum? Example

Explanation: Yes. $|A| + |B|$, this is so when the two vectors are parallel.

27. Can the magnitude of the difference between two vectors ever be greater than the magnitude of:
    a) One of the vectors.
    b) Both vectors.
    c) Their sum

Explanation: a) Yes b) Yes c) Yes

For the three cases, consider the vectors $\vec{A} = 5i$ and $\vec{B} = -2i$.

28. Is it possible to add a vector quantity to a scalar quantity? Explain

Explanation: No. Because, it does not make sense to add a 5 kg and a 5N due North. In the same manner, we cannot add two different vector quantities like velocity vector and force vector. Hence, 5 m/s due North +5N due North, Wrong!

29. The magnitude of two vectors are A = 4, B = 3. Which of the following could be the magnitude of resultant of the two?

A. 9 B. C. 5 D. −1

Explanation: The resultant vector magnitude of two vectors is always in the range.

$|A - B| \le R \le |A + B|$

In our case, $|4 - 3| \le R \le |4 + 3|$

$1 \le R \le 7$

Answer: C

30. If $\vec{A} + \vec{B} = 0$ What can you say about the components of the two vectors?

Explanation: $\vec{A} + \vec{B} = 0 \Rightarrow \vec{A} = -\vec{B}$ or $A_x + A_y = -(B_x + B_y)$

Hence, $A_x = -B_x$ and $A_y = -B_y$

31. When two vectors are added, the resultant has components $R_x = 4$ and $R_y = 3$. If the first vector is twice of the second vector, what are the values of the x and y components of each vector?

Explanation: $R = R_x + R_y = (A_x + B_x) + (B_x + B_y)$

Let $A = 2B \Rightarrow A_x = 2B_x$ and $A_y = 2B_y$

$A_x + B_x = 4$ $\qquad$ $A_y + B_y = 3$

$2B_x + B_x = 4$ $\qquad$ $2B_y + B_y = 3$

$3B_x = 4$ $\qquad$ $3B_y = 3$

$B_x = 4/3$ $\qquad$ $B_y = 1$

$A_x = 8/3$ $\qquad$ $A_y = 2$

Illustrative Examples

32. Two vectors are given by $\vec{A} = i - 3j + k$ and $\vec{B} = -i + j - k$. Find

a) $\vec{A} + \vec{B}$ b) $\vec{A} - \vec{B}$

c) vectors $\vec{C}$ such that $3(\vec{A} - \vec{B}) + \vec{C} = 0$

Solution: a) $\vec{A} + \vec{B} = (i - 3j + k) + (-i + j - k)$

$= (1 - 1)i + (-3 + 1)j + (1 - 1)k = -2j$

b) $\vec{A} - \vec{B} = (i - 3j + k) - (-i + j - k)$

$= (1 - (-1))i + (-3 - 1)j + (1 - (-1))k = 2i - 4j + 2$

c) $3(\vec{A} - \vec{B}) + \vec{C} = 0 \Rightarrow \vec{C} = -3(\vec{A} - \vec{B})$

$\vec{C} = -3(2i - 4j + 2k) = -6i + 12j - 6k$

33. If $\vec{A} + \vec{B} = 3i + j$ and $\vec{A} - \vec{B} = -2i + 5j$. Find the magnitude of $\vec{A}$

Solution: From, $\vec{A} + \vec{B} = 3i + j$

and $\underline{\vec{A} - \vec{B} = -2i + 5j}$

$2\vec{A} = i + 6j \Rightarrow \vec{A} = \frac{1}{2}i + 3j$

$A = \sqrt{\left(\frac{1}{2}\right)^2 + (3)} \simeq 3$ units

34. Given that: $\vec{A} + \vec{B} = |A|\,i$ and $\vec{A} = -5i + 12j$. Find $\vec{B}$

Solution: $A = \sqrt{(-5)^2 + (12)^2} = 13$

$\Rightarrow \vec{A} + \vec{B} = 13i$ or $-5i + 12j + \vec{B} = 13i$

$\Rightarrow \vec{B} = 18i - 12j$

35. If $\vec{A} = 2i - 2j$, $\vec{B} = -2i + j$, $\vec{C} = 3i - j$. Determine m and n such that, $m\vec{A} + n\vec{B} + \vec{C} = 0$

Solution: $m\vec{A} + n\vec{B} + \vec{C} = 0$

$\Rightarrow m(2i-2j)+n(-2i+j)+3i-j=0$

$2mi-2mj-2ni+nj+3i-j=0$

$\Rightarrow 2mi-2ni+3i=0$

$2m-2n+3=0$ ..... (*)

And, $-2mj+nj-j=0$

$-2m+n-1=0$ ..... (**)

From equation (*) and (**)

$2m-2n+3=0$

$\underline{-2m+n-1=0}$

$-n+2=0 \Rightarrow n=2$

and, $-2m+2-1=0 \Rightarrow m=\frac{1}{2}$

36. If $\vec{P}=-4i+5j$ and $\vec{Q}=5i-7j$ and $\vec{P}-\vec{Q}+3\vec{C}=0$ is true. What are the components of vector $\vec{C}$?

Solution: $\vec{P}-\vec{Q}+3\vec{C}=0$

$\Rightarrow(-4i+5j)-(5i-7j)+3\vec{C}=0$

$(-4-5)i+(5+7)j+3\vec{C}=0$

$-9i+12j+3\vec{C}=0$

$\Rightarrow 3\vec{C}=9i-12j$

Or, $\vec{C}=3i-4j$

37. For the two vectors $\vec{A}=3i-2j$ and $\vec{B}=-i-4j$. Find the ratio $\frac{|\vec{A}+\vec{B}|}{|\vec{A}-\vec{B}|}$.

Solution: $\vec{A}+\vec{B}=(3i-2j)+(-i-4j)=(3-1)i+(-2-4)j=2i-6j$

$|\vec{A}+\vec{B}|=\sqrt{(2)^2+(-6)^2}=\sqrt{40}$

$\vec{A}-\vec{B}=(3i-2j)-(-i-4j)=(3-(-1))i+(-2(-4))j=4i-2j$

$|\vec{A}-\vec{B}|=\sqrt{(4)^2+(2)^2}=\sqrt{20}$

Then, $\frac{|\vec{A}+\vec{B}|}{|\vec{A}-\vec{B}|}=\frac{\sqrt{40}}{\sqrt{20}}=\frac{\sqrt{2}}{1}=\sqrt{2}:1$

38. If $\vec{A}-\vec{B}=3\vec{C}$ and $\vec{A}+\vec{B}=5\vec{C}$, where $\vec{C}=2i+4j$ then, what are the values of $\vec{A}$ and $\vec{B}$?

Solution: $\vec{A}-\vec{B}=3\vec{C}$

$\Rightarrow \vec{A} - \vec{B} = 3(2i + 4j)$ or $\vec{A} - \vec{B} = 6i + 12j$ ..... (*)

$\vec{A} + \vec{B} = 5\vec{C}$

$\Rightarrow \vec{A} + \vec{B} = 5(2i + 4j)$ or $\vec{A} + \vec{B} = 10i + 20j$ ..... (**)

Solving for the two equations, we get

$\vec{A} = 8i + 16j$ and $\vec{B} = 2i + 4j$

39. What must be vector $\vec{C}$ which when added with $\vec{A} = 8i - 2j$ and $\vec{B} = i + 9j$ gives rise to, $\vec{R} = 6i + 3j$.

Solution: $\vec{A} + \vec{B} + \vec{C} = \vec{R}$

$\Rightarrow (8i - 2j) + (i + 9j) + \vec{C} = 6i + 3j$

$9i + 7j + \vec{C} = 6i + 3j$

$\Rightarrow \vec{C} = -3i + 4j$

40. Is the vector $(i + j + k)$ a unit vector? Justify your answer.

Solution: Check the magnitude of the given vector, which is $\sqrt{(1)^2 + (1)^2 + (1)^2} = \sqrt{3}$

Hence, the given vector is NOT a unit vector since $\sqrt{3} \neq 1$.

41. The x and y components of vector $\vec{A}$ are 4 and 6 units respectively. If the x and y components of $(\vec{A} + \vec{B})$ are 10 and 9 units respectively. Find the vector $\vec{B}$.

Solution: $\vec{A} = 4i + 6j$, and

$\vec{A} + \vec{B} = 10i + 9j$

$\Rightarrow (4i + 6j) + \vec{B} = 10i + 9j$

$\vec{B} = 6i + 3j$

42. What vector must be added to $10i - 7j$ and $4i + 2j$ to get a resultant vector of $7j$?

Solution: $(10i - 7j) + (4i + 2j) + \vec{C} = 7j$

$14i - 5j + \vec{C} = 7j$

$\Rightarrow \vec{C} = -14i + 12j$

43. Which of the following is a vector quantity? (UEE 2003)

A. Power C. Momentum

B. Temperature D. Electric current **Answer: C**

44. Which pair of forces CANNOT give a resultant of 5N? (UEE 2003)

A. 1N and 4N C. 3N and 4N

B. 1N and 6N D. 1N and 7N

Explanation: The magnitude of the resultant of any two vectors $\vec{A}$ and $\vec{B}$ lies in the range. $|\vec{A}-\vec{B}| \le R \le |\vec{A}+\vec{B}|$

For example, 1 and 7 gives, $|1-7| \le R\ |1+7| \Rightarrow 6 \le R \le 8$

Thus, 5 is not in this range. **Answer: D**

45. Suppose, $\vec{A} = 3i - 2j$, $\vec{B} = -i - 4j$ and $\vec{C}$ are vectors in xy plane with the property $\vec{A}+\vec{B}-\vec{C}=0$. Which of the following is a unit vector along $\vec{C}$? (UEE 2003)

A. $2i - 6j$ C. $-2i + 6j$

B. $\sqrt{\frac{1}{10}}i - \frac{6}{\sqrt{10}}j$ D. $\sqrt{\frac{1}{10}}i - \frac{3}{\sqrt{10}}j$

Solution: $\vec{A}+\vec{B}-\vec{C}=0$

$(3i-2j)+(-i-4j)-\vec{C}=0$

$\vec{C} = 2i - 6j$

The unit vector along $\vec{C}$ is

$$\vec{C} = \frac{\vec{C}}{|\vec{C}|} = \frac{2i-6j}{\sqrt{(2)^2+(-6)^2}} = \frac{2i-6j}{\sqrt{40}} = \frac{2i-6j}{2\sqrt{10}} = \frac{1}{\sqrt{10}}i - \frac{3}{\sqrt{10}}j$$

Answer: D

46. Suppose $\vec{A} = 3i - 2j$ and $\vec{B} = -i - aj$ are two vectors in xy plane. What is the value of a such that $\vec{A}+\vec{B} = 2i$ (UEE 2004)

A. 3 B. 2 C. -2 D. -3

Solution: $\vec{A}+\vec{B} = 2i$

$(3i-2j)+(-i-aj) = 2i$

$2i + (-2-a)j = 2i$

Hence, $-2 - a = 0 \Rightarrow a = -2$ **Answer: C**

47. A vector that represents the position of an object in relation to another object is called (UEE 2005)

A. Unit vector C. Position vector

B. Coplanar vector D. Collinear vector

Answer: C

48. A vector of magnitude 20 is added to a vector of magnitude 25. The magnitude of their sum can be (UEE 2006)

A. 12 B. 3 C. 9 D. 47

Solution: $|A-B| \le R \le |A+B|$

$\Rightarrow |20-25| \le R \le |20+25|$

$5 \le R \le 45$ **A and C can be Answers**

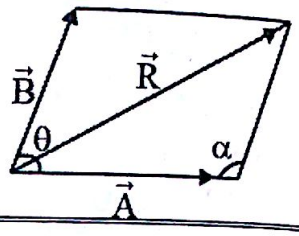
Both are within the range.

Cosine or parallelogram Law

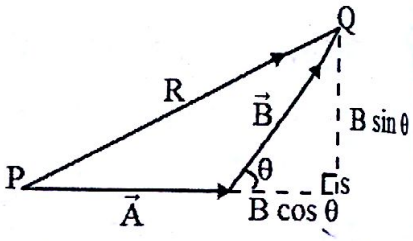
The resultant of two vectors $\vec{A}$ and $\vec{B}$ having an angle θ between them given by

$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

Or $$R = \sqrt{A^2 + B^2 - 2AB\cos\alpha}$$



That is, if we have two vectors $\vec{A}$ and $\vec{B}$ connected head to tail, as shown, then we can have the following:



From triangle PQS and Pythagoras theorem,

$R^2 = (A + B\cos\theta)^2 + (B\sin\theta)^2$

$= A^2 + 2AB\cos\theta + B^2\cos^2\theta + B^2\sin^2\theta = A^2 + 2AB\cos\theta + B^2(\cos^2\theta + \sin^2\theta)$

But, $\cos^2\theta + \sin^2\theta = 1$

$R^2 = A^2 + B^2 + 2AB\cos\theta$

$\Rightarrow \boxed{R = \sqrt{A^2 + B^2 + 2AB\cos\theta}}$

Conceptual examples

49. What should be the angle between two vectors $\vec{A}$ and $\vec{B}$ of equal magnitude for the resultants of their difference to be equal to magnitude of $\vec{A}$?

Explanation: $R^2 = A^2 + B^2 - 2AB\cos\theta$, but $R = A = B$

$$A^2 = A^2 + A^2 - 2A^2\cos\theta$$

$$\cos\theta = \frac{1}{2} \text{ or } \theta = 60°$$

50. The resultant of two vectors $\vec{P}$ and $\vec{Q}$ is 3P when the angle between them is 60° and it is P when the angle between them is 120°. What is the relationship between $\vec{P}$ and $\vec{Q}$?

Explanation: $R^2 = P^2 + Q^2 + 2PQ\cos\theta$

When $\theta = 60°$, $(3P)^2 = P^2 + Q^2 + 2PQ\cos 60°$

$9P^2 = P^2 + Q^2 + PQ$ ..... (*)

When $\theta = 120°$, $(P)^2 = P^2 + Q^2 + 2PQ\cos 120°$

$P^2 = P^2 + Q^2 - PQ$ ..... (**)

Solving equation (*) and (**) at a time,

$8P^2 = 2PQ$

$\Rightarrow P = \frac{1}{4}Q$

51. If $|P + Q| = |P - Q|$, what can you say about the two vectors?

Explanation: $|P + Q|^2 = P^2 + Q^2 + 2PQ\cos\theta$ ..... (*)

$|P - Q|^2 = P^2 + Q^2 - 2PQ\cos\theta$ ..... (**)

Since the two equations are equal,

$P^2 + Q^2 + 2PQ\cos\theta = P^2 + Q^2 - 2PQ\cos\theta$

$\Rightarrow \cos\theta = 0$ or $\theta = 90°$, $\vec{P}$ and $\vec{Q}$ are perpendicular.

52. The resultant of two vectors 3P and 2P is R. If the first vector is doubled, the resultant vector also becomes double. Find the angle between the vectors.

Explanation: First case, $R^2 = (3P)^2 + (2P)^2 + 2(3P)(2P)\cos\theta$

or $R^2 = 13P^2 + 12P^2\cos\theta$ ..... (*)

Second case, $(2R)^2 = (6P)^2 + (2P)^2 + 2(6P)(2P)\cos\theta$

or $R^2 = 10P^2 + 6P^2\cos\theta$ ..... (**)

From (*) and (**), we get

$13P^2 + 12P^2\cos\theta = 10P^2 + 6P^2\cos\theta$

$\cos\theta = -\frac{1}{2}$ or $\theta = 120°$

53. If $\vec{a} + \vec{b} = \vec{c}$ and $a + b = c$. What is the angle between $\vec{a}$ and $\vec{b}$?

Explanation: $c = \sqrt{a^2 + b^2 + 2ab\cos\theta}$ but $c = a + b$

$a + b = \sqrt{a^2 + b^2 + 2ab\cos\theta}$

$(a + b)^2 = a^2 + b^2 + 2ab\cos\theta$

$a^2 + b^2 + 2ab = a^2 + b^2 + 2ab\cos\theta$

$\Rightarrow \cos\theta = 1$ or $\theta = 0°$

Illustrative Example

54. Two non zero vectors D and E have equal magnitudes. For the magnitude of D + E to be 3 times larger than the magnitude of D – E, , what must be the angle between D and E? (UEE 2006)

A. 30° B. 37° C. 53° D. 60°

Solution: $D + E = \sqrt{D^2 + E^2 + 2DE\cos\theta}$ ..... (*)

$D - E = \sqrt{D^2 + E^2 - 2DE\cos\theta}$ .....(**)

but $D + E = 3(D - E)$

take ratio

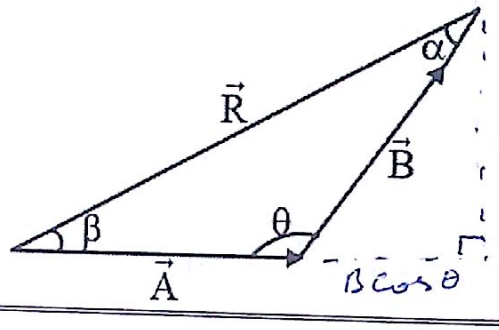
$$\frac{3(D-E)}{D-E}=\frac{\sqrt{D+E+2DE\cos\theta}}{\sqrt{D^2+E^2-2DE\cos\theta}}, \text{ but } D=E$$

$$\Rightarrow 9D^2+9E^2-18DE\cos\theta = D^2+E^2+2DE\cos\theta$$

$$\Rightarrow \cos\theta=\frac{4}{5} \text{ or } \theta=37^\circ$$

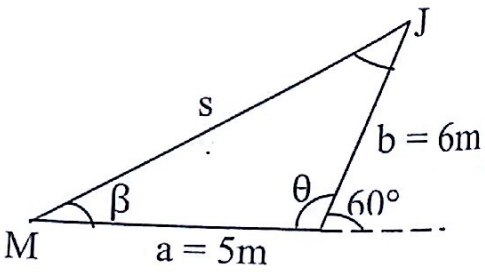
Answer: B

Sine Law: states that "The ratio of the sine of an angle to its opposite side is constant". That is, if we have a triangle with its corresponding sides and angles then: $\frac{\sin\theta}{R}=\frac{\sin\beta}{B}=\frac{\sin\alpha}{A}$



Illustrative Examples

5. Micheal Jackson once made his "Moon Walking" starting from point M for 5m horizontally and then for 6m to point J at an angle of 60° from the horizontal. Find



a) Micheal's displacement

b) His direction from the horizontal

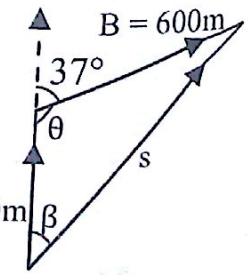
Solution: a) from cosine law

$$s=\sqrt{5^2+6^2+2(5)(6)\cos 60^\circ}=9.5\text{m}$$

From sine law

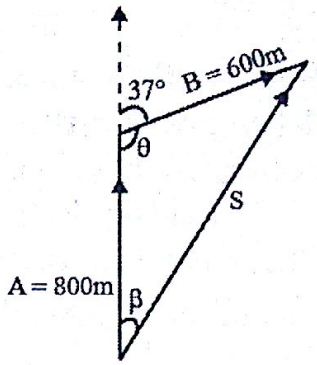
$$\frac{\sin\theta}{s}=\frac{\sin\beta}{b}\Rightarrow \sin\beta=\frac{b}{s}\sin\theta=\frac{6}{9.5}\sin 120^\circ$$

$$\Rightarrow \beta=33^\circ$$



A boy walks 800m to the North and then 0.6km at 37° East of North. What is?

a) The displacement, and
b) The direction of the boy



Solution: a) From cosine law

$$s = \sqrt{(800)^2 + (600)^2 + 2(800)(600)\cos 37°}$$

$$s = 1329 \text{ m}$$

b) From sine law

$$\frac{\sin\theta}{s} = \frac{\sin\beta}{B} \Rightarrow \sin\beta = \frac{B}{S}\sin\theta = \frac{600}{1329}\sin 143° \Rightarrow \beta = 15.8°$$

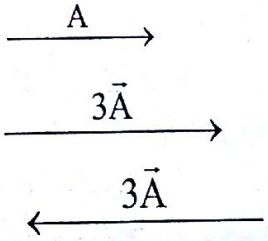
2.6 Multiplication of Vectors

A. Multiplying a Vector by a Scalar

When a vector is multiplied by a scalar, the result is a vector with the following properties.

> The product of a scalar k and a vector $\vec{A}$ is a new vector whose magnitude is KA and whose direction is the same as that of $\vec{A}$ if k is positive and opposite to $\vec{A}$ if k is negative.

For example, multiplying vector $\vec{A}$ by 3 increases its magnitude by a factor of 3, but does not change its direction.
Multiplying it by -3 increases the magnitude by a factor of 3 and reverses its direction.



Illustrative Examples

57. What will be the new vector if we multiply vector $\vec{A} = 6i + 2j + 12k$ by:

A. $k = 2$ B. $k = -2$ C. $k = \frac{1}{2}$

Solution: a) $k\vec{A} = 2(6i + 2j + 12k) = 12i + 4j + 24k$

b) $k\vec{A} = -2(6i + 2j + 12k) = -12i - 4j - 24k$

c) $k\vec{A} = \frac{1}{2}(6i + 2j + 12k) = 3i + j + 6k$

58. Vector $\vec{P}$ has magnitude of 5N pointing south. What are the magnitude and direction of:

A. $3\vec{P}$ B. $-3\vec{P}$

Solution: a) $3\vec{P} = 3(5) = 15\text{N south}$

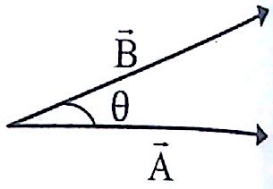
b) $-3P = 3(5) = 15N$ north

B. Multiplying A vector by Another Vector

When a vector is multiplied by a vector the result may be a scalar or a vector. We have scalar product and vector product.

I. Scalar or Dot product

The scalar product of two non – zero vectors $\vec{A}$ and $\vec{B}$ is defined by



$$\boxed{\vec{A}.\vec{B} = |A||B|\cos\theta}$$

This is the same as the product of the size of one vector with the component of the other in the direction of the first

- The dot or scalar product of the unit vectors take the following forms.
  - $i.i = j.j = k.k = (1)(1)\cos 0° = 1$
  - $i.j = j.k = k.i = (1)(1)\cos 90° = 0$

Consequently, the dot product of vector $\vec{A}$ and $\vec{B}$ in terms of their components becomes as follows.

If we have the space vectors $\vec{A}$ and $\vec{B}$ such that;

$\vec{A} = A_x i + A_y j + A_z k$, and

$\vec{B} = B_x i + B_y j + B_z k$, then

$$\vec{A}.\vec{B} = (A_x i + A_y j + A_z k).(B_x i + B_y j + B_z k)$$
$$= A_x B_x (i.i) + A_x B_y (i.j) + A_x B_z (i.k) + A_y B_x (j.i) + A_y B_y (j.j) + A_y B_z (j.k) + A_z B_x (k.i) + A_z B_y (k.j) + A_z B_z (k.k)$$

$$\Rightarrow \boxed{\vec{A}.\vec{B} = A_x B_x + A_y B_y + A_z B_z}$$

Scalar Projection

Consider vector $\vec{A} = A_x i + A_y j + A_z k$ and $\vec{B} = B_x i + B_x j + B_z k$ then we have,

The scalar projection of vector $\vec{A}$ and $\vec{B}$ is given by

$$|A|\cos\theta = \frac{\vec{A}.\vec{B}}{|B|} = \frac{A_x B_x + A_x B_y + A_z B_z}{|B|}$$

The scalar projection of vector $\vec{B}$ to $\vec{A}$ is given by

$$|B|\cos\theta = \frac{\vec{A}.\vec{B}}{|A|} = \frac{A_x B_x + A_y B_y + A_z B_z}{|A|}$$

Important points to remember

- Because of the notation $\vec{A}.\vec{B}$ scalar product is also known as *dot product* and read as, "$\vec{A}$ dot $\vec{B}$".
- The Result of scalar or dot product is scalar quantity.
- The scalar product obeys commutative and distributive laws. That is,
  - $\vec{A}.\vec{B} = \vec{B}.\vec{A}$, since $\cos\theta = \cos(-\theta)$
  - $\vec{A}.(\vec{B}+\vec{C}) = \vec{A}.\vec{B} + \vec{A}.\vec{C}$
- $\vec{A}.\vec{B} = AB\cos\theta$ implies
  - $A(B\cos\theta) = A$ (projection of $\vec{B}$ along $\vec{A}$), or
  - $B(A\cos\theta) = B$ (projection of $\vec{A}$ along $\vec{B}$)
- Two non-zero vectors are perpendicular if their dot product gives zero. That is

  $\vec{A}.\vec{B} = 0$, implies they are orthogonal.

Conceptual Examples

59. If $\vec{A}.\vec{B} = \vec{A}.\vec{C}$, can you conclude that $\vec{B}$ must equal $\vec{C}$? Explain

Explanation: No. If we take, $\vec{A} = i + 3j + 2k$, $\vec{B} = -2i + 9j + 5k$ and $\vec{C} = 5i + 10j$. It is true that, $\vec{A}.\vec{B} = \vec{A}.\vec{C}$ but $\vec{B} \neq \vec{C}$.

60. If $\vec{A}.\vec{B} = 0$. Does it necessarily follow that $\vec{A} = 0$ or $\vec{B} = 0$? Explain

Explanation: Not necessarily. $\vec{A}.\vec{B} = 0$ if $\vec{A} = 0$ or $\vec{B} = 0$ or if $\vec{A}$ and $\vec{B}$ are mutually perpendicular.

61. How can you verify that two non zero vectors are orthogonal?

Explanation: Two vectors are orthogonal or perpendicular if their dot product is zero. $\vec{A}.\vec{B} = 0 \Rightarrow \vec{A} \perp \vec{B}$

62. If the dot product of two vectors is positive does this imply that the vectors must have positive components?

Explanation: Not necessarily. Consider $\vec{A} = -2i - j - 3k$ and $\vec{B} = -i - 3$. We see $\vec{A}.\vec{B} = 5$, but the components are all negative.

63. What is the direction associated with the result of a dot product?

Explanation: None. Dot product gives rise to scalar quantity.

Illustrative Examples

64. If $\vec{A} = 2i + 4j - k$ and $\vec{B} = 3i - 2j - 3k$. Calculate $\vec{A}.\vec{B}$

Solution: $\vec{A}.\vec{B}\,(2i + 4j - k).\ (3j - 2j - 3k) = (2)(3) + (4)(-3)$
$(-1)(-3) = 1$

65. Given the vectors

$\vec{A} = 3i + 4j + 5k$ and $\vec{B} = 3i + 4j - 5k$

Find the angle between them

Solution: $\vec{A}.\vec{B} = AB\cos\theta$

$$\cos\theta = \frac{\vec{A}.\vec{B}}{AB} = \frac{A_xB_x + A_yB_y + A_zB_z}{\left(\sqrt{A_x^2 + A_y^2 + A_z^2}\right) + \left(\sqrt{B_x^2 + B_y^2 + B_z^2}\right)}$$

$$\cos\theta = \frac{(3)(3) + (4)(4) + (5)(-5)}{\left(\sqrt{3^2 + 4^2 + 5^2}\right)\left(\sqrt{3^2 + 4^2 + (-5)^2}\right)} = \frac{0}{\left(\sqrt{50}\right)\left(\sqrt{50}\right)}$$

$\cos\theta = 0 \Rightarrow \theta = 90°$

66. Given that $\vec{A} = 5i + 4j - 3k$ and $\vec{B} = 2i + 3j + 4k$. Find the sca projection of

A. $\vec{A}$ on to $\vec{A}$ B. $\vec{B}$ to $\vec{A}$

Solution:

a) $\vec{A}$ on to $\vec{B} = \dfrac{\vec{A}.\vec{B}}{|B|} = \dfrac{A_xB_x + A_yB_y + A_zB_z}{\left(\sqrt{B_x^2 + B_y^2 + B_z^2}\right)}$

$$\vec{A} \text{ on to } \vec{B} = \frac{(5)(2) + (4)(3) + (-3)(4)}{\sqrt{2^2 + 3^2 + 4^2}} = \frac{10}{\sqrt{29}}$$

b) $\vec{B}$ to $\vec{A} = \dfrac{\vec{A}.\vec{B}}{|A|} = \dfrac{A_xB_x + A_yB_y + A_zB_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}}$

$$\vec{B} \text{ on to } \vec{A} = \frac{(5)(2) + (4)(3) + (-3)(4)}{\sqrt{5^2 + 4^2 + (-3)^2}} = \frac{10}{\sqrt{50}}$$

67. For $\vec{A} = 3i + j - k$, $\vec{B} = -i + 2j + 5k$ and $\vec{C} = 2j - 3k$
Find $\vec{C}.(\vec{A} - \vec{B})$

Solution: $\vec{A} - \vec{B} = (3 - (-1))i + (1 - 2)j + (-1 - 5)k = 4i - j - 6k$

$\vec{C}.(\vec{A} - \vec{B}) = (2j - 3k).\ (4i - j - 6k)$

$\vec{C}.(\vec{A} - \vec{B}) = (0)(4) + (2)(-1) + (-3)(-6) = 16$

68. For what value of $\vec{C}$ lying along the x axis does $\vec{A}.(\vec{B}+\vec{C})=0$. Given that; $\vec{A}=3i-2j+k$ and $\vec{B}=4i+5j+7k$.

Solution: $\vec{C}=C_x i$

$\vec{B}+\vec{C}=(4+C_x)\,i+5j+7k$

$\vec{A}.(\vec{B}+\vec{C})=0$

$\Rightarrow(3)(4+C_x)+(-2)(5)+(1)(7)=0$

$\Rightarrow 12+3C_x-10+7=0$

$C_x=-3$

Therefore, $\vec{C}=-3i$

69. Three vectors are given by: $\vec{P}=3i+3j-2k$, $\vec{Q}=-i-4j+2k$ and $\vec{S}=2i+2j+k$. Find $3\vec{P}.(\vec{Q}+\vec{S})$

Solution: $3\vec{P}=9i-9j-6k$ and $\vec{Q}+\vec{S}=i-2j+3k$

$3\vec{P}.(\vec{Q}+\vec{S})=(9)(1)+(9)(-2)+(-6)(3)=-27$

70. Find the value of m such that the vectors $\vec{A}=4i-8j+10k$ and $\vec{B}=6i+mj-4k$ are perpendicular.

Solution: $\vec{A}.\vec{B}=0$

$(4)(6)+(-8)(m)+(10)(-4)=0$

$\Rightarrow m=-2$

71. Given that; $\vec{A}=5i-6j$ and $\vec{B}=-2i+3j$ and $\vec{C}$ lies in the xy plane and is perpendicular to $\vec{A}$. If the dot product of $\vec{C}$ and $\vec{B}$ is 9. Find $\vec{C}$

Solution: $\vec{A}.\vec{C}=0\Rightarrow(5)\,(C_x)+(-6)\,(C_y)=0$

And, $\vec{B}.\vec{C}=9\Rightarrow(-2)\,(C_x)+(3)(C_y)=9$

Solving Simultaneously, we get

$\vec{C}=18i+15j$

72. Find m such that; the angle between $\vec{A}=i+mj$ and $\vec{B}=i+j$ is 45°.

Solution: $\vec{A}.\vec{B}=AB\cos\theta\Rightarrow(1)(1)+(m)(1)$

$=\left(\sqrt{1^2+m^2}\right)\left(\sqrt{1^2+1^2}\right)\cos 45°$

$1+m=\left(\sqrt{1+m^2}\right)\left(\sqrt{2}\right)\left(\frac{1}{\sqrt{2}}\right)$

$$1+m=\sqrt{1+m^2}$$
$$(1+m^2)=1+m^2$$
$$1+2m+m^2=1+m^2 \Rightarrow 2m=0 \text{ or } m=0$$

73. Find $\vec{C}=xi+yj+k$ which is orthogonal to both $\vec{A}=3i+j$ and $\vec{B}=-3i+2j+2k$

Solution: $(x)(3)+(y)(1)+(1)(-1)=0$, and
$(x)(-3)+(y)(2)+(1)(2)=0$

$$\Rightarrow x=4 \text{ and } y=-\frac{1}{3}$$

$$\text{Thus, } \vec{C}=4i-\frac{1}{3}j+k$$

Application of dot product

The dot product of two vectors is applied while dealing with work done, power and energy etc.

Illustrative Example

74. Work done, A force of $(6i-2j)N$ displaced a particle to $(3i+j)$. Find
   a) The work done
   b) The angle between the force and the displacement.

Solution: a) $W=\vec{F}.\vec{S}=F_xS_x+F_yS_y+F_zS_z$
$W=(6)(3)+(-2)(1)=16J$

b) $\vec{F}.\vec{S}=FS\cos\theta \Rightarrow \cos\theta=\dfrac{\vec{F}.\vec{S}}{FS}$

$$\cos\theta=\frac{F_xS_x+F_yS_y+F_zS_z}{\left(\sqrt{F_x^2+F_y^2+F_z^2}\right)\left(\sqrt{S_x^2+S_y^2+S_z^2}\right)}$$

$$\cos\theta=\frac{16}{\left(\sqrt{6^2(-2)^2}\right)\left(\sqrt{3^2+1^2}\right)}=\frac{16}{\sqrt{400}}=\frac{16}{20}$$

$$\theta=37°$$

75. Power output: The displacement of a block subjected to a force of 100N at an angle of 60° is 15m in every 5 sec. What is the power output of the force?

Solution: $P=\vec{F}.\vec{V}=FV\cos\theta=F\left(\dfrac{s}{t}\right)\cos\theta$

$$P = 100\left(\frac{15}{5}\right)\cos 60° = 140W$$

76. Suppose $\vec{A} = 3i - 2j + k$, $\vec{B} = -i - 4j + 3k$ and $\vec{C}$ is a vector lying along x – axis with the property that $\vec{A}.(\vec{B} + \vec{C}) = 0$. Then, vector $\vec{C}$ is given by (UEE 2003)

A. $\frac{3}{\sqrt{10}}$ B. $-\frac{11}{3}i$ C. $\frac{6}{\sqrt{10}}i$ D. $-\frac{8}{3}i$

Solution: $\vec{A}.(\vec{B} + \vec{C}) = 0$

Where, $\vec{C} = C_x i$ and $\vec{B} + \vec{C} = (-1 + C_x)j - 4j + 3k$

$\Rightarrow \vec{A}.(\vec{B} + \vec{C}) = (3)(-1 + C_x) + (-2)(-4) + (1)(3) = 0$

$-3 + 3C_x + 8 + 3 = 0$

$C_x = -\frac{8}{3}$

$\Rightarrow \vec{C} = -\frac{8}{3}i$ **Answer: D**

77. Two non – zero vectors $\vec{A}$ and $\vec{B}$ are related by $\vec{A} = C\vec{B}$, where C is a scalar. If the two vectors have opposite directions, then one of the following is true about C? (UEE 2005)

A. C is positive C. $C = 1$

B. C is negative D. $C = 0$ **Answer: B**

78. The vector $2i + j + k$ is perpendicular to $i - 4j + \lambda k$. If λ is equal to (UEE 2006)

A. 0 B. -1 C. 2 D. 3

Solution: $\vec{A}.\vec{B} = 0 \Rightarrow (A_xB_x) + (A_yB_y) + (A_zB_z) = 0$

$\Rightarrow (2)(1) + (1)(-4) + (1)(\lambda) = 0 \Rightarrow \lambda = 2$ **Answer: C**

79. What is the angle between $A = (ai + a\sqrt{3}j)$ and $\vec{B} = (a\sqrt{3}i + aj)$ (UEE 2007)

A. 90° B. 60° C. 37° D. 30°

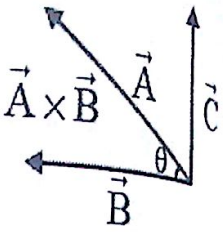
Solution: $\vec{A}.\vec{B} = AB\cos\theta \Rightarrow \cos\theta = \frac{A.B}{AB}$

$$\cos\theta = \frac{a^2\sqrt{3} + a^2\sqrt{3}}{\left(\sqrt{a^2 + 3a^2}\right)\left(\sqrt{3a^2 + a^2}\right)} = \frac{2a^2\sqrt{3}}{4a^2} = \frac{\sqrt{3}}{2}$$

$\Rightarrow \theta = 30°$ **Answer: D**

II. Vector or Cross Product

The vector product of two non – zero vectors $\vec{A}$ and $\vec{B}$ is defined by $\vec{C} = \vec{A} \times \vec{B}$



$$\boxed{\vec{A} \times \vec{B} = |A||B| \sin\theta}$$

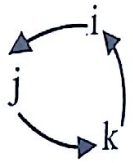
The direction of the vector $\vec{A} \times \vec{B}$ is determined by the right hand rule and it is perpendicular to both vector $\vec{A}$ and vector $\vec{B}$.

- The cross or vector product of the unit vectors take the following form.
  - $i \times i = j \times j = k \times k = (1)(1) \sin 0° = 0$
  - $i \times j = k$, but $j \times i = -k$

    $j \times k = i$, but $k \times j = -i$

    $k \times i = j$, but $i \times k = -j$

Consequently, the cross product of vector $\vec{A}$ and $\vec{B}$ in terms of their component becomes as follows.



Positive direction

If we have the space vectors $\vec{A}$ and $\vec{B}$ such that:

$\vec{A} = A_x i + A_y j + A_z k$, and

$\vec{B} = B_x i + B_y j + B_z k$, then

$$\vec{A} \times \vec{B} = (A_x i + A_y j + A_z k) \times (B_x i + B_y j + B_z k)$$
$$= A_x B_x (i \times i) + A_x B_y (i \times j) + A_x B_z (i \times k) +$$
$$A_y B_x (j \times i) + A_y B_y (j \times j) + A_y B_z (j \times k) +$$
$$A_z B_x (k \times i) + A_z B_y (k \times j) + A_z B_z (k \times k)$$

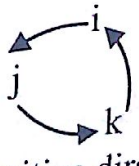
$$\Rightarrow \boxed{\vec{A} \times \vec{B} = (A_y B_z - A_z B_y)i + (A_z B_x - A_x B_z)j + (A_x B_y - A_y B_x)k}$$

- A better way to perform a vector product is to use the determinant form. That is, assuming

  $\vec{A} = A_x i + A_y j + A_z k$,

  $\vec{B} = B_{xi} + B_{yj} + B_{zk}$ And,

  $\vec{A} \times \vec{B} = \vec{C} = C_{xi} + C_{yj} + C_{zk}$, then



Positive direction

$$\vec{C} = \vec{A} \times \vec{B} = \begin{vmatrix} i & j & k \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

  - For the x-component in the i-direction.

$$C_x = \begin{vmatrix} A_y & A_z \\ B_y & B_z \end{vmatrix} i = (A_y B_z - A_z B_y) i$$

- For the y – component in the j – direction

$$C_y = \begin{vmatrix} A_x & A_z \\ B_x & B_z \end{vmatrix} j = (A_z B_x - A_x B_z) j$$

- For the z – direction in the k – direction

$$C_z = \begin{vmatrix} A_x & A_y \\ B_x & B_y \end{vmatrix} k = (A_x B_y - A_y B_x) k$$

Finally, we have $\vec{A} \times \vec{B} = \vec{C} = C_x i + C_y j + C_z k$.

$$\boxed{\vec{A} \times \vec{B} = (A_y B_z - A_z B_y) i + (A_z B_x - A_x B_z) j + (A_x B_y - A_y B_x) k}$$

Important points to remember

- Because of the notation $\vec{A} \times \vec{B}$, vector product is also known as *cross product* and read as " $\vec{A}$ cross $\vec{B}$ ".
- The result of vector or cross product is vector quantity.
- Vector product is not commutative
  - $\vec{A} \times \vec{B} \neq \vec{B} \times \vec{A}$, but $\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A})$
- Vector product is Distributive over addition
  - $\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$
- The vector product of two vectors $\vec{A}$ and $\vec{B}$ is perpendicular to both vectors.
  - If $\vec{A} \times \vec{B} = \vec{C}$, then $\vec{C}$ is perpendicular to both vectors $\vec{A}$ and $\vec{B}$.
- Two non – zero vectors are collinear, that is parallel or anti parallel to each other if their cross product gives zero. That is, $\vec{A} \times \vec{B} = 0$, implies they are parallel or anti parallel.

Conceptual Examples

80. How can we verify that two non zero vectors are collinear?

Explanation: If $\vec{A} \neq 0$, and $\vec{B} \neq 0$, then $\vec{A} \times \vec{B} = AB \sin\theta = 0$ iff $\theta = 0°$ or $\theta = n180°$, where n is whole number.

81. What is the vector product of a vector with itself?

Explanation: Let $\vec{A}$ be a non – zero vector, then $\vec{A} \times \vec{A} = |A||A| \sin 0° =$ zero.

82. Is the triple product given by $\vec{A}.(\vec{B}\times\vec{C})$ a scalar or a vector? Explain

Explanation: Let $\vec{B}\times\vec{C}=\vec{D}$, then the triple product $\vec{A}.(\vec{B}\times\vec{C})=\vec{A}.\vec{D}$, the result of a dot product is scalar.

83. If $|\vec{A}\times\vec{B}|=\vec{A}.\vec{B}$. What can you say about the angle between vector $\vec{A}$ and $\vec{B}$?

Explanation: $|\vec{A}\times\vec{B}|=A.B$

$AB\sin\theta = AB\cos\theta$

$\tan\theta = 1$ or $\theta = 45°$

84. No matter what $\vec{A}$ and $\vec{B}$ may be, $\vec{A}.(\vec{A}\times\vec{B})=0$. Is this claim true or false? Explain

Explanation: It is true, Because, $\vec{A}\times\vec{B}$ is perpendicular to $\vec{A}$ and $\vec{B}$. Hence, $\vec{A}.(\vec{A}\times\vec{B})=(A)(A\times B)\cos 90°=0$.

Illustrative Example

85. Let $\vec{A}=2i-j+3k$ and $\vec{B}=-i-2j+4k$. Find $\vec{A}\times\vec{B}$.

Solution: $\vec{A}\times\vec{B} = (A_yB_z - A_zB_y)i + (A_zB_x - A_xB_z)j + (A_xB_y - A_yB_x)k = ((-1)(4) - (3)(-2))i + ((3)(-1)-(2)(4))j + ((2)(-2) - (-1)(-1))k = 2i - 11j - 5k$

86. If $\vec{A}=2i+3j$ and $\vec{B}=i+j$. Find $(\vec{A}+\vec{B})\times(-\vec{B})$.

Solution: $\vec{A}+\vec{B}=3i+4j$ and $-\vec{B}=-i-j$

Hence, $(\vec{A}+\vec{B})\times(-\vec{B})=\begin{vmatrix} i & j & k \\ 3 & 4 & 0 \\ -1 & -1 & 0 \end{vmatrix} = ((3)(-1)-(4)(-1))k = k$

87. If $\vec{A}=-5i+5j$, $\vec{A}\times\vec{B}=-2k$ and $\vec{A}.\vec{B}=10$. Find $\vec{B}$

Solution: $\vec{A}\times\vec{B}=-2k$, $(A_xB_y - A_yB_x)k$

$\Rightarrow A_xB_y - A_yB_x = -2$ or $-5B_y - 5B_x = -2$ .....(*)

$\vec{A}.\vec{B}=10=A_xB_x + A_yB_y$

$\Rightarrow -5B_x + 5B_y = 10$, or $-B_x + B_y = 2$ .....(**)

Solve equation (*) and (**)

$B_x = -2.8$ and $B_y = -0.8$

88. Find a unit vector perpendicular to both vector $\vec{A}$ and vector $\vec{B}$. Given that: $\vec{A}=3i+j+2k$ and $\vec{B}=2i-2j+4k$

Solution: $\vec{u} = \dfrac{\vec{A}\times\vec{B}}{|\vec{A}\times\vec{B}|}$

$$\vec{A}\times\vec{B} = \begin{vmatrix} i & j & k \\ 3 & 1 & 2 \\ 2 & -2 & 4 \end{vmatrix} = ((1)(4)-(2)(-2))i+((2)(2)-(3)(4))j + ((3)(-2)-(1)(2))k$$

$$\vec{A}\times\vec{B} = 8i - 8j - 8k$$

$$|\vec{A}\times\vec{B}| = \sqrt{(8)^2+(-8)^2+(-8)^2} = 8\sqrt{3}$$

$$\therefore \vec{u} = \frac{\vec{A}\times\vec{B}}{|\vec{A}\times\vec{B}|} = \frac{8i-8j-8k}{8\sqrt{3}} = \frac{i-j-k}{\sqrt{3}}$$

89. Vector $\vec{A}$ has magnitude of 6 units along the positive x axis. Vector $\vec{B}$ has magnitude of 4 units and lies in xy plane making an angle of 30° with the positive x-axis. Find $\vec{A}\times\vec{B}$

Solution: From the given information

$$\vec{A} = 6i$$

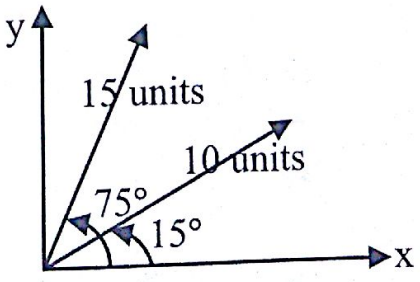
$$\vec{B} = 4\cos 30°i + 4\sin 30°j = 4\left(\frac{\sqrt{3}}{2}\right)i + 4\left(\frac{1}{2}\right)j$$

$$\vec{B} = 2\sqrt{3}i + 2j$$

$$\vec{A}\times\vec{B} = \begin{vmatrix} i & j & k \\ 6 & 0 & 0 \\ 2\sqrt{3} & 2 & 0 \end{vmatrix} = 12k$$

90. Two vectors $\vec{A}$ and $\vec{B}$ are given as shown. Find the magnitude of

A. $\vec{A}.\vec{B}$ B. $\vec{A}\times\vec{B}$



Solution: $\theta = 75° - 15° = 60°$

a) $\vec{A}.\vec{B} = AB\cos 60°$

$= (10)(15)\left(\frac{1}{2}\right) = 75$ units

b) $|\vec{A}\times\vec{B}| = AB\sin 60° = (10)(15)\dfrac{\left(\sqrt{3}\right)}{2} = 75\sqrt{3}$ units

91. Two vectors have equal magnitude of A = 3 units and $\vec{B}$ = 3 units. If $\vec{A}\times\vec{B} = 4i + 2k$. Find the angle between the vectors.

Solution: $|\vec{A}\times\vec{B}| = AB\sin\theta$

$$\sin\theta = \frac{|\vec{A}\times\vec{B}|}{AB} = \frac{\sqrt{(4)^2+(2)^2}}{(3)(3)} = 0.497 \simeq 0.5 \Rightarrow \theta = 30^\circ$$

92. Suppose vectors $\vec{A} = 2(i + j)$ and $\vec{B} = 2i + 3j + k$. Which of the following is a unit vector perpendicular to both vectors? (UEE 2001)

A. $\frac{1}{\sqrt{3}}(-i+j+k)$ C. $\frac{1}{\sqrt{3}}(i+j+k)$

B. $\frac{1}{\sqrt{3}}(i-j-k)$ D. $\frac{1}{\sqrt{3}}(i-j+k)$

Solution: $\vec{u} = \frac{\vec{A}\times\vec{B}}{|\vec{A}\times\vec{B}|}$

$$\vec{A}\times\vec{B} = \begin{vmatrix} i & j & k \\ 2 & 2 & 0 \\ 2 & 3 & 1 \end{vmatrix} = 2i - 2j + 2k$$

$$\vec{A}\times\vec{B} = \sqrt{(2)^2+(-2)^2+(2)^2} = 2\sqrt{3}$$

$$\therefore \vec{u} = \frac{\vec{A}\times\vec{B}}{|\vec{A}\times\vec{B}|} = \frac{2i-2j+2k}{2\sqrt{3}} = \frac{1(i-j+k)}{\sqrt{3}}$$ **Answer: D**

93. Which of the following expression is NOT correct in a clock wise system (UEE 2001)?

A. $k \times i = j$ C. $i \times j = -j \times i$

B. $i \times i = 1$ D. $i \times k = -j$ **Answer: B**

94. Which of the following is NOT true of vector product (UEE 2002)

A. $\vec{A}\times\vec{B} = -\vec{B}\times\vec{A}$ C. $\vec{A}\perp\vec{B} = |\vec{A}\times\vec{B}| = AB$

B. $\vec{A}\times(\vec{B}+\vec{C}) = \vec{B}\times\vec{A} + \vec{A}\times\vec{C}$ D. $\vec{A}//\vec{B} \Rightarrow |\vec{A}\times\vec{B}| = 0$

Answer: B

95. If $\vec{A} = 8i + j - 2k$ and $\vec{B} = 5i - 3j + k$. Then, what is the value of $\vec{B}\times\vec{A}$? (UEE 2005)

A. $3i + 4j - 3k$ C. $-5i - 18j - 29k$

B. $-3i-4j+3k$ D. $5i+18j+29k$

Solution: $\vec{B}\times\vec{A}=\begin{vmatrix} i & j & k \\ 5 & -3 & 1 \\ 8 & 1 & -2 \end{vmatrix}$

$=((-3)(-2)-(1)(1))i+((1)(8)-(5)(-2))j+((5)(1)-(-3)(8))k$

$=5i+18j+29k$ **Answer: D**

Application of Cross product

Cross product is applicable in the computation of Torque, Angular momentum, magnetic force, area of parallelogram, Area of triangle, volume of parallel piped.

Illustrative Examples

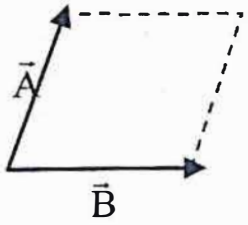
96. Torque developed: what is the torque about the origin if a force of $\vec{F}=(i+2j+3k)N$ acts at a position vector of $\vec{r}=(2i+3j-k)m$.

Solution: Torque, $\tau=\vec{r}\times\vec{F}=(11i-7j+k)\ N.m$

97. Magnetic force: If a particle of charge 2C is moving at a velocity of $(i+2j+3k)m/s$ in a magnetic field of $(4k)N/C$. What is the magnetic force acting on the particle?

Solution: $\vec{F}=q\vec{v}\times\vec{B}=2(8i-4j)=(16i-8j)N$

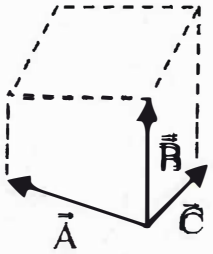
98. Area of parallelogram: What is the area of the parallelogram formed by vectors $\vec{A}=3i+4j$ and $\vec{B}=i-3k$



Solution: Area of parallelogram $=|\vec{A}\times\vec{B}|$

$a=\vec{A}\times\vec{B}=\sqrt{(-12)^2+(9)^2+(-4)^2}=15.5$ unit square

99. Volume of parallelepiped: Let $\vec{A}=i-j$, $\vec{B}=2i+3j-k$ and $\vec{C}=-i+2k$. Find the volume of the parallelepiped.



Solution: Volume: $\vec{A}.(\vec{B}\times\vec{C})$

$\vec{B}\times\vec{C}=6i-3j-3k$

$V=\vec{A}.(\vec{B}\times\vec{C})=(i-j).(6i-3j-3k)=9$ units cube

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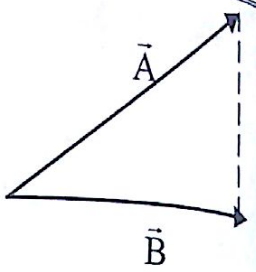
100. Find the area of a triangle formed by vectors $\vec{A} = i + 5j$ and $\vec{B} = 3i - j$.



Solution: Area $= \frac{1}{2}|\vec{A} \times \vec{B}| = 8$ unit square

Additional Examples

General on vectors

101. Which of the following is not a vector quantity?
   A. Density
   B. Electric filed intensity
   C. Displacement
   D. Angular Momentum

Answer: A

102. What is the magnitude of the vector $3i + 2j + \hat{k}$?
   A. $\sqrt{5}$ B. $\sqrt{6}$ C. $\sqrt{14}$ D. $\sqrt{24}$

Solution: $|A| = \sqrt{3^2 + 2^2 + 1^2} = \sqrt{14}$

Answer: C

103. Which of the following is a vector quantity?
   A. Electric current
   B. Electrostatic potential
   C. Charge
   D. moment of a spinning body

Answer: D

104. A vector is not changed if
   A. It is displaced parallel to itself.
   B. It is rotated through an arbitrary angle.
   C. It is cross – multiplied by a unit angle.
   D. It is multiplied by an arbitrary scalar.

Explanation: When a vector is displaced parallel to itself neither its magnitude nor its direction changes.

Answer: A

105. Which of the following statement is correct?
   A. Displacement, force and electric field are vector quantities.
   B. Velocity, length and acceleration are vector quantities.
   C. Relative velocity, mass and work are vector quantities.
   D. Energy, weight and force are vector quantities.

Answer: A

106. Indentify the vector quantity
   A. Work
   B. Time
   C. Heat
   D. Angular momentum

Answer: D

107. Identify the scalar quantity
A. Work B. Force C. Acceleration D. Displacement
Answer: A

108. Which of the following is neither a scalar nor a vector?
A. Torque C. Angular momentum
B. Force D. Dielectric constant
Explanation: Dielectric constant is a tensor, it has different values in different directions. **Answer: D**

109. Which of the following is a vector quantity?
A. Electric current C. Gravitational potential
B. Current density D. Angle **Answer: B**

110. The vector joining the points (1, 1, −1) and (2, −3, 4) is:
A. $-i + 4j - 5k$ C. $i - 4j - 5k$
B. $i + 4j + 5k$ D. $-i - 4j - 5k$
Explanation: $\vec{P} = (2-1)i + (-3-1)j + (4-(-1))k$
$\vec{P} = i - 4j + 5k$ **Answer: C**

111. The co-ordinate of the end points of a vector are (4, −4, 0) and (−2, −2, 0). The magnitude of this vector is:
A. $2\sqrt{10}$ units C. $4\sqrt{10}$ units
B. $3\sqrt{10}$ units D. $5\sqrt{10}$ units
Explanation: $\vec{P} = (-2-4)i + (-2-(-4))j + (0-0)k$
$\vec{P} = -6i + 2j$
$|P| = \sqrt{-6^2 + 2^2} = \sqrt{40} = 2\sqrt{10}$ units **Answer: A**

112. Which of the following is a scalar quantity?
A. Electric current C. Velocity
B. Force D. Linear momentum
Answer: A

113. Which of the following is a vector quantity?
A. Electric current C. Temperature
B. Flux density D. Magnetic field intensity
Answer: D

Composition and Resolution of vectors

114. The position vectors of a moving particle at two different points are $\vec{r}_1$ and $\vec{r}_2$. The corresponding displacement vector is
A. $\vec{r}_1 + \vec{r}_2$ B. $\vec{r}_1 - \vec{r}_2$ C. $\sqrt{r_1^2 + r_2^2}$ D. $\sqrt{r_1^2 - r_2^2}$

Explanation: Displacement vector is equal to difference of position vectors. **Answer: B**

115. If $\vec{A}=\vec{B}+\vec{C}$ and $\vec{B}=\vec{C}+\vec{A}$, then vector $\vec{C}$ is equal to:

A. $\vec{A}+\vec{B}$ B. $\vec{B}-\vec{A}$ C. $\vec{A}-\vec{B}$ D. Null vector

Solution: $\vec{A}=\vec{C}+\vec{A}+\vec{C}\Rightarrow 2\vec{C}=0$ or $\vec{C}=0$ **Answer: D**

116. When can two vectors be divided?

A. If they represent like physical quantities.
B. If they represent unlike physical quantities.
C. If they do not represent physical quantities.
D. Never

Explanation: Vectors can never be divided. **Answer: D**

117. If $\vec{P}+\vec{Q}=\vec{P}-\vec{Q}$ and θ is the angle between $\vec{P}$ and $\vec{Q}$ then,

A. $\theta=0$ B. $\theta=90°$ C. $P=0$ D. $Q=0$

Explanation: $\vec{P}+\vec{Q}=\vec{P}-\vec{Q}\Rightarrow 2\vec{Q}=0$ or $Q=0$ **Answer: D**

118. The maximum value of resultant of two vectors $\vec{P}$ and $\vec{Q}$ is?

A. $P+Q$ B. $P-Q$ C. $\sqrt{P^2+Q^2}$ D. $\sqrt{P^2-Q^2}$

Explanation: The maximum is P + Q and the minimum is P – Q. **Answer: A**

119. What is the angle between two vectors (x + y) and (x – y) if their resultant is $\sqrt{2(x^2+y^2)}$?

A. 0° B. 30° C. 60° D. 90°

Solution: $C=\sqrt{A^2+B^2+2AB\cos\theta}$

$\Rightarrow\left(\sqrt{2(x^2+y^2)}\right)=\sqrt{(x+y)^2+(x-y)^2+2(x+y)(x-y)\cos\theta}$

$2(x^2+y^2)=x^2+y^2+2xy+x^2+y^2-2xy+2(x^2-y^2)\cos\theta$

$\Rightarrow\cos\theta=0$ or $\theta=90°$ **Answer: D**

120. The minimum number of vectors in different planes, which can be added to give zero resultant is

A. 5 B. 4 C. 3 D. 2

Answer: B

121. Two vectors $\vec{A}$ and $\vec{B}$ lie in a plane. Another vector $\vec{C}$ lies outside this plane. The resultant $\vec{A}+\vec{B}+\vec{C}$ of these three vectors.

A. Can be zero
B. Cannot be zero
C. lies in the plane of $\vec{A}+\vec{B}$
D. lies in the plane of $\vec{A}-\vec{B}$

Explanation: The vector sum of $\vec{A}$ and $\vec{B}$ may be equal in magnitude to $\vec{C}$ but can never be opposite. So, $(\vec{A}+\vec{B})+\vec{C}$ can never be zero. **Answer: B**

122. The resultant of two vectors, each P, acting at an angle θ is

A. $2P\sin\frac{\theta}{2}$ B. $2P\cos\frac{\theta}{2}$ C. $2P\cos\theta$ D. $P\sqrt{2}$

Solution: $R=\sqrt{P^2+P^2+2P^2\cos\theta}=\sqrt{2P^2+2P^2\cos\theta}$

$R=\sqrt{2P^2(1+\cos\theta)}$ but $1+\cos\theta=1+2\cos^2\frac{\theta}{2}-1$

$R=\sqrt{4P^2\cos^2\frac{\theta}{2}}=2P\cos\frac{\theta}{2}$ **Answer: B**

123. A zero resultant cannot be obtained by combining:

A. Two vectors of different magnitudes.
B. Three vectors of different magnitudes.
C. Four vectors of different magnitudes.
D. Five vectors of different magnitudes. **Answer: A**

124. Which of the following sets of displacements might be capable of bringing a car to its initial point?

A. 5, 10, 30 and 50 km C. 40, 40, 90 and 200 km
B. 5, 7, 9 and 16 km D. 10, 20, 40 and 90km

Explanation: The resultant of 5 and 7 varies from 2 to 12, so 9 is within the range, similarly 7 and 9. **Answer: B**

125. When two vectors $\vec{a}$ and $\vec{b}$ are added, the magnitude of the resultant vector is always?

A. Greater than (a + b) C. less than (a + b)
B. Less than or equal to (a+b) D. equal to (a + b)

Explanation: The maximum possible value of resultant of $\vec{a}$ and $\vec{b}$ is (a + b). **Answer: B**

126. The resultant of two vectors of magnitude 2A and $\sqrt{2}$A acting at an angle θ is $\sqrt{10}$A. The correct value of θ is

A. 30° B. 45° C. 60° D. 90°

Solution: $C^2=A^2+B^2+2AB\cos\theta$

$\Rightarrow\left(\sqrt{10}A\right)^2=(2A)^2+\left(\sqrt{2}A\right)^2+2(2A)\left(\sqrt{2}A\right)\cos\theta$

$$\cos\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45°$$

Answer: B

127. Given: $\vec{A} = 2i + 3j$ and $\vec{B} = 5i - 6j$. The magnitude of $\vec{A} + \vec{B}$ is

A. 4 units B. 10 units C. $\sqrt{58}$ units D. $\sqrt{61}$ units

Solution: $\vec{A} + \vec{B} = (2i + 3j) + (5i - 6j) = 7i - 3j$

$$|\vec{A} + \vec{B}| = \sqrt{7^2 + 3^2} = \sqrt{58}$$

Answer; C

128. A set of vectors taken in a given order gives a closed polygon. Then the resultant of these vectors is

A. Scalar quantity C. Unit vector
B. Pseudo vector D. Null vector **Answer: D**

129. The sum of two vectors $\vec{A}$ and $\vec{B}$ is maximum when the angle θ between their positive directions is

A. 0° B. $\pi/4$ C. $\pi/2$ D. π

Answer: A

130. A vector is represented by $i + 3j + 5k$. Its length in xy plane is

A. 1 B. 3 C. 5 D. $\sqrt{10}$

Solution: $\ell = \sqrt{1^2 + 3^2} = \sqrt{10}$ **Answer: D**

131. Two vectors $\vec{a}$ and $\vec{b}$ inclined at an angle θ have a resultant $\vec{C}$ which makes an angle β with $\vec{a}$. If the direction of $\vec{a}$ and $\vec{b}$ are interchanged, then the resultant will have the same

A. Magnitude C. Magnitude as well as direction
B. Direction D. Neither magnitude nor direction

Explanation: Since the magnitude and the angel between the vectors is not changed, the magnitude of the resultant remain the same but its direction will be changed. **Answer: A**

132. Given: $\vec{C} = \vec{A} + \vec{B}$, and the magnitude of $\vec{A}, \vec{B}$ and $\vec{C}$ are 12, 5 and 13 respectively. The angle between $\vec{A}$ and $\vec{B}$ is

A. 0° B. $\pi/4$ C. $\pi/2$ D. π

Solution: $C^2 = A^2 + B^2 + 2AB\cos\theta$

$$13^2 = 12^2 + 5^2 + 2(12)(5)\cos\theta$$

$$\Rightarrow \cos\theta = 0 \text{ or } \theta = 90° = \pi/2$$

Answer: C

133. The resultant of $\vec{P}$ and $\vec{Q}$ makes an angle θ_1, with $\vec{P}$ and θ_2 with $\vec{Q}$. Hence,

A. $\theta_1 < \theta_2$ C. $\theta_1 < \theta_2$ if $\vec{P} > \vec{Q}$

B. $\theta_1 < \theta_2$ if $\vec{P} > \vec{Q}$ D. $\theta_1 < \theta_2$ if $\vec{P} = \vec{Q}$

Explanation: The resultant is more inclined towards the bigger vector. **Answer: C**

134. What is the component of $3i + 9j$ along $z-$axis?

A. 0 B. 3 C. 9 D. $\sqrt{3^2+9^2}$

Answer: A

135. Two vectors $\vec{A}$ and $\vec{B}$ are such that $\vec{A}+\vec{B}=\vec{C}$ and $A^2 + B^2 = C^2$. Which of the following is correct?

A. $\vec{A}$ is parallel to $\vec{B}$

B. $\vec{A}$ is anti-parallel to $\vec{B}$

C. $\vec{A}$ is perpendicular to $\vec{B}$

D. $\vec{A}$ and $\vec{B}$ are equal in magnitude

Explanation: The condition, $C^2 = A^2 + B^2$ is satisfied for perpendicular vectors. **Answer: C**

136. Given: $\vec{A} = i + 2j - 3k$. When vector $\vec{B}$ is added to $\vec{A}$, we get a unit vector along x−axis. Then B is?

A. $-2j + 3k$ C. $-i + 3k$

B. $-i - 2j$ D. $2j - 3k$

Solution: $\vec{B} + (i - 2j - 3k) = i \Rightarrow \vec{B} = -2j + 3k$ **Answer: A**

137. Given: $\vec{R} = \vec{A} + \vec{B}$ and $R = A = B$. The angle between $\vec{A}$ and $\vec{B}$ is

A. 60° B. 90° C. 120° D. 180°

Solution: $R^2 = A^2 + B^2 + 2AB\cos\theta$

$$R^2 = R^2 + R^2 + 2R^2\cos\theta$$

$$\cos\theta = -\frac{1}{2} \text{ or } \theta = 120°$$

Answer: C

138. Given: $\vec{C} = \vec{A} + \vec{B}$ and $|\vec{C}| = |\vec{A}| + |\vec{B}|$. The angle between $\vec{A}$ and $\vec{B}$ is

A. 0° B. 45° C. 60° D. 90°

Explanation: The resultant is A + B when they act in the same direction. **Answer: A**

139. Given: $\vec{P} = \vec{A} + \vec{B}$ and $P = A + B$. The angle between $\vec{A}$ and $\vec{B}$ is

A. $0°$ B. $\pi/4$ C. $\pi/2$ D. π

Solution: $P^2 = A^2 + B^2 + 2AB\cos\theta$

$(A+B)^2 = A^2 + B^2 + 2AB\cos\theta$

$A^2 + B^2 + 2AB = A^2 + B^2 + 2AB\cos\theta \Rightarrow \cos\theta = 1$ or $\theta = 0°$

Answer: A

140. Two vectors $\vec{a}$ and $\vec{b}$ are such that $|\vec{a}+\vec{b}| = |\vec{a}-\vec{b}|$. What is the angle between $\vec{a}$ and $\vec{b}$?

A. $0°$ B. $90°$ C. $60°$ D. $180°$

Solution: $a^2 + b^2 + 2ab\cos\theta = a^2 + b^2 - 2ab\cos\theta$

$\cos\theta = 0 \Rightarrow \theta = 90°$

Answer: B

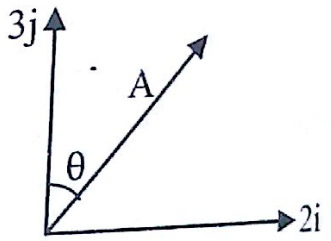
141. Given: $\vec{A} = 2i + 3j$. The angle between $\vec{A}$ and y – axis is:

A. $\sin^{-1} 2/3$

B. $\cos^{-1} 2/3$

C. $\tan^{-1} 2/3$

D. $\tan^{-1} 3/2$



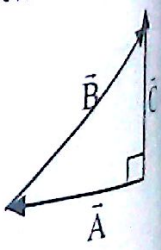
Solution: $\tan\theta = \dfrac{2}{3} \Rightarrow \theta = \tan^{-1}\left(\dfrac{2}{3}\right)$

Answer: C

142. In the given vector diagram, what is the angel between $\vec{A}$ and $\vec{B}$ if $C = \dfrac{B}{2}$

A. $30°$ C. $120°$

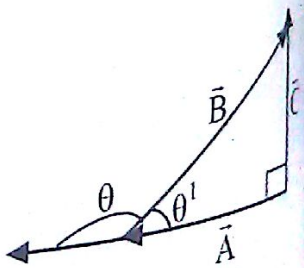
B. $60°$ D. $150°$



Solution: $\sin\theta^1 = \dfrac{C}{B}$

$\sin\theta^1 = \dfrac{B/2}{B}$

$\sin\theta^1 = \dfrac{1}{2}$ or $\theta^1 = 30°$



$\theta + \theta^1 = 180° \Rightarrow \theta = 180° - 30° = 150°$ **Answer: D**

143. What is the unit vector along $i + j$?

A. $\frac{i+j}{\sqrt{2}}$ B. $\sqrt{2}(i+j)$ C. $i+j$ D. k

Solution: $\vec{A} = \frac{\vec{A}}{A} = \frac{i+j}{|j+j|} = \frac{i+j}{\sqrt{2}}$ **Answer: A**

144. Given: $0.3i + 0.4j + nk$ is a unit vector. The value of n is

A. $\sqrt{0.25}$ B. $\sqrt{0.50}$ C. $\sqrt{0.75}$ D. $\sqrt{0.80}$

Solution: $\sqrt{(0.3)^2 + (0.4)^2 + n^2} = 1$

$0.09 + 0.16 + n^2$

$n^2 = 1 - 0.25 = 0.75 \Rightarrow n = \sqrt{0.75}$ **Answer: C**

145. The resultant of two vectors of magnitudes 3 units and 4 units, is $\sqrt{37}$. The angle between the two vectors is

A. 0° B. 30° C. 60° D. 90°

Solution: $C^2 = A^2 + B^2 + 2AB\cos\theta$

$37 = 9 + 16 + 2 \times 3 \times 4\cos\theta$

$\cos\theta = \frac{1}{2}$ or $\theta = 60°$ **Answer: C**

146. If $|\vec{A} + \vec{B}| = |\vec{A}| = |\vec{B}|$, then the angle between $\vec{A}$ and $\vec{B}$ is

A. 0° B. 60° C. 90° D. 120°

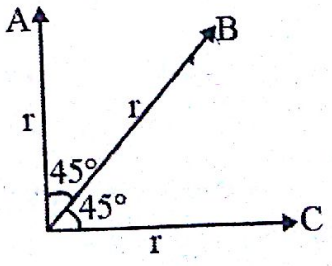
Solution: $A^2 = A^2 + A^2 + 2A^2\cos\theta$

$\Rightarrow \cos\theta = -\frac{1}{2}$ or $\theta = 120°$ **Answer: D**

147. The resultant of the three vectors $\vec{A}, \vec{B}$ and $\vec{C}$ is

A. r B. 2r C. $r(1+\sqrt{2})$ D. $r(\sqrt{2}-1)$

Explanation: $\vec{A}$ and $\vec{C}$ are perpendicular and their resultant $R^2 = A^2 + C^2 = r^2 + r^2 \Rightarrow R = \sqrt{2}r$



The resultant of $\vec{A}$ and $\vec{C}$ acts in the same direction as $\vec{B}$. Thus, total resultant = $\vec{R} + \vec{B} = \sqrt{2}r + r = r(1+\sqrt{2})$

Answer: C

148. If $\vec{A}+\vec{B}=\vec{C}$ and A = $\sqrt{3}$, B = $\sqrt{3}$ and C = 3, then the angle between $\vec{A}$ and $\vec{B}$ is

A. 0° B. 30° C. 60° D. 90°

Solution: $C^2 = A^2 + B^2 + 2AB\cos\theta$

$3^2 = 3 + 3 + 2(3)\cos\theta$

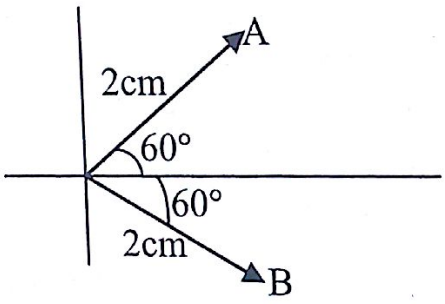
$\cos\theta = \frac{1}{2} \Rightarrow \theta = 60°$ **Answer: C**

149. Vector $\vec{A}$ is 2 cm long and is 60° above the x-axis in the first quadrant. Vector $\vec{B}$ is 2 cm long is 60° below the x-axis in the fourth quadrant. The sum $\vec{A}+\vec{B}$ is a vector of magnitude.

A. 2 along + y – axis C. 2 long + x – axis

B. 1 along – x – axis D. 2 along – x – along

Solution: Resolve the vectors and then add



$A_x = A\cos\theta = 2\cos 60°$

$A_y = A\sin\theta = 2\sin 60°$

$B_x = B\cos\theta = 2\cos 60°$

$B_y = B\sin 60 = -2\sin 60°$

$\vec{C} = \vec{A} + \vec{B} = (A_x + B_x)i + (A_y + B_y)j$

$\vec{C} = (2\cos 60° + 2\cos 60°)i + (2\sin 60 - 2\sin 60)j = 2i$ **Answer: C**

150. If the magnitude of the sum of two vectors is equal to the difference of their magnitude, then the angle between vectors is:

A. 0° B. 45° C. 90° D. 180°

Solution: $|A + B| = A - B$

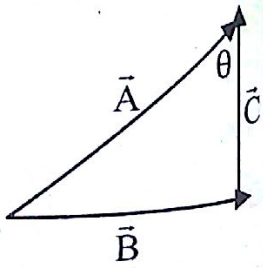
$A^2 + B^2 + 2AB\cos\theta = A^2 + B^2 - 2AB \Rightarrow \cos\theta = -1$ or $\theta = 180°$

Answer: D

151. If $\vec{A} = \vec{B} + \vec{C}$, and the magnitude of $\vec{A}, \vec{B}, \vec{C}$ are 5, 4 and 3 units, then the angle between $\vec{A}$ and $\vec{C}$ is

A. $\cos^{-1}\left(\frac{3}{5}\right)$ C. $\sin^{-1}\left(\frac{3}{4}\right)$

B. $\cos^{-1}\left(\frac{4}{5}\right)$ D. $\frac{\pi}{2}$



Solution: $\cos\theta = \frac{C}{A} = \frac{3}{5}$

$\theta = \cos^{-1}\left(\frac{3}{5}\right)$ **Answer: A**

152. The square of the resultant of two equal vectors is three times the product of the magnitudes of the vectors. The angle between the vectors is:

A. 0° B. 60° C. 90° D. 120°

Solution: $A^2 + A^2 + 2A^2 \cos\theta = 3A^2$, let $A = B$

$$\cos\theta = \frac{1}{2} \Rightarrow \theta = 60°$$ **Answer: B**

153. Each components of vector A making equal angles with x, y and z axes is:

A. $A/\sqrt{3}$ B. $\sqrt{3}A$ C. $\sqrt{2}A$ D. $A/\sqrt{2}$

Solution: $\cos\alpha = \frac{A_x}{A}, \cos\beta = \frac{A_y}{A}, \cos\gamma = \frac{A_z}{A}$

But, $\cos\alpha = \cos\beta = \cos\gamma$

Now, $\cos\alpha = \frac{A_x}{\sqrt{A_x^2 + A_y^2 + A_z^2}} = \frac{A_x}{\sqrt{3A_x^2}} = \frac{1}{\sqrt{3}}$

$$\Rightarrow A_x = A\cos\alpha = A\left(\frac{1}{\sqrt{3}}\right) = \frac{A}{\sqrt{3}}$$

Similarly, $A_y = A_z = \frac{A}{\sqrt{3}}$ **Answer: A**

154. The vector which must be added to the sum of the two vectors $i + 2j - k$ and $i - 2j + 2k$ to get a resultant of unit vector along z-axis is

A. $2i + j$ B. $-2i$ C. $i + j + k$ D. $i - j - k$

Solution: $\vec{A} + (i + 2j - k) + (i - 2j + 2k) = k$

$\Rightarrow \vec{A} = -2i$ **Answer: B**

155. If the sum of two unit vectors is also a unit vector, then the magnitude of their difference is

A. $\sqrt{2}$ B. $\sqrt{3}$ C. $\sqrt{5}$ D. $\sqrt{7}$

Explanation: The magnitude of the sum of two vectors is equal to the magnitude of either vector, then the angle between the two vectors is 120°. Difference of unit vectors is then

$$\sqrt{1^2 + 1^2 - 2\times1\times1\times\cos120°} = \sqrt{1+1-2\left(-\frac{1}{2}\right)} = \sqrt{3}$$ **Answer: B**

156. Which vector should be added to $2i + 4j - 3k$ and $3i - 5j + 7k$ to get a unit vector along the y-axis?

A. $5i + 9j + k$ C. $-3k$

B. i D. $-5i + 2j - 4k$

Solution: $\vec{A} + (2k + 4j - 3k) + (3i - 5j + 7k) = j$

Or, $\vec{A} = -5i + 2j - 4k$ **Answer: D**

157. The ratio of maximum and minimum magnitudes of the resultant of two vectors $\vec{a}$ and $\vec{b}$ is 3: 1 then $|a|$ is equal to:

A. $|b|$ B. $2|b|$ C. $3|b|$ D. $4|b|$

Solution: $\frac{a+b}{a-b} = \frac{3}{1} \Rightarrow a + b = 3a - 3b$ or $a = 2b$ **Answer: B**

158. Given: $\vec{P} = \vec{A} - \vec{B}$ and $P = A + B$. The angle between $\vec{A}$ and $\vec{B}$ is

A. $0°$ B. $90°$ C. $180°$ D. $270°$

Solution: $|\vec{P}| = A + B$

$|\vec{A} - \vec{B}|^2 = (A + B)^2$

$A^2 + B^2 - 2AB\cos\theta = A^2 + B^2 + 2AB$

$\cos\theta = 1$ or $\theta = 180°$ **Answer: C**

Dot Product

159. If $\vec{A}$ is perpendicular to $\vec{B}$, then

A. $\vec{A}.(\vec{A} + \vec{B}) = A^2 + AB$ C. $\vec{A}.(\vec{A} + \vec{B}) = A^2$

B. $\vec{A}.\vec{B} = 0$ D. $\vec{A}.\vec{B} = AB$

Solution: $\vec{A}.(\vec{A} + \vec{B}) = \vec{A}.\vec{A} + \vec{A}.\vec{B} = A^2 + 0 = A^2$ **Answer: C**

160. Given: $\vec{P} = 3j + 4k$ and $\vec{Q} = 2i + 5k$. The magnitude of the scalar product of these vectors is

A. 20 B. 23 C. 26 D. $5\sqrt{33}$

Solution: $\vec{P}.\vec{Q} = (3j + 4k).(2i + 5k) = (2.0) + (3.0) + (4.5) = 20$

Answer: A

161. The numerical value of $\vec{P}.\vec{P}$ is:

A. P^2 B. P C. 1 D. 0

Solution: $\vec{P}.\vec{P} = (P)(P)\cos\theta$, but $\theta = 0$ and $\cos 0° = 1$

$\Rightarrow \vec{P}.\vec{P} = P^2$ **Answer: A**

162. $|\vec{b}|$ can be written as

A. $\sqrt{\vec{b}.\vec{b}}$ B. $\vec{b}.\vec{b}$ C. $(\vec{b}.\vec{b})^2$ D. $(\vec{b}.\vec{b})^{3/2}$

Solution: $\sqrt{\vec{b}.\vec{b}} = \sqrt{b^2} = |b|$ **Answer: A**

163. The dot product of $(i + j + k)$ and $(i - j + k)$ is

A. 1 B. 2 C. 4 D. -1

Solution: $(i + j + k).(i - j + k) = i.i - j.j + k.k = 1$

Answer: A

164. The dot product of two vectors is commutative because

A. $\cos\theta = \cos(-\theta)$ C. $\sin\theta = \sin(-\theta)$

B. $\tan\theta = \tan(-\theta)$ D. $\text{cosec}\,\theta = \text{cosec}\,(-\theta)$

Answer: A

165. If the angle between $\vec{A}$ and $\vec{B}$ is 270°, then $\vec{A}.\vec{B}$ is

A. 0 B. 1 C. -1 D. $\frac{1}{3}$

Solution: $\vec{A}.\vec{B} = AB\cos 270 = 0$ **Answer: A**

166. Which of the following is perpendicular to $4i - 3j$?

A. $4i + 3j$ B. $6i$ C. $7k$ D. $3i - 4j$

Explanation: The given vector is in the xy plane. And z-axis is perpendicular to xy plane. **Answer: C**

167. If $\vec{A}.\vec{B} = AB$, then the angel between $\vec{A}$ and $\vec{B}$ is

A. 0° B. 45° C. 90° D. 180°

Solution: $\vec{A}.\vec{B} = AB\cos\theta = AB \Rightarrow \cos\theta = 1$ or $\theta = 0°$

Answer: A

168. The angle between the z-axis and the vector $i + j + \sqrt{2}k$ is

A. 30° B. 45° C. 60° D. 90°

Solution: $k.(i + j + \sqrt{2}k) = (1)(2)\cos\theta$

$$\cos\theta = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45°$$ **Answer: B**

169. The angle between $i + j + k$ and $2i + 2j + 2k$ is

A. 0° B. 30° C. 60° D. 90°

Solution: $(i + j + k).(2i + 2j + 2k) = \left(\sqrt{3}\right)\left(\sqrt{12}\right)\cos\theta$

$$\cos\theta = \frac{2+2+2}{\left(\sqrt{3}\right)\left(\sqrt{12}\right)} = \frac{6}{6} = 1 \text{ or } \theta = 0°$$ **Answer: A**

170. If two vectors $\vec{a}$ and $\vec{b}$ are perpendicular to each other then

A. $\vec{a}+\vec{b}=0$ C. $\vec{a}\times\vec{b}=0$

B. $\vec{a}-\vec{b}=0$ D. $\vec{a}.\vec{b}=0$

Explanation: The dot product of perpendicular vectors is zero.

Answer: D

171. Given $\vec{a}.\vec{b}=\frac{\sqrt{3}}{2}ab$. The angle between $\vec{a}$ and $\vec{b}$ is

A. 0° B. 30° C. 60° D. 90°

Solution: $\vec{a}.\vec{b}=ab\cos\theta=\frac{\sqrt{3}}{2}ab\Rightarrow\cos\theta=\frac{\sqrt{3}}{2}$ or $\theta=30°$

Answer: B

172. What is the angle between $i+j+\sqrt{2}k$ and the x-axis

A. 0° B. 30° C. 60° D. 90°

Solution: $\cos\theta=\frac{(i+j+\sqrt{2}k).i}{\left(\sqrt{1^2+1^2+\left(\sqrt{2}\right)^2}\right)(1)}=\frac{1}{2}$ or $\theta=60°$

Answer: C

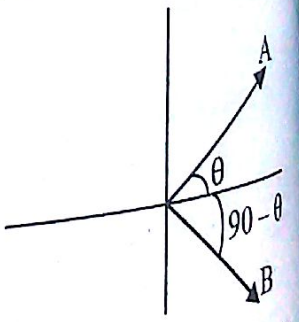
173. Given: $\vec{A}$ = A cos θi + A sin θj. A vector $\vec{B}$ which is perpendicular to A is given by

A. B cos θ i – B sin θ j C. B cos θ i + B sin θ j

B. B sin θ i – B cos θ j D. B sin θ i + B cos θ j

Solution:

$\vec{B}$ should make 90° – θ with the x–axis. Hence, $\vec{B}$ = B cos (90 – θ)i – B sin (90 – θ)j = B sin θ i – B cos θ j



Answer: B

174. The angle between the vectors (i + j) and (i – j) is

A. 30° B. 60° C. 45° D. 90°

Solution: $\cos\theta=\frac{(i+j)(i-j)}{\left(\sqrt{1^2+1^2}\right)\left(\sqrt{1^2+(-1)^2}\right)}=0$ or $\theta=90°$

Answer: D

175. The angle between the vectors (i + j) and (j + k) is

A. 30° B. 60° C. 45° D. 90°

Solution: $\cos\theta = \dfrac{(i+j)(j+k)}{\left(\sqrt{1^2+1^2}\right)\left(\sqrt{1^2+1^2}\right)} = \dfrac{1}{2}$ or $\theta = 60°$

Answer: B

176. If the resultant of two vectors having magnitudes of 6 and 3 is 9, then the dot product of the vectors is

A. 6 B. 3 C. 9 D. 18

Solution: $R = A + B \Rightarrow 3 + 6 = 9$, hence $\theta = 0°$

$\vec{A}.\vec{B} = AB\cos\theta = 6 \times 3 = 18$ **Answer: D**

177. Given: $\vec{A} = 3i - 4j - 2k$ and $\vec{B} = 2i - 4j - 5k$. The angle which $\vec{A} + \vec{B}$ makes with y-axis is

A. 0° B. 45° C. 60° D. 90°

Solution: $\vec{A} + \vec{B} = (3i - 4j - 2k) + (2i - 4j - 5k) = 5i - 7k$, since $\vec{A} + \vec{B}$ is in the xz plane, the angle with the y−axis is 90°.

Answer: D

178. If a vector 2i + 3j + 8k is perpendicular to the vector 4j − 4i + αk, then the value of α is

A. −1 B. $-\dfrac{1}{2}$ C. $\dfrac{1}{2}$ D. 1

Solution: condition for the perpendicularly of vectors is

$(2i + 3j + 8k).(4j - 4i + \alpha k) = 0$

$(2)(-4) + (3)(4) + (8)(\alpha) = 0$

$-8 + 12 + 8\alpha = 0 \Rightarrow \alpha = -\dfrac{1}{2}$ **Answer: B**

Cross Product

179. $\vec{A}$ is directed vertically downwards and $\vec{B}$ is directed along the north. What is the direction of $\vec{B} \times \vec{A}$?

A. East B. West C. north east D. north west

Hint: use the right hand rule **Answer: B**

180. Given: $\vec{A} = 3i + 2j + k$ and $\vec{B} = 6i + 4j + 2k$. The vector

A. Are parallel C. Make angle of 30° to each other

B. Are perpendicular D. Make angle of 60° to each other

Explanation: $\vec{B} = 2\vec{A}$, thus they are parallel **Answer: A**

181. Given: $\vec{C} = \vec{a} \times \vec{b}$. The angle which $\vec{a}$ makes with $\vec{C}$ is
A. 0° B. 90° C. 45° D. 180° **Answer: B**

182. If the magnitudes of two vectors $\vec{a}$ and $\vec{b}$ are a and b respectively, then the vector product of the two vectors cannot be
A. Equal to zero C. equal to ab
B. Less than ab D. greater than ab
Explanation: $\vec{a} \times \vec{b} = ab \sin\theta$. Since $\sin\theta$ cannot be greater than 1, $|\vec{a} \times \vec{b}|$ cannot be greater than ab. **Answer: D**

183. The numerical value of $\vec{A} \times \vec{A}$ is
A. 0 B. 1 C. A^2 D. A
Explanation: $\vec{A} \times \vec{A} = (A)(A) \sin\theta$, but $\theta = 0°$ and $\sin 0° = 0$
Answer: A

184. Vector $\vec{A}$ is a long the positive z – axis and its vector product with vector $\vec{B}$ is zero, then $\vec{B}$ could be
A. $i - j$ B. $4j$ C. $-9k$ D. $j - k$
Hint: $\vec{A}$ and $\vec{B}$ must be along the same axis. **Answer: C**

185. What is the angle between $\vec{A} \times \vec{B}$ and $\vec{B} \times \vec{A}$?
A. 0° B. 180° C. 45° D. 90°
Explanation: $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$, they are oppositely directed.
Answer: B

186. What is the angle between $\vec{c} \times \vec{d}$ and $\vec{c} + \vec{d}$?
A. 0° B. 45° C. 60° D. 90°
Explanation: $\vec{c} + \vec{d}$ is in the plane of $\vec{c}$ and $\vec{d}$. Also, $\vec{c} \times \vec{d}$ is perpendicular to the plane of $\vec{c}$ and $\vec{d}$. So, angle between $\vec{c} + \vec{d}$ and $\vec{c} \times \vec{d}$ is 90°. **Answer: D**

187. Vector $\vec{A}$ is along the positive x –axis. If its vector product with vector $\vec{B}$ is zero, then $\vec{B}$ could be?
A. $4j$ B. $-4i$ C. $j + k$ D. $-i - j$
Hint: the two vectors may be parallel. **Answer: B**

188. If $\vec{A} \times \vec{B} = 0$ and $\vec{B} \times \vec{C} = 0$, then the value of $\vec{A} \times \vec{C}$ is
A. 0 B. AC sin θ C. AC cos θ D. AB tan θ
Explanation: $\vec{A} \times \vec{B} = 0 \Rightarrow \vec{A} \parallel \vec{B}$
$\vec{B} \times \vec{C} = 0 \Rightarrow \vec{B} \parallel \vec{C}$
$\therefore \vec{A} \parallel \vec{C}$, thus $\vec{A} \times \vec{C} = 0$ **Answer: A**

189. What is the angle between $\vec{A}-\vec{B}$ and $\vec{A}\times\vec{B}$?

A. $0°$ B. $\pi/2$ C. π D. $3\pi/2$

Explanation: $\vec{A}\times\vec{B}$ is perpendicular to the plane of $\vec{A}\times\vec{B}$ also, $\vec{A}-\vec{B}$ is in the plane of $\vec{A}$ and $\vec{B}$. So, $\vec{A}\times\vec{B}$ is perpendicular to $\vec{A}-\vec{B}$, $\therefore \theta=\frac{\pi}{2}$ **Answer: B**

190. The value of the scalar triple product $\vec{A}.\vec{B}\times\vec{A}$ is

A. A^2B B. Zero C. $A^2B\sin\theta$ D. $A^2B\cos\theta$

Hint: $\vec{B}\times\vec{A}$ is perpendicular to $\vec{A}$. **Answer: B**

191. For two vectors $\vec{A}$ and $\vec{B}$, $\vec{A}.\vec{B}=|\vec{A}\times\vec{B}|$, the magnitude of $\vec{C}=\vec{A}+\vec{B}$ is

A. $\sqrt{A^2+B^2}$ C. $\sqrt{A^2+B^2+\frac{1}{\sqrt{2}}AB}$

B. $A+B$ D. $\sqrt{A^2+B^2+\sqrt{2}AB}$

Solution: $AB\cos\theta = AB\sin\theta \Rightarrow \tan\theta = 1$ or $\theta = 45°$

$C^2 = A^2 + B^2 + 2AB\cos 45° = A^2 + B^2 + \sqrt{2}aAB$ or $C=\sqrt{A^2+B^2+\sqrt{2}AB}$ **Answer: D**

192. The magnitude of the vector product of two vector is $\sqrt{3}$ times their scalar product. The angle between vectors is:

A. $\pi/2$ B. $\pi/6$ C. $\pi/3$ D. $\pi/4$

Solution: $AB\sin\theta = \sqrt{3}\,AB\cos\theta \Rightarrow \tan\theta=\sqrt{3}$ or $\theta = 60° = \pi/3$

Answer: C

193. If $\vec{P}.\vec{Q}=0$, then $|\vec{P}\times\vec{Q}|$ is

A. $|\vec{P}||\vec{Q}|$ B. Zero C. 1 D. $\sqrt{PQ}$

Solution: $\vec{P}.\vec{Q}=0\Rightarrow \vec{P}\perp Q$ or $\theta = 90°$

$|\vec{P}\times\vec{Q}|=PQ\sin 90° = PQ = |P||Q|$ **Answer: A**

194. $(\vec{A}.\vec{B})^2+|\vec{A}\times\vec{B}|^2$

A. Zero B. A^2B^2 C. AB D. $\sqrt{AB}$

Solution: $A^2B^2\cos^2\theta + A^2B^2\sin^2\theta = A^2B^2(\cos^2\theta+\sin^2\theta) = A^2B^2$

Answer: B

195. Two vectors $\vec{A}$ and $\vec{B}$ are parallel to each other, if

A. $\vec{A}\times\vec{B}=1$ C. $\vec{A}.\vec{B}=1$

B. $\vec{A}\times\vec{B}=0$ D. $\vec{A}.\vec{B}=\text{Zero}$

Explanation: For parallel vectors, $\theta = 0°$

$\Rightarrow \vec{A}\times\vec{B} = AB\sin\theta = 0$, since $\sin 0° = 0$

Answer: B

196. Given: $\vec{A}.\vec{B}=0$ and $\vec{A}\times\vec{C}=0$. The angle between $\vec{B}$ and $\vec{C}$ is

A. 0° B. 90° C. 180° D. 270°

Solution: $\vec{A}.\vec{B}=0\Rightarrow\vec{A}\perp\vec{B}$

$\vec{A}\times\vec{C}=0\Rightarrow\vec{A}//\vec{B}$

$\therefore\vec{B}\perp\vec{C}\therefore\theta=90°$

Answer: B

197. If two vectors $\vec{a}$ and $\vec{b}$ are perpendicular to each other, then

A. $\vec{a}+\vec{b}=0$ C. $\vec{a}\times\vec{b}=0$

B. $\vec{a}-\vec{b}=0$ D. $\vec{a}.\vec{b}=0$

Explanation: Dot product of perpendicular vectors is zero.

Answer: D

198. The value of $j \times (i \times k)$ is

A. Zero B. 1 C. 2 D. $\sqrt{2}$

Solution: $\vec{A}.(\vec{B}\times\vec{C}) = (\vec{A}.\vec{C})\vec{B} - (\vec{A}.\vec{B})C$

$\Rightarrow j\times(i\times k) = (j.k)i - (j.i)k = 0i - 0k = 0$

Answer: A

199. If $\vec{A}$ is parallel to $\vec{B}$, then

A. $\vec{A}.(\vec{A}+\vec{B}) = A^2 + AB$ C. $\vec{A}.\vec{B}=0$

B. $\vec{A}\times(\vec{A}+\vec{B}) = AB$ D. $\vec{A}\times(\vec{A}\times\vec{B}) = 9$

Solution: $\vec{A}.(\vec{A}+\vec{B}) = \vec{A}.\vec{A} + \vec{A}.\vec{B} = A^2 + AB\cos°$

$= A^2 + AB$

Answer: A

200. Given: $|\vec{A}\times\vec{B}| = |\vec{A}.\vec{B}|$. The angle between $\vec{A}$ and $\vec{B}$ is

A. 0° B. 30° C. 45° D. 90°

Solution: $AB\sin\theta = AB\cos\theta \Rightarrow \tan\theta = 1$ or $\theta = 45°$ **Answer: C**

201. $\dfrac{|\vec{A}\times\vec{B}|}{|\vec{A}.\vec{B}|}$

A. $\tan\theta$ B. $\cot\theta$ C. $\sin\theta$ D. $\cos\theta$

Solution: $\frac{AB\sin\theta}{AB\cos\theta} = \tan\theta$ **Answer: A**

202. If the resultant of two vectors having magnitudes of 7 and 4 is 3, then the magnitude of the cross product of the two vectors is

A. 28 B. Zero C. 3 D. 11

Explanation: The given vectors are collinear and $\theta = 180°$ so their cross product is zero. **Answer: B**

203. The adjacent sides of a parallelogram are represented by $2i + 3j$ and $i + 4j$. The area of the parallelogram is

A. 5 units along z-axis
B. 5 units in xy plane
C. 3 units in xz plane
D. 3 units in yz plane

Solution: $\vec{A}\times\vec{B} = (2i + 3j) \times (i + 4j) = 5k$ **Answer: A**

204. If $\vec{P}$ and $\vec{Q}$ denote the sides of a parallelogram and its area is $\frac{1}{2}PQ$, then the angle between $\vec{P}$ and $\vec{Q}$

A. 0° B. 30° C. 45° D. 60°

Solution: $\frac{1}{2}PQ = |\vec{P}\times\vec{Q}| = PQ\sin\theta \Rightarrow \sin\theta = \frac{1}{2}$ or $\theta = 30°$

Answer: B

205. Two vectors of magnitude 2 units and 3 units have the same direction. Their vector product is:

A. A null vector
B. A unit vector
C. 9 units
D. 6 units

Solution: $\vec{A}\times\vec{B} = AB\sin\theta = (2)(3)\sin 0° = 0$ **Answer: A**

206. If $|\vec{A}\times\vec{B}| = \vec{A}.\vec{B}$, then the angle between $\vec{A}$ and $\vec{B}$ is

A. $\pi/4$ B. $\pi/2$ C. $3\pi/4$ D. $3\pi/2$

Solution: $AB\sin\theta = AB\cos\theta \Rightarrow \tan\theta = 1$ or $\theta = \pi/4$

Answer: B

207. The angles which $i + j + \sqrt{2}k$ makes with x-axis, y-axis and z-axis respectively are

A. 60°, 60° and 45°
B. 60°, 45° and 45°
C. 45°, 45° and 45°
D. 60°, 60° and 60°

Solution: $\vec{A} = i + j + \sqrt{2}k$ and $A = \sqrt{1^2 + 1^2 + (\sqrt{2})^2} = 2$

$$\cos\beta = \frac{A_y}{A} = \frac{1}{2} \Rightarrow \alpha = 60°$$

$$\cos\gamma = \frac{A}{A} = \frac{\sqrt{2}}{2} \Rightarrow \alpha = 45°$$

Answer: A

208. Given: $\vec{A} = 4i + 6j$ and $\vec{B} = 2i + 3j$ Which of the following is correct?

A. $\vec{A} \times \vec{B} = 0$ C. $\frac{|\vec{A}|}{|\vec{B}|} = \frac{1}{2}$

B. $\vec{A}.\vec{B} = 24$ D. $\vec{A}$ and $\vec{B}$ are anti-parallel

Answer: A

209. If $\vec{A}.\vec{B} = 0$ and $\vec{A} \times \vec{B} = 1$, then $\vec{A}$ and $\vec{B}$ are

A. Perpendicular unit vectors C. Parallel

B. Parallel unit vector D. Perpendicular

Solution: $\vec{A}.\vec{B} = 0 \Rightarrow \vec{A} \perp \vec{B}, \theta = 90°$

$\vec{A} \times \vec{B} = AB \sin\theta = 1 \Rightarrow AB = 1 \Rightarrow A = 1$ and $B = 1$

Answer: A

210. If $\vec{P}, \vec{Q}$ and $\vec{R}$ are coplanar vectors, then which one of the following is definitely correct?

A. $(\vec{P} \times \vec{Q}).\vec{R} = 0$ C. $\vec{P} \times \vec{Q} \times \vec{R} = 0$

B. $(\vec{P}.\vec{Q}) \times \vec{R} = 0$ D. $(\vec{P}.\vec{Q}).\vec{R} = 0$

Solution: $\vec{P} \times \vec{R}$ is perpendicular to $\vec{R}$ **Answer: A**

211. What is the scalar product of $\vec{A}$ and $(\vec{A} + \vec{B})$ if the vector product of $\vec{A}$ and $\vec{B}$ is zero?

A. $A^2 + AB$ B. AB C. A^2 D. Zero

Solution: $\vec{A} \times \vec{B} = 0 \Rightarrow \vec{A} // \vec{B}$ or $\theta = 0°$

$\vec{A}.(\vec{A} + \vec{B}) = \vec{A}.\vec{A} + \vec{A}.\vec{B} = A^2 + AB\cos 0° = A^2 + AB$

Answer: A

212. The adjacent sides of a parallelogram are represented by $i + 4j$ and $7i + 3j$. The area of the parallelogram is

A. 10 units B. 15 units C. 25 units D. 30 units

Solution: $\vec{A} \times \vec{B} = \begin{vmatrix} i & j & k \\ 1 & 4 & 0 \\ 7 & 3 & 0 \end{vmatrix} = -25k \Rightarrow |\vec{A} \times \vec{B}| = 25$ **Answer: C**

213. The area of the triangle formed by vectors $3i + 4j$ and $-3i + 7j$ is

A. 8.5 units C. 14.5 units

B. 1.25 units D. 16.5 units

Solution: $\vec{A}\times\vec{B} = \begin{vmatrix} i & j & k \\ 3 & 4 & 0 \\ -3 & 7 & 0 \end{vmatrix} = (0-0)i + (0-0)k + (21+12)k = 33k$

Area of triangle $= = \frac{1}{2}|\vec{A}\times\vec{B}| = \frac{33}{2} = 16.5$ units **Answer: D**

214. Three vectors satisfy the relation $\vec{A}.\vec{B} = 0$ and $\vec{A}.\vec{C} = 0$. Then $\vec{A}$ is parallel to

A. $\vec{C}$ B. $\vec{B}$ C. $\vec{B}\times\vec{C}$ D. $\vec{B}.\vec{C}$

Solution: $\vec{A}\times\vec{B} = 0 \Rightarrow \vec{A} \perp \vec{B}$

$\vec{A}.\vec{C} = 0 \Rightarrow \vec{A} \perp \vec{C}$

$\therefore \vec{A}$ is parallel to $\vec{B}\times\vec{C}$

Answer: C

215. If $|\vec{A}\times\vec{B}| = \sqrt{3}\vec{A}.\vec{B}$, then the value of $|\vec{A}\times\vec{B}|$ is

A. $A^2 + B^2$ C. $A^2 + B^2 + \sqrt{3}AB$

B. $A + B$ D. $\sqrt{A^2 + B^2 + AB}$

Solution: $AB\sin\theta = \sqrt{3}\,AB\cos\theta$

$\tan\theta = \sqrt{3}$ or $\theta = 60°$

$|\vec{A}+\vec{B}|^2 = A^2 + B^2 + 2\,AB\cos\theta$

$|\vec{A}+\vec{B}| = \sqrt{A^2 + B^2 + AB}$

Answer: D

216. If $\vec{A}\times\vec{B} = \vec{B}\times\vec{A}$, then angle between $\vec{A}$ and $\vec{B}$ **is**

A. $\pi/2$ B. $\pi/3$ C. π D. $\pi/4$

Solution: $\vec{A}\times\vec{B} = \vec{B}\times\vec{A}$

$\vec{A}\times\vec{B} - \vec{B}\times\vec{A} = 0$

$\vec{A}\times\vec{B} + \vec{A}\times\vec{B} = 0$

$2(\vec{A}\times\vec{B}) = 0 \Rightarrow \vec{A}\times\vec{B} = 0$

$\vec{A}$ and $\vec{B}$ are collinear **Answer: C**

217. If the angle between vector $\vec{A}$ and $\vec{B}$ is θ, the value of the product $(\vec{B}\times\vec{A}).\vec{A}$ is equal to:

A. Zero
B. $BA^2\sin\theta\cos\theta$
C. $BA^2\sin\theta$
D. $BA^2\cos\theta$

Explanation: $\vec{B}\times\vec{A}$ is perpendicular to $\vec{A}$, thus $(\vec{B}\times\vec{A}).\vec{A}=0$

Answer: A

218. The value of $(i+k)\times i$ is

A. 0 B. -j C. j D. 1

Solution: $(i+k)\times i = i\times i + k\times i = 0 + j = j$

Answer: C

219. Vectors $\vec{a},\vec{b}$ and $\vec{c}$ are such that $\vec{a}.\vec{c}=0$. Then the vector $\vec{a}$ is parallel to:

A. $\vec{b}$ B. $\vec{c}$ C. $\vec{b}.\vec{c}$ D. $\vec{b}\times\vec{c}$

Explanation: $\vec{a}.\vec{b}=0\Rightarrow\vec{a}\perp\vec{b}$

$\vec{a}.\vec{c}=0\Rightarrow\vec{a}\perp\vec{c}$

$\therefore \vec{b}\times\vec{c}\perp$ b and c in the plane of $\vec{a}$

Answer: D

220. If $\vec{a}.\vec{b}=|\vec{a}\times\vec{b}|$, then the value of $|\vec{a}+\vec{b}|$ is

A. $\sqrt{a^2+b^2}$
B. $a+b$
C. $\sqrt{a^2+b^2+\sqrt{2}ab}$
D. $\sqrt{a^2+b^2+\dfrac{ab}{\sqrt{2}}}$

Solution: $\vec{a}.\vec{b}=|\vec{a}\times\vec{b}|$

$ab\cos\theta = ab\sin\theta \Rightarrow \tan\theta = 1$ or $\theta = 45°$

$|a+b| = \sqrt{a^2+b^2+2ab\cos 45°} = \sqrt{a^2+b^2+\sqrt{2}ab}$

Answer: C

221. If $\vec{a}\neq\vec{b}$ and $|\vec{a}|\neq 0, |\vec{b}|\neq 0$, then it is possible to have

A. $\vec{a}.\vec{b}=0$
B. $\vec{a}-\vec{b}=0$
C. $\vec{a}.\vec{a}=\vec{b}.\vec{b}$
D. $\vec{a}\times\vec{b}=0$ or $\vec{a}.\vec{b}=0$

Explanation: If $\theta=0°$, $\vec{a}\times\vec{b}=0$ and If $\theta=0°$, $\vec{a}.\vec{b}=0$

Answer: D

222. $|\vec{a}\times\vec{b}|^2+(\vec{a}.\vec{b})^2$ is equal to

A. $(\vec{a}+\vec{b})^2$
B. $(\vec{a}-\vec{b})^2$
C. a^2+b^2
D. a^2b^2

Solution: $|\vec{a}+\vec{b}|^2+(\vec{a}+\vec{b})^2=(ab\sin\theta)^2+(ab\cos\theta)^2$

$=a^2b^2\sin^2\theta+a^2b^2\cos^2\theta=a^2b^2(\sin^2\theta+\cos^2\theta)=a^2b^2$

Answer: D

223. If the angle between A and B is θ, the value of the triple product, $\vec{A}.(\vec{B}\times\vec{A})$ is

A. A^2B B. $A^2B\sin\theta$ C. $A^2B\cos\theta$ D. zero

Solution: Let $\vec{C}=\vec{B}\times\vec{A}\Rightarrow\vec{C}\perp\vec{A}$ and $\vec{B}$

So, $\vec{A}.\vec{C}=0$ or $\vec{A}.(\vec{B}\times\vec{A})=0$ **Answer: D**

224. Which of the following is not correct

A. $j\times i=-k$ C. $i\times k=-j$

B. $k\times j=-i$ D. $k\times i=-j$

Hint: Using the direction

as positive. **Answer: D**

225. Which of the following is not correct?

A. $\vec{A}\times\vec{B}=-\vec{B}\times\vec{A}$ C. $\vec{A}\times(\vec{B}\times\vec{C})=\vec{A}\times\vec{B}+\vec{A}\times\vec{C}$

B. $\vec{A}\times\vec{B}\neq\vec{B}\times\vec{A}$ D. $\vec{A}\times(\vec{B}+\vec{C})=(\vec{A}\times\vec{B})+\vec{C}$

Answer: D

226. Which of the following is correct?

A. $\vec{A}\times(\vec{B}\times\vec{C})=(\vec{A}\times\vec{B})\times\vec{C}$ C. $\vec{A}\times(\vec{B}\times\vec{C})\neq(\vec{A}\times\vec{B})\times\vec{C}$

B. $\vec{A}\,(\vec{B}\,.\,\vec{C})=(\vec{A}.\vec{B})\vec{C}$ D. $\vec{A}\times\vec{B}=\vec{B}\times\vec{A}$

Answer: C

227. Which of the following is correct?

A. $\vec{A}.\vec{B}\neq\vec{B}.\vec{A}$ C. $\vec{A}.(\vec{B}+\vec{C})\neq\vec{A}.\vec{B}+\vec{A}.\vec{C}$

B. $\vec{A}.(\vec{B}+\vec{C})=\vec{A}.\vec{B}+\vec{A}.\vec{C}$ D. $\vec{A}\times\vec{B}=\vec{B}\times\vec{A}$

Answer: B

Unit 2: Conceptual Questions

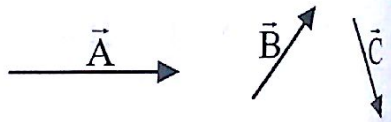
1. Under what circumstance would a vector possess components that are equal in magnitude?
2. Two vectors have unequal magnitudes. Can their sum be zero? Explain
3. Can the magnitude of a vector have a negative value? Explain
4. If $\vec{A}$ and $\vec{B}$ are non-zero vectors, is it possible for $\vec{A}.\vec{B}$ and $\vec{A}\times\vec{B}$ both to be zero? Explain

5. What is
   a. The dot product of a vector with itself.
   b. The cross product of a vector with itself.
6. If $\vec{A}\times\vec{B}=0$, does it follow that $\vec{A}=0$ or $\vec{B}=0$? Explain
7. When will $\vec{A}\times(\vec{B}\times\vec{C})=(\vec{A}\times\vec{B})\times\vec{C}$? Explain
8. How can you obtain a vector that is perpendicular to two non zero vectors?
9. For what angle between vectors $\vec{A}$ and $\vec{B}$ will $|\vec{A}\times\vec{B}|=12$. Given that $|A|=3$ and $|B|=4$.
10. Three non-zero vectors have the relation, $\vec{A}.\vec{B}=0$ and $\vec{A}.\vec{C}=0$. What can you say about the direction of vector $\vec{A}$?
11. If thirty position vectors are represented by the sides of a regular polygon when taken in order of head to tail arrangement, what can you say about their resultant?

Unit 2: Problems

Adding vectors graphically

12. For the given vectors, Find
    a. $\vec{A}+\vec{B}+\vec{C}$ c. $\vec{A}+\vec{B}-\vec{C}$
    b. $\vec{A}-\vec{B}+\vec{C}$

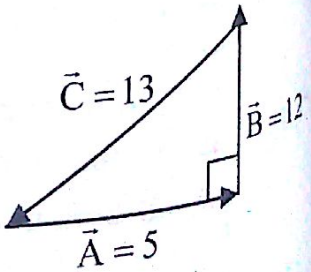


Adding Vectors analytically

13. If you are given $\vec{A}=i+j$ and $\vec{B}=-i+3j-5k$. Find $\vec{A}+\vec{B}$
14. Given that: $\vec{A}=i-j$, $\vec{B}=2i+3j$ and $\vec{C}=-5k$. Find $\vec{A}+\vec{B}-\vec{C}$
15. If $\vec{A}=7i-6j$, $\vec{B}=-3i+12j$ and $\vec{A}+\vec{B}+\vec{C}=8i+2j$. Find $\vec{C}$
16. Find the magnitude of vector $\vec{A}$. Given that $\vec{A}=2i+4j+7k$.
17. Find a unit vector in the direction of vector $\vec{A}$, $\vec{A}=2i+4j+7k$

Multiplying Vectors

18. If you are given a vector of magnitude 5m directed north like A = 5m north. Find
    a. 4A b. -4A
19. If $\vec{A}=2i-3j$ and $\vec{B}=i+j-k$ are given. Find
    a. $\vec{A}.\vec{B}$
    b. The angle between them



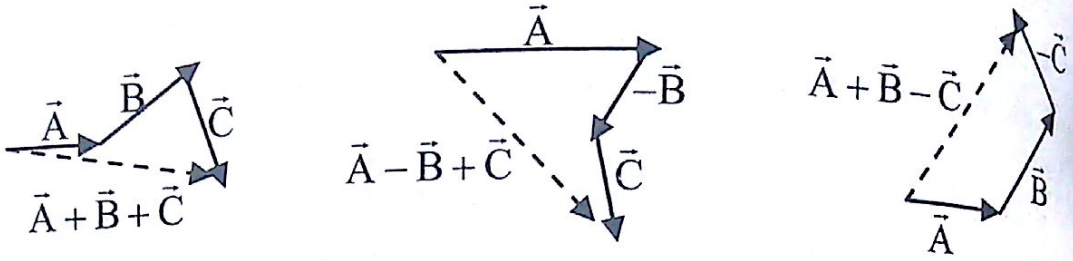
20. We are given, $\vec{A} = 2i - 3j$ and $\vec{B} = i + j - k$. Find $|\vec{A} \times \vec{B}|$
21. For the vectors show, find
    a. $\vec{A}.\vec{B}, \vec{A}.\vec{C}, \vec{B}.\vec{C}$ b. $\vec{A} \times \vec{B}, \vec{A} \times \vec{C}, \vec{B} \times \vec{C}$
22. Calculate $\vec{A} \times \vec{B} \times \vec{C}$ if $\vec{A} = 2i + k$, $\vec{B} = 3i + j + k$ and $\vec{C} = i + j + 4k$.
23. If $\vec{A} = i + j + 2k$, $\vec{B} = 2i - j + k$ and $\vec{C} = 2i - j + 4k$ and if $\vec{D} = \vec{C} - x\vec{A} + y\vec{B}$ is perpendicular to $\vec{A}$ and $\vec{B}$, find the constants x and y.
24. Find a vector perpendicular to both vectors $\vec{A} = i - 3j + 2k$ and $\vec{B} = -2i + j - 5k$.
25. Find m such that $i - 2j + 3k$ and $5i + 2j + mk$ are perpendicular.
26. If $\vec{A} = i + 2j - k$ and $\vec{A} \times \vec{B} = 3i - j + 5k$. Find $\vec{B}$.
27. Given that: $\vec{A} = 2i - 3j + 6k$ and $\vec{B} = i - 2j + 2k$. Find the scalar projection of $\vec{A}$ on to $\vec{B}$ and $\vec{B}$ on to $\vec{A}$.

Unit 2: Explanation to Questions

1. i) For vectors in xy plane, $A_x = A_y$ when $\theta = 45°$.
   ii) For vectors in space, $A_x = A_y = A_z$ when $\alpha = \beta = \gamma = 54.7°$
2. No. They must have equal magnitude and opposite direction to give zero sum.
3. No. Magnitude of say vector $\vec{A} = A_{xi} + A_{yj} + A_{zk}$ is obtained by $|A| = \sqrt{A_x^2 + A_y^2 + A_z^2}$. Hence, no chance of being negative.
4. No. Two vectors cannot be mutually parallel and perpendicular at the same time. This is so if $\vec{A} = \vec{B} = 0$.
5. a) $\vec{A}.\vec{A} = |A||A|\cos\theta$, here $\theta = 0° \Rightarrow \vec{A}.\vec{A} = A^2(1) = A^2$
   b) $\vec{A} \times \vec{A} = |A||A|\sin\theta$, here $\theta = 0° \Rightarrow \vec{A} \times \vec{A} = A^2(0) = 0$
6. $\vec{A} \times \vec{B} = 0$, if $\vec{A} = \vec{B} = 0$ or $\vec{A} // \vec{B}$
7. Only when $|B| = 1$ or 0
8. By taking the cross product of both vectors. That is, if $\vec{C} = \vec{A} \times \vec{B}$, then $\vec{C} \perp \vec{A}$ and $\vec{C} \perp \vec{B}$.
9. $\vec{A} \times \vec{B} = AB\sin\theta$, in our case, θ must be 90°.
10. $\vec{A} \cdot \vec{B} = 0$ and $\vec{A} \cdot \vec{C} = 0$, implies $\vec{A}$ is perpendicular to $\vec{B}$ & $\vec{C}$.
11. Zero. Because, it forms a closed loop by itself. That means, it came back to its initial position.

Unit 2: Solution to problem

12. a) b) c)



13. $\vec{R}=\vec{A}+\vec{B}=(i+j)+(-i+3j-5k)=4j-5k$
14. $\vec{R}=\vec{A}+\vec{B}-\vec{C}=(i-j)+(2i+3j)-(-5k)=3i+2j+5k$
15. $\vec{C}=4i-4j$
16. $|\vec{A}|=\sqrt{2^2+4^2+7^2}=\sqrt{69}$
17. $\hat{u}=\dfrac{\vec{A}}{|A|}=\dfrac{2i-j-3k}{\sqrt{2^2+(-1)^2+(-3)^2}}=\dfrac{1}{\sqrt{14}}(2i-j-3k)$
18. a) 20m North b) 20m south
19. a) $\vec{A}.\vec{B}=-1$ b) $\theta=99°$
20. $\vec{A}\times\vec{B}=4i+2j+5k$ and $|\vec{A}\times\vec{B}|=\sqrt{4^2+2^2+5^2}=\sqrt{45}$
21. a) 0, 25, 144 b) 60k, 60k, −60k
22. $|\vec{A}\times\vec{B}\times\vec{C}|=8$
23. $\vec{A}.\vec{D}=0$ and $\vec{B}.\vec{D}=0\Rightarrow x=-1$ and $y=-2$
24. $\vec{C}=\vec{A}\times\vec{B}=13i+j-5k$ is perpendicular to $\vec{A}$ and $\vec{B}$
25. $\vec{A}.\vec{B}=0\Rightarrow m=-\dfrac{1}{3}$
26. $\vec{B}=-3i-j+4k$
27. $\vec{A}$ on to $\vec{B}=\dfrac{\vec{A}.\vec{B}}{|\vec{B}|}=\dfrac{20}{3}$ and $\vec{B}$ on to $\vec{A}=\dfrac{\vec{A}.\vec{B}}{|\vec{A}|}=\dfrac{20}{7}$

Unit 3

Kinematics

The universe and everything in it is in motion. From the smallest particles within atoms to the largest galaxies, from living things to non-living things all move in one or the other way.

3.1 Motion in a Straight Line

The beauty of physics lies in its attempt to understand the laws of nature mainly through mechanics and electromagnetism so as to make life better and simple.

Every time you walk, stand up, sit down, throw a ball or open a door, your actions are governed by the laws of mechanics.

Mechanics: Is part of physics that studies the relationship among force, matter and motion.

Basically, mechanics is the study of how objects move, how they respond to external forces and how other factors such as size, mass and mass distribution affect their motion. That means, mechanics deals with kinematics and dynamics.

Kinematics: If part of mechanics that studies the relationship among displacement, velocity, acceleration and time of a moving body.

- kinematics come from the Greek word "kinema" meaning motion. Therefore, kinematics describes motion without concerning with its cause that is force.
- To describe motion, we use some fixed reference point called *frame of reference*, and without which motion is undefined, since motion is a continuous change in position.

Displacement and Distance

Displacement: Is the change in position of a moving body. It tells the shortest path between two points and it is a vector. However,
Distance: Is the actual path length between two points about which the body moved. It is a scalar.

For example, if a body moved from A to B along paths 1, 2 and 3 then, path 1 represents its displacement and path 1, 2 or 3 represent the distance covered by the body.

Conceptual Example

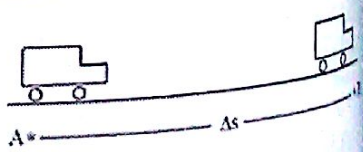
1. Tirunesh Dibaba perform her training in a circular track of radius r. After ten complete revolutions, what can you say about her
   A. displacement B. distance moved
   Explanation: A. zero B. 10 (2πr)
2. A particle starts from the origin and moved 2cm to the right and then 9 cm to the left. What is its
   A. displacement B. distance moved
   Explanation: A. $\Delta\vec{s} = \vec{s_2} - \vec{s_1} = 7\text{cm}$ to the left
   B. S = 11 cm
3. A boy starts from his home moved 400m due east and then 300m due north to a supermarket. What is his
   A. displacement B. distance covered
   Explanation:
   A. $\vec{S} = \sqrt{(400)2+(300)2} = 500\text{m}$, and $\tan\theta = \frac{S_y}{S_x} = \frac{300}{400} = 37°$
   B. S = 700m

3.2 Average and Instantaneous Velocity

In most practical cases, motion of a body may not be uniform. So we take the average value

Average velocity: Is displacement divided by time taken

Suppose a car moved from point A to point B which are ΔS apart in a time Δt then the average velocity of the car is obtained by.



$$\overrightarrow{V_{av}} = \frac{\overrightarrow{\Delta S}}{\Delta t}$$

Average speed: Is total distance travelled divided by the total time taken

That is,

$$V_{av} = \frac{S^T}{t^T}$$

Instantaneous velocity: Is the velocity of a body at a particular instant of time, or at a specific point.

Mathematically, the instantaneous velocity is obtained by

$$\vec{V}_{inst} = \frac{\Delta\vec{S}}{\Delta t}, \text{ as } \Delta t \to 0$$

$\Delta t \to 0$ is read as "as change of t approaches zero"

- In calculating the instantaneous velocity, it is good to follow the following steps
  - First, find the displacement at t and t + Δt. i.e, $\vec{s}(t)$ $8\vec{s}(t+\Delta t)$
  - Then, find the change in displacement, $\Delta\vec{s}$. $\Delta\vec{s} = \vec{s}(t+\Delta t) - \vec{s}(t)$
  - Finally, calculate $\vec{V}_{inst} = \frac{\Delta\vec{s}}{\Delta t}$, as $\Delta t \to 0$
- **Important points to remember**
  - The magnitude of displacement can be positive or negative
  - Distance is always positive. distance ≥ displacement
  - Velocity can be positive, negative or zero.
  - Decreasing speed is NOT negative speed.
  - Speed is always positive or zero.
  - When a body moves with constant velocity, the motion of the body is said to be uniform motion

Conceptual Example

4. Under what condition is the average velocity of a body equal to the instantaneous velocity?
   Explanation: When the body moves with constant velocity.
5. Speed is always positive while velocity may be positive or negative. Why? Explain.
   Explanation: Speed is the magnitude of velocity and magnitude is always positive.
6. If the average velocity is nonzero for sometime interval, does this means that the instantaneous velocity is never zero during this time interval? Explain

Explanation: No, A city but for example, moving from Bole to Gullele stops at stations. Thus, its instantaneous velocity is zero at stations.

7. A car travels half the distance with velocity of 30 km/h and the remaining half distance with velocity of 40 km/h. Then the average velocity of the car is 35 km/h.

Explanation: $V_{av} = \frac{S_T}{t_T} = \frac{s_1 + s_2}{t_1 + t_2} = \frac{v_1 t_1 + v_2 t_2}{t_1 + t_2}$

$V_{av} = \frac{2v_1 v_2}{v_1 + v_2} = 34.3$ km/h. . So, it is false!

Illustrative Example

8. A body moves along the $x-$ axis in such a way that its distance as a function of time is, $\vec{s}(t) = 12t - t^2$ .What is

A. Distance moved when $t = 3$ sec

B. The average velocity during the 4^{th} sec.

C. The instantaneous velocity at $t = 4$ sec.

D. When will the body change it, direction of motion.

Solution:

A. $\vec{s} = (t) = 12t - t^2$, at $t = 3$sec, $\vec{s} = 12(3) - (3)^2 = 27$m

B. $\vec{V}_{av} = \frac{\Delta\vec{s}}{\Delta t} = \frac{\vec{s}(t=4) - \vec{s}(t=3)}{4-3} = \frac{12(4) - (4)^2 - 12(3) - (3)^2}{1} = 5$m/s

C. $\vec{V}_{inst} = \frac{\Delta\vec{s}}{t}$, as $\Delta t \to 0$

$\Delta\vec{s} = \vec{s}(t + \Delta t) - \vec{s}(t) = 12(t + \Delta t) - (t + \Delta t)^2 - (12t - t^2)$

$\Delta\vec{s} = 12\Delta t - 2t\Delta t - \Delta t^2\Delta$

$\vec{V}_{inst} = \frac{\Delta s}{\Delta t}, \Delta t \to 0 = \frac{12\Delta t - 2t\Delta t - \Delta t^2}{\Delta t} = 12 - 2t - \Delta t, \Delta t \to 0$

$\boxed{V_{inst} = 12 - 2t}$, at $t = 4$sec, $V_{inst} = 12 - 2(4) = 4$m/s

D. The body change direction when $\vec{V}_{inst} = 0$

$V_{inst} = 12 - 2t = 0 \Rightarrow t = 6$ sec

9. The motion of a body is described by $\vec{S}(t) = 25t + 6t^2$

a. What is the displacement in the interval; $t = 2$sec and $t = 5$ sec.

b. What is the instantaneous velocity at $t = 2$ sec?

Solution: a. $\Delta\vec{S}=\vec{S}(t=5)-\vec{s}(t=2)=25(5)+6(5)^2-25(2)-6(2)^2$

$= 227\text{m}$

b. $\vec{V}_{inst}=\dfrac{\Delta\vec{s}}{\Delta\vec{t}}$ as$\Delta t\to 0$

$$\Delta\vec{S}=\vec{S}(t+\Delta t)-\vec{s}(t)=25(t+\Delta t)^2-\left(25t+6t^2\right)$$

$$\Delta\vec{s}=25\Delta t+12t\Delta t+6\Delta t^2,\ \frac{\Delta s}{\Delta t}=25+12t+6\Delta t,\ \Delta t\to 0$$

$V_{inst} = 25 + 12t$. At $t = 2$sec, $v_{inst} = 25 + 12\ (2) = 49$ m/s

10. A train travels from one station to another at a speed of 40 km/h and returns to the first station at a speed of 60 km/h. Calculate.
    a. The average speed
    b. The average velocity

Solution: a. $V_{av}=\dfrac{S_T}{t_T}=\dfrac{s_1+s_2}{t_1+t_2}=\dfrac{s+s}{\frac{s_1}{v_1}+\frac{s_2}{v_2}}=\dfrac{s+s}{\frac{s}{40}+\frac{s}{60}}=48\text{km/h}$

b. $\vec{V}_{av}=\dfrac{\Delta s}{\Delta t}=\dfrac{\Delta s}{\Delta t}=\dfrac{0}{t_1+t_2}=\text{zero}$

11. A man walks for 1 minute at a speed of 1m/s and then runs for 1 minute at 3m/s along a straight road. What is the average speed of the man?

Solution: $V_{av}=\dfrac{S_T}{t_T}=\dfrac{s_1+s_2}{t_1+t_2}=\dfrac{v_1t_1+v_2t_2}{t_1+t_2}=\dfrac{(11)(60)+(3)(60)}{60+60}=2\text{m/s}$

12. The motion of a particle is described by the equation. $\vec{S}(t)=6+4t^2-t^4$. Find the position and velocity at $t = 2$ second.

Solution: $\vec{S}(t)=6+4t^2-t^4,\Delta t\to 0$ at $t = 2$, $s = 6+ 4(2)^2 - (2)^4$

$= 6$ m and the velocity of the particle can be found as follows

Method 1: $V_{inst}=\dfrac{\Delta\vec{s}}{\Delta t}$ as $\Delta t\to 0$

$$=\frac{6+4(t+\Delta t)^2-(t+\Delta t)^4-\left(6+4t^2-t^4\right)}{\Delta t},\ \Delta t\to 0$$

$= 8t - 4t^3 = -16$ m/s at $t = 2$ sec

Method 2: Using the concept of derivation

$$\vec{V} = \frac{d\vec{s}}{dt} = \frac{d}{dt}(6 + 4t^2 - t^4) = 8t - 4t^3 = -16 \text{m/s at } t = 2 \text{ sec}$$

13. A car travels along a straight line at a constant speed of 20 m/s for a distance d and then another distance in the same direction at a constant speed of 36 km/hr. What is the average speed for the entire trip? (UEE: 2007)

A. 23 km/h B. 48 km/h C. 54 km/h D. 60 km/h

Solution: $V_{av} = \frac{2v_1v_2}{v_1 + v_2} = \frac{2(72)36}{72 + 36} = 48$ km/h.

Note: 20m/s = 72 km/hr

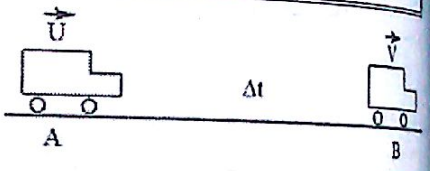
Answer: B

3.3 Average and Instantaneous Acceleration

If the velocity of a moving body is changing in magnitude in direction or both, the body is said to be *accelerating*.

Acceleration: Is the rate of increase of velocity with time.

Suppose a car with initial velocity $\vec{U}$ at point A is observed to have a velocity $\vec{V}$ at some other point B with in a time interval of Δt.



Then the acceleration of the car can be obtained by the equation.

$$\vec{a} = \frac{\Delta v}{\Delta t} = \frac{\vec{v} - \vec{u}}{t}$$ Its unit is m/s^2.

Instantaneous Acceleration: Is the acceleration of a body at a specific point or at a particular time

The instantaneous acceleration of a body when the change in time is infinitesimal or very small is obtained by

$$\vec{a}_{inst} = \frac{\Delta \vec{V}}{\Delta t}, \text{as } \Delta t \to 0$$

Important points to remember

- A body is said to be Accelerating when it is speeding up. i.e., velocity increasing at some rate and Decelerating when it is slowing down, velocity decreasing at some rate.
- When a body is in uniform motion, its velocity is constant and its acceleration is zero.
- Decreasing acceleration is NOT called retardation

- An acceleration of $5m/s^2$ means that the velocity increases at the rate of 5m/s every second. Thus, the velocity is 10m/s in 2sec.
- Acceleration can be negative or positive. Negative acceleration or retardation means that the body is slowing down.

Conceptual Examples

14. If a body is traveling East ward, can its acceleration be west ward? Explain.
Explanation: Yes. This happens when the body is decelerating

15. If the velocity of a body is non-zero, can its acceleration be zero? Explain.
Explanation: Yes. This happens when the body is moving at constant velocity.

16. Can a body have zero velocity and still be accelerating.
Explanation: Yes. Consider a free fall as an example.

17. If a body is slowing down, can its acceleration be positive?
Explanation: Yes. This happens when the motion is in the negative direction.

18. A car traveling along a straight road is decelerating. Does the car is acceleration necessarily have a negative value?
Explanation. Deceleration has nothing to do with whether the acceleration is positive or negative. It only means that the acceleration points opposite to the velocity and indicates that the moving object is slowing down. For example if a car traveling to the right is slowing down, its acceleration is to the left, which is negative. But if the car travelling to the left slows down, its acceleration is to the right, which is positive. We see, then, that a decelerating object does not necessarily have a negative acceleration.

Illustrative Examples

19. The velocity of a car is given by $\vec{V}(t)=15t+4t^2$. Find the instantaneous acceleration of the car at t = 3sec

Solution: $\vec{a}_{inst} = \dfrac{\Delta\vec{v}}{\Delta t}$ as $\Delta t \to 0$.

$$\Delta\vec{v} = \vec{v}(t+\Delta t) - \vec{v}(t) = 15(t+\Delta t) + 4(t+\Delta t)^2 - (15t+4t^2)$$
$$= 15\Delta t + 8t\,\Delta t + 4\Delta t^2$$

$\frac{\Delta\vec{v}}{\Delta t} = 15 + 8t + 4\Delta t$, as $\Delta t \to 0$

$\vec{a}_{inst} = 15 + 8t$, at $t = 3$, $\vec{a}_{inst} = 15 + 8(3) = 39 m/s^2$

or using derivation

$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}(15t + 4t^2) = 15 + 8t = 39\ m/s^2$ at $t = 3sec$

20. The velocity of a particle is given by $\vec{v}(t) = 20t^2 - 100t + 50$

a. When will the acceleration of the particle become zero?

b. What is its velocity when the acceleration is zero?

Solution: a. Using derivation

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}(20t^2 - 100t + 50) = 40t - 100$$

$0 = 40t - 100 \Rightarrow t = 2.5$ sec

b. $\vec{v}(t = 2.5) = 20(2.5)^2 - 100(2.5) + 50 = 75 m/s$

21. Albert Einstein cycling in a relating mood in such a way that the distance he traveled is proportional to the time cubed. If he traveled 19m in the first 3 seconds, what is his acceleration at $t = 3$ sec

Solution: $\vec{s}(t) = ct^3$

$19 = c\,(3)^3 \Rightarrow c = 0.7$

$\vec{s}(t) = 0.7t^3$

$\vec{V}_{inst} = \frac{d\vec{s}}{d\vec{t}} = 2.1t^2$

$\vec{a}_{inst} = \frac{d\vec{v}}{dt} = 4.2t$ and at $t = 3sec$

$\vec{a} = 4.2(3) = 12.6\ m/s^2$

22. Which one of the following statement is correct (UEE 2007)

A. An object moving toward east cannot have acceleration towards west

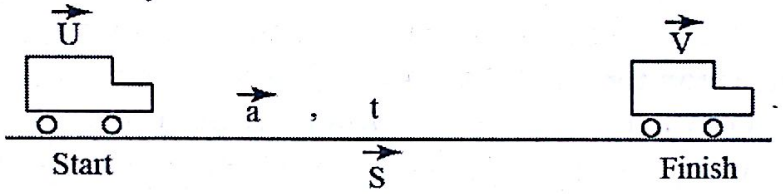
B. If the average velocity of an object is zero in sometime interval, the average speed of the object for that time interval is also zero.

C. The $\vec{v} - t$ graph of the object is parallel to the time axis

D. An object having zero velocity can have acceleration different from zero **Answer: D**

3.4 Motion with Constant Acceleration

This is a special type of motion, which occurs when the velocity change at the same rate throughout the motion. The equations of motion for a body moving in a straight line with constant acceleration are the following. Now consider a body that starts with an initial velocity $\vec{u}$ moved a displacement $\vec{s}$ with a constant acceleration $\vec{a}$ for a time t and reached a final velocity $\vec{v}$.



Average velocity, $\vec{V}_{av} = \dfrac{\vec{v}+\vec{u}}{2}$ .....(1)

Acceleration, $\vec{a} = \dfrac{\Delta\vec{v}}{\Delta t} = \dfrac{\vec{v}-\vec{u}}{t}$ ..... (2)

Displacement, $\Delta\vec{s} = \left(\vec{V}_{av}\right)(t) = \left(\dfrac{\vec{v}+\vec{u}}{2}\right)(t)$ ..... (3)

Substituting $\vec{v} = u + a\,t$ into equation (3), we have

$$\boxed{\Delta s = ut + \frac{1}{2}a\,t^2 .....(4)}$$

Substituting $t = \dfrac{v-u}{a}$ into equation (3), we get

$$\boxed{v^2 = u^2 + 2as .....(5)}$$

Substituting $u = v - at$ into equation (3), we obtain

$$\boxed{\Delta S = Vt - \frac{1}{2}at^2 .....(6)}$$

Distance traveled by a body in the n^{th} second

$$S_n^{th} = S_n - S_{n-1} = \left[ut_n + \frac{1}{2}at_n^2\right] - \left[u(t_n - 1) + \frac{1}{2}a(t_n - 1)^2\right]$$

$$S_n^{th} = u + a\left(t - \frac{1}{2}\right) = u + \frac{a}{2}(2t-1)$$

Conceptual Examples

23. Two Cars A and B are moving in the same direction. At some instant, the velocity of car A exceeds the velocity of car B. Does this mean that the acceleration of A is greater than of B? Explain.
Explanation: Not necessarily. In fact, Car A is moving faster than car B at that instant, but its rate of change of velocity may be less than that of B.

24. A taxi driver in Merkato penalized for "speeding". The traffic police man reported that, the taxi was moving alongside a car that was going above the speed limit, so the traffic man said, if two cars were side by side, they were both speeding. If you were a lawyer, how could you argue this case?
Explanation: The decision is wrong. Because, two cars may be alongside either when they have the same velocity or when one pass the other.

25. The velocity of a ball changes from 6m/s to 14m/s when it travels for 20m along a straight line with uniform acceleration. In the next 2 sec, it will cover a distance of 36m.
Explanation: True. $v^2 = u^2 + 2as \Rightarrow a = 4m/s^2$

$$S = ut + \frac{1}{2}at^2, \text{ when } t = 2\text{sec. } s = 36m$$

Illustrative Examples

26. A bullet moving with a speed of 200m/s can penetrate a plank 4cm thick. What should be the speed to penetrate the same plank up to 10 cm?
Solution: $V^2 = u^2 - 2a\,s \Rightarrow 0 = u^2 - 2\,as$

$u = \sqrt{2\,as}$, Therefore,

$u_1 = \sqrt{2\,as_1}$ .....(*) and

$u_2 = \sqrt{2as_2}$ .....(**)

Taking ratio, $\frac{u_2}{u_1} = \sqrt{\frac{s_2}{s_1}} \Rightarrow u_2 = \left(\sqrt{\frac{10}{4}}\right)(200) = 316m/s$

27. Two cars are travelling towards each other on a straight road at velocities 10m/s and 12m/s respectively. When they are 150m apart, both drivers apply their brakes and each decelerate at $2m/s^2$. How far apart will they be when they stop?

Solution: Distance moved by the first car before rest

$$V^2 = u_1^2 + 2a\,s_1 \Rightarrow 0 = (10)^2 - (2)\,(2)s_1$$

$$S_1 = 25m$$

Distance moved by the second car before test

$$V^2 = u_2^2 + 2a\,s_2 \Rightarrow 0 = (12)^2 - (2)\,(2)s_2$$

$$S_2 = 36m$$

$$S = s_1 + s_2 = 25 + 36 = 61m$$

Final separation = 150 – 61 = 89m

28. The reaction time of an automobile driver is 0.7sec. If the car was moving at initial velocity of 15m/s and decelerate at a rate of $5m/s^2$. What distance will it move before stopping?

Solution: distance moved by the first car before rest.

$$s_1 = vt = (15)\,(0.7) = 10.5m$$

distance moved while decelerating

$$S_2 = \frac{u^2}{2a} = \frac{(15)^2}{2(5)} = 22.5m$$

$$S = s_1 + s_2 = 10.5 + 22.5 = 33m$$

29. At the instant the traffic light turns green, a car starts with a constant acceleration of $2m/s^2$. At the same instant a truck, travelling with a constant speed of 10m/s, overtakes and passes the car.

a. How far beyond the starting point the car overtake the truck

b. How fast will the car be travelling at instant.

Solution: a. $s_c = s_t \Rightarrow ut = \frac{1}{2}at^2 \Rightarrow t = 10\,sec.$

$$S_c = s_t = ut = (10)\,(10) = 100m$$

b. $v = u + at = 0 + (2)\,(10) = 20m/s$

30. A taxi driver travelling at 20m/s sees a child on the road 50m ahead. He instantly applies the brakes so as to stop the car within 10m from the child. What is the retardation of the taxi?

Solution: $v^2 = u^2 + 2\,a\,s \Rightarrow a = \frac{-u^2}{2s} = \frac{(20)^2}{2(40)} = -5m/s^2$

31. A bus takes 10 sec to cover 30m and 12sec to cover 42m. Find the initial velocity and acceleration of the bus.

Solution: $S_1 = ut_1 + \frac{1}{2}at_1^2 \Rightarrow 30 = 10u + 50a$ ..... (*)

$S_2 = ut_2 + \frac{1}{2}a\,t_2^2 \Rightarrow 42 = 12u + 72a$ ..... (**)

From equation (*) and (**) $u = 0.5$ m/s and $a = 0.5$ m/s^2

32. A Zebra pass by a lion at a constant speed of 10m/s. After 5 sec. the lion starts chasing the zebra at a constant acceleration of 2m/s^2. When will the lion catch its food?

Solution: distance of lion is same as distance of zebra

$$S_L = S_z$$

$$\frac{1}{2}a\,t_L^2 = vt_z$$

$$\frac{1}{2}a\,t_L^2 = v(t_L + 5) \Rightarrow t_L^2 = 10t_L + 50 \text{ or } t_L = 137 \text{ sec.}$$

33. A bus and a taxi behind it start moving from rest at the same instant. The bus maintains an acceleration of 2m/s^2. If the taxi overtake the bus, after the bus moved 49m.

a. How much time does the taxi take to overtake the bus?

b. Initially, how far was the taxi behind the bus?

Solution: a. $\frac{1}{2}a\,t^2 = S \Rightarrow \frac{1}{2}(2)t^2 = 49$ or $t = 7$sec

b. $\Delta S = S_{taxi} - S_{bus} = \frac{1}{2}a_t\,t^2 - \frac{1}{2}a_b\,t^2$

$$\Delta s = \frac{1}{2}t^2(a_t - a_b) = \frac{1}{2}(7)^2(3-2) = 24.5\text{m}$$

34. A car moving with constant acceleration covered the distance between two points that are 60m apart in 6sec. Its speed as it passed the second point was 15m/s.

a. What was the speed of the car at the first point?

b. What was the acceleration of the car?

c. How far behind the first point was the car at rest?

Solution: a. $S = \left(\frac{v+u}{2}\right)t \Rightarrow u = \frac{2s}{t} - v = \frac{2(60)}{6} - 15 = 5\text{m/s}$

b. $a = \frac{v-u}{t} = \frac{15-u}{6} = 1.7\ m/s^2$

c. $s_o = \frac{1}{2}at^2$, but $t = \frac{u}{a} = \frac{5}{1.7} = 2.9\ sec$

$$s_o = \frac{1}{2}(1.7)(2.9)^2 = 7.15m$$

35. A train starting from rest moves with uniform acceleration and covers 500m in 5sec. Where was the train 3 sec after the beginning of its motion?

Solution: $S_1 = \frac{1}{2}at_1^2$ ..... (*)

$S_2 = \frac{1}{2}at_2^2$ ..... (**)

Taking ratio,

$$\frac{S_2}{S_1} = \frac{t_2^2}{t_1^2} \Rightarrow S_2 = S_1\left(\frac{t_2}{t_1}\right)^2 = 500\left(\frac{3}{5}\right)^2 = 180m$$

36. A taxi driver moving at 10m/s sees a dog walking across a road 32m ahead. If the driver's reaction time is 0.2 sec, and the taxi decelerates at $2m/s^2$. Find the distance the taxi stop from the dog.

Solution: Distance moved before stop = reaction distance + deceleration distance

$S = S_r + S_d$

$S_r = Vt = (10)(0.2) = 2m$

$S_d = \frac{u^2}{2a} = \frac{10^2}{2(2)} = 25m$

$S = 2 + 25 = 27m$

$\Delta S = S_o - s = 32 - 27 = 5m$

37. Two cars A and B are in the same direction and accelerate uniformly from rest. After 1sec, A is 2m ahead of B. What will be their separation after 1 sec.

Solution: $S_A = \frac{1}{2}a_A t^2$ and $S_B = \frac{1}{2}a_B t^2$

$$\Delta S = S_A - S_B = \frac{1}{2}a_A t^2 - \frac{1}{2}a_B t^2 = \frac{1}{2}(1)^2(a_A - a_B) = 2,\ a_A - a_B = 4$$

$$\text{at } t = 10 \text{ sec},\ \Delta s = \frac{1}{2}t^2(a_A - a_B) = \frac{1}{2}(10)^2(4) = 200\text{m}$$

38. A car starting from rest accelerates at 4m/s^2. If it travelled 20m in the last second. What is the total distance covered by the car?

Solution: $S_n^{th} = u + \frac{a}{2}(2t-1) = 0 + \frac{4}{2}(2t-1) \Rightarrow t = 5.5 \text{ sec}$

$$S = ut + \frac{1}{2}at^2 = \frac{1}{2}(4)(5.5)^2 = 60.5\text{m}$$

39. Two motor cycles move at 0.4m/s^2 starting from rest. If one starts moving 1 minute after the other, how long after the starting of the first motor will their separation be 4.2 km

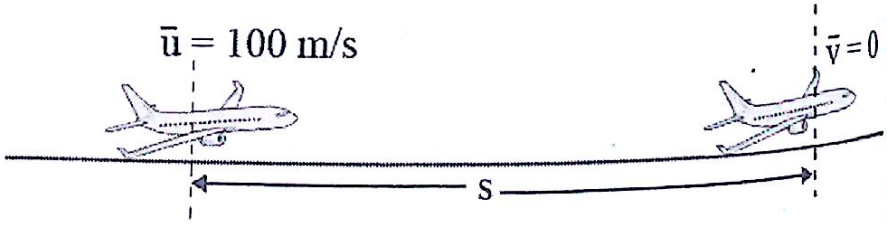
Solution: $S = ut + \frac{1}{2}at^2,\ t_2 = t_1 - 60$

$$\Delta S = S_1 - S_2 = \frac{1}{2}at_1^2 - \frac{1}{2}at_2^2$$

Substituting and solving $t_1 = 145$ sec.

40. An airplane come for landing at a speed of 10m/s. If it decelerates at 5m/s^2 on the runway.
a. What is the minimum time it needs before coming to rest?
b. Can this plane land at an airport where the runway is 0.9 km long?

Solution:



a. $v = u + at \Rightarrow 0 = 100 - 5t$ or $t = 20$ sec

b. $v^2 = u^2 + 2as \Rightarrow 0 = (100)^2 - 2(5)(s) \Rightarrow S = 1000\text{m} = 1\text{km}$ It needs 1km runway.

41. A boy runs at constant speed of 5m/s to catch a city bus. When he is 40m behind the bus, the bus starts moving at a constant acceleration of 0.2m/s^2. When and where will the boy catch the bus?

Solution: $S_{boy} = S_{bus} + 40$

$$Vy = 40 + \frac{1}{2}at^2$$

$$5t = 40 + \frac{1}{2}(0.2)t^2$$

$0.1\ t^2 - 5t + 40 = 0 \Rightarrow t = 10$sec shorter time

and $S = vt = (5)\ (10) = 50$m.

42. Bolt, the world class athlete can reach his maximum speed in 4 sec, and maintain this speed for the remained of a 100m race, finishing with a total time of 9.1 sec. what is his acceleration?

a. During the first 4 sec

b. During the last 5.1 sec

c. During the whole racing distance

Solution: $S_1 + S_2 = 100$

$$\left(\frac{u+v}{2}\right)(t_1)+(v)(t_2)=100,\ u=0$$

$$\frac{v}{2}(4)+v(5.1)=100 \Rightarrow v=14\text{m/s}$$

a. $a = \dfrac{\Delta v}{\Delta t} = \dfrac{v-u}{t} = \dfrac{14-0}{4} = 3.5\text{m/s}^2$

b. $a = \dfrac{\Delta v}{\Delta t} = \dfrac{0}{5.1} = 0$

c. $a = \dfrac{\Delta v}{\Delta t} = \dfrac{v-u}{t} = \dfrac{14}{9.1} = 1.5\ \text{m/s}^2$

43. An object starts from rest and moved with a constant acceleration of 4m/s^2 along a straight line.

Find. a. The average velocity in a 5 sec interval

b. The distance moved in the 5^{th} sec

Solution: a. $V_{av} = \dfrac{v+u}{2}, V = u + at \Rightarrow V_{av} = \dfrac{(4)(5)}{2} = 10\text{m/s}$

$$\text{b. } S^{th} = u+\frac{a}{2}(2t-1)= 0+\frac{4}{2}(2(5)-1)=18\text{m}$$

44. A body moved along the x – axis with an acceleration of 6m/s^2. Initially, the body is at $x = 5$m and has a velocity of 4m/s. Where is the body when its velocity is 8m/s?

Solution: $\Delta s = \dfrac{v^2 - u^2}{2a} \Rightarrow s - s_o = \dfrac{v^2 - u^2}{2a}$

$$S = S_o + \frac{v^2 - u^2}{2a} = 5 + \frac{8^2 - 4^2}{2(6)} = 9m$$

45. An object moving at 24m/s slows uniformly at a rate of 3m/s each second for 5 sec.
a. What distance it covers during the given time?
b. How long will it take for the object to stop?

Solution: a. $S = ut + \frac{1}{2}at^2 = (24)(5) - \frac{1}{2}(-3)(5)^2 = 82.5m$

b. $V = u + at \Rightarrow 0 = 24 + (-3)t \Rightarrow t = 8$ sec

46. A motorist drives along a straight road at a constant speed of 15m/s. A police officer starts to accelerates at $2m/s^2$ to overtake the motorist. What time it takes the officer to reach the motorist?

Solution: $S_p = S_m \Rightarrow \frac{1}{2}at^2 = vt \Rightarrow t = \frac{2v}{a} = \frac{2(15)}{2} = 15sec$

47. A car is travelling at 50 km/h. the driver sees a child run out into the road 5m a head. She applies the breaks and the car stops in 5 sec. the driver's thinking time is 1.5 sec. will the car stop in time?

Solution: $S = \left(\frac{v+u}{2}\right)(t), t = 1.5 + 5 = 6.5 \Rightarrow S = 4.5m$. Yes.

48. An automobile driver applied on the brakes when he sees a tree blocking the road. The car slows with $5m/s^2$ for 4sec, leaving 62m long skid marks on the road. With what speed will the car hit the tree?

Solution: $S = vt - \frac{1}{2}at^2 \Rightarrow 62 = 4(v) - \frac{(-5)(4)^2}{2}, v = 5.5m/s$

49. A particle has a constant acceleration of $4m/s^2$. At a certain instant, its velocity is 10m/s. What it its velocity
a. 2 sec earlier b. 2sec later

Solution: a. $v = u - at = 10 - 2(4) = 2m/s$

b. $v = u + at = 10 + 2(4) = 18$ m/s

50. An object has a constant acceleration of $3.2\ m/s^2$. At a certain instant, its velocity is 9.6 m/s. What is the ratio of its velocities at 2.5 second later to 2.5 sec earlier. (UEE: 2001)
A. 111 B. 11 C. 11.1 D. 1.1

Solution: The velocity 2.5 second later

$$V = u + at = 9.6 + (3.2)(2.5) = 17.6 \text{ m/s}$$

The velocity 2.5 second earlier $V = u - at = 9.6 - (3.2)(2.5) = 1.6$ m/s

$$\frac{17.6}{1.6} = 11$$

Answer: B

51. If the x and y coordinates of a particle at any time t is given by $x = 7t + 4t^2$ and $y = 5t$. What is the acceleration of the particle at $t = 5$ sec. (UEE: 2001)

A. zero B. 8 m/s^2 C. 20 m/s^2 D. 40 m/s^2

Solution: $\vec{v} = \frac{d\vec{r}}{dt} = (7 + 8t)i + 5j$

$$a = \frac{d\vec{v}}{dt} = 8 \text{ m/s}^2$$

Answer: B

52. A cart on air track is moving at 0.5m/s when the air is suddenly turned off. The cart comes to rest after traveling 1m. The experiment is repeated, but now the cart is moving at 1m/s when the air is turned off. How far does the cart travel before coming to rest? (UEE: 2004)

A. 1m B. 2m C. 3m D. 4m

Solution: For the first experiment

$$a = \frac{v^{2-}u^2}{2s} = \frac{o^2 - (0.5)^2}{2(1)} = -\frac{0.25}{2} \text{ m/s}^2$$

For the second experiment

$$S = \frac{v^2 - u^2}{2a} = \frac{0 - (1)^2}{-2\frac{(0.25)}{2}} = 4\text{m}$$

Answer: D

53. An object moving with uniform acceleration has a velocity of 12m/s in the positive x direction when its x coordinate is 3cm. if its x coordinate 2 sec later is –5cm. what is its acceleration (UEE: 2004)

A. -12 m/s^2 B. -13 m/s^2 C. -16 m/s^2 D. 12 m/s^2

Solution: $\Delta s = ut + \frac{1}{2}at^2$, $\Delta s = -5 - 3 = -8\text{m}$

$$-8 = 12(2) + (a)(2)^2$$

$$\Rightarrow a = -16 \text{ m/s}^2$$

Answer: C

54. The coordinate of a particle is given by $x(t) = 25t - 3t^3$. At what value t will the particle become momentarily at rest? (UEE: 2005)

A. 2.78s B. 1.67s C. 0.65 D. 0.365

Solution: $v = \frac{dx}{dt} = 25 - 9t^2 \Rightarrow 0, t = \frac{5}{3} = 1.67\text{sec}$ **Answer: B**

55. A driver moving at a constant speed of 20m/s sees an accident up ahead and hits the brakes. If the car decelerates at a constant rate of -5m/s^2, how far does the car go before it comes to stop (UEE: 2005)

A. 120 m B. 100 m C. 40 m D. 20 m

Solution: $S = \frac{v^2 - u^2}{2a} = \frac{0^2 - 20^2}{2(-5)} = 40$ **Answer: C**

56. A body moving with constant acceleration covers the distance between two points 60m a part in 5 sec. Its velocity as it passes the second point is 15m/s. What is the acceleration? (UEE: 2006)

A. $3\ \text{m/s}^2$ B. $1.8\ \text{m/s}^2$

C. $2.4\ \text{m/s}^2$ D. $1.2\ \text{m/s}^2$

Solution: $S = \left(\frac{v+u}{2}\right)t \Rightarrow 60 = \left(\frac{15+u}{2}\right)(5)$ or $u = 9\text{m/s}$

$$a = \frac{v-u}{t} = \frac{15-9}{5} = 1.2\text{m/s}^2$$ **Answer: D**

3.5 Freely Falling Bodies

The most famous example of motion with constant acceleration is free fall – the motion of an object falling freely under the influence of gravity.

Free Fall: Is the motion of an object subjected only to the influence of gravity.

- The word *Free* in free fall means free from any effects other than gravity. For example, in free fall we assume that an object's motion is not influenced by any form of friction or air resistance.
- The word *Fall* in free fall does not mean the object is necessarily moving downward. By free fall we mean any motion under the influence of gravity alone. If you drop a ball, it is in free fall. If you throw a ball upward or downward, it is in free fall as soon as it leaves your hand.

An object is in free fall as soon as it is released, whether it is dropped from rest, thrown downward, or thrown upward.

The equations for uniformly accelerated straight line motion are used for freely falling bodies with some changes in notations. Here we use (y) or (h) for (s) and (g) for (a). Moreover, the acceleration due to gravity $g = 9.8m/s^2 \cong 10m/s^2$ and is assumed as positive for downward motion and negative for upward motion.

Equations of a Freely Falling Body

Let's consider the equations of a body thrown upward, thrown down ward and dropped from some height.

Thrown upward	Thrown downward	Dropped
$v = u - gt$	$V = u + g\,t$	$V = gt$
$y = ut - \frac{1}{2}gt^2$	$y = ut + \frac{1}{2}gt^2$	$y = \frac{1}{2}gt^2$
$v^2 = u^2 - 2gy$	$v^2 = u^2 + 2gy$	$v^2 = 2gy$
$t_T = \frac{2u}{g}$		

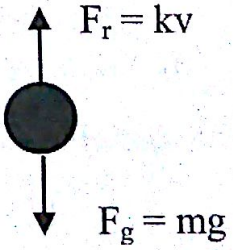
- For a body thrown upward, its velocity at the maximum height is zero, v = 0, and its displacement is zero y = 0 when the body return to its initial position.
- For a body dropped from some height, the initial velocity is zero, u = 0

Terminal Velocity

When a body falls from a very high altitude, the air resistance balances the weight of the body and attains constant velocity.

Terminal Velocity: Is the velocity attained by a falling body at the instant when air resistance prevents further increase in velocity due to gravity.

Therefore, at terminal velocity, the resistive force is equal to gravitational force or the weight of the body.



That is $F_r = F_g = W$

$\boxed{Kv = mg}$

Where k = proportionality constant

V = terminal velocity

Illustrative Examples

57. A boy on a building throws a ball upward and another downward with the same speed. Which ball will hit the ground with larger speed? Explain

Explanation: For the first ball; $v^2 = -u^2 - 2gh$

for the second ball, $v^2 = u^2 + 2gh$

$\Rightarrow$ Both do hit the ground with same speed

58. Can a body have zero velocity and still be accelerating? Explain.

Explanation: Yes. A ball thrown upward for example do have zero velocity at the top most but is still accelerating at $g = 9.8\ m/s^2$

59. A boy throws a ball upward. While it is in free fall, what happens to its

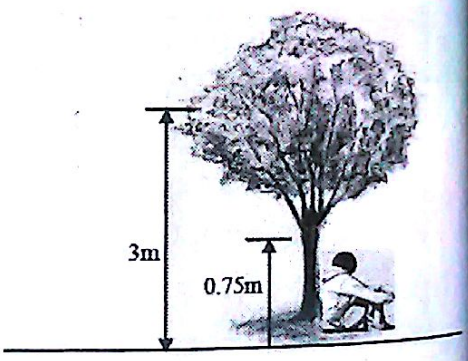
a. velocity b. acceleration

Explanation: a . Its velocity decreases by 9.8 m/s each second of its motion and become zero at the maximum height.

b. Remains constant, $g = 9.8\ m/s^2$

Illustrative Examples

60. Once upon a time Sir Issac Newton was sitting underneath an apple tree when an apple dropped off the tree from 3m height. If Newton's head was 0.75 m from the ground, what was the apple's speed as it reached Newton's head?



Solution: $v^2 = u^2 + 2g\Delta y$, $\Delta y = 3 - 0.75 = 2.25m$

$v^2 = 2g\Delta y \Rightarrow v = \sqrt{2g\Delta y} = \sqrt{(2)(10)(2.25)} = 6.7\ m/s$

61. A boy throws an orange to his sister who is in a window 10m above. Her out stretched hand caught the orange 2.5 sec later. Find

a. The initial velocity of the orange

b. The velocity of the orange as it reached the girl's hand

Solution: a. $y = ut - \frac{1}{2}g t^2 \Rightarrow 10 = u(2.5) - 5(2.5)^2 \Rightarrow u = 16.5 m/s$

b. $v = u - g t = 16.5 - (10)(2.5) = -8.5\ m/s$

62. A ball is thrown vertically upward from the ground with a speed of 20m/s. calculate
a. The time the ball takes to reach the maximum height
b. The maximum height reached
c. The velocity and acceleration at the maximum height

Solution: a. $v = u - gt$, at h_{max}, $v = 0$

$$o = u - gt \Rightarrow t = \frac{v}{g} = \frac{20}{10} = 2\,sec$$

b. $v^2 = u^2 - 2\,gh$, $v = 0$ at h_{max}

$$\Rightarrow h_{max} = \frac{u^2}{2g} = \frac{(20)^2}{2(10)} = 20m$$

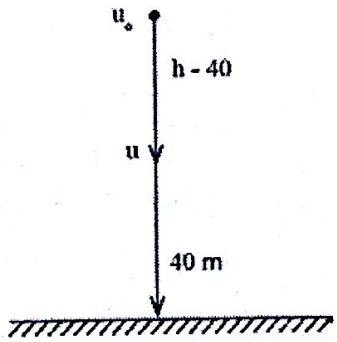
c. $v = u - gt = 20 - (10)\,(20) = 0$, $g = 10m/s^2$

63. A ball is thrown upward with a speed of 25 m/s.
a. How long the ball take to return to the ground?
b. What is its velocity just when it hit the ground?

Solution: a. $y = ut - \frac{1}{2}gt^2 \Rightarrow 0 = 25t - 5t^2$ or $t = 5\,sec$

b. $v = u - gt = 25 - (10)\,(5) = -25m/s = 25$ m/s down

64. A certain free fall requires 2 sec to travel the last 40m before it hits the ground. From what height above the ground did it fall?



Solution: $y = ut + \frac{1}{2}gt^2$, lower part

$40 = u(2) + (5)\,(2)^2 \Rightarrow u = 10m/s$

$u^2 = u_o^2 + 2g\,(h-40)$, upper part

$(10)^2 = 0 + 2\,(10)\,(h - 40) \Rightarrow h = 45m$

65. A mailbox is released from a helicopter that is descending at 1.5 m/s. after 2 sec.
a. What is the speed of the box?
b. How far is it below the helicopter?
c. What are the values of part (a) and (b) if the helicopter is rising at 1.5 m/s?

Solution: a. $v = u + gt = 1.5 + (10)\,(2) = 21.5$ m/s

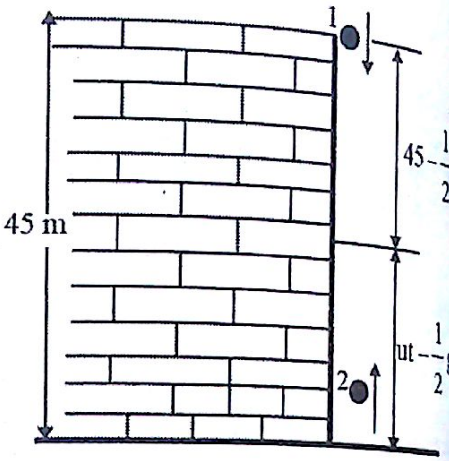
b. $y_{box} = ut + \frac{1}{2}gt^2$ and $Y_{helicopter} = ut$

$$\Delta y = Y_{box} - Y_{heli} = ut + \frac{1}{2}gt^2 - ut = \frac{1}{2}(10)(2)^2 = 20m$$

c. $v = u - gt = 1.5 - (10)(2) = 18.5$ m/s

$$\Delta y = ut - \frac{1}{2}gt^2 - ut = \frac{1}{2}gt^2 = \frac{1}{2}(10)(2)^2 = 20m$$

66. A ball is thrown upward from the ground with an initial speed of 25m/s. At the same instant, another ball is dropped from the building. If the building is 45m high, after how long will the ball be at the same height?



Solution: Displacement of ball 1 = Displacement of ball 2

$$ut - \frac{1}{2}gt^2 = 45 - \frac{1}{2}gt^2$$

$ut = 45 \Rightarrow 25t = 45$ or $t = 1.8$ sec

67. Two boys are on a balcony 20m above the ground. One of them throws a ball downward at 15m/s and the other throws a ball vertically upward at the same speed and at the same moment.
a. What is the difference in their time of flight?
b. What is the velocity of each ball as it hit the ground?
c. How far are the balls 0.8sec after they are thrown?

Solution: a. $y_1 = ut_1 + \frac{1}{2}gt_1^2$, for ball one

$$20 = 15t_1 + 5t_1^2 \Rightarrow t_1 = 1\text{sec}$$

$$y_2 = 20 + ut_2 - \frac{1}{2}gt_2^2, \text{ for ball two}$$

$$-20 = 15\,t_2 - 5t^2 \Rightarrow t_2 = 4\text{sec}$$

$$\Delta t = t_2 - t_1 = 4 - 1 = 3\text{sec}$$

b. $v_1 = v_2 = \sqrt{u_2 + 2gh} = \sqrt{(15)^2 + (2)(10)(20)} = 25\text{m/s}$

c. $y_1 = ut + \frac{1}{2}gt^2$ and $y_2 = 20 + ut - \frac{1}{2}gt^2$

$$\Delta y = y_1 + y_2 = 2ut + 20 = 2(15)\,(0.8) + 20 = 44\text{m}$$

68. In south America, the ultimate test of a young man's courage used to be to jump off a cliff 45m into a river.
a. How long did the man fly in air?
b. How fast was the jumper upon impact with the river?

Solution: a. $y = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{\frac{2y}{g}} = \sqrt{\frac{2(45)}{10}} = 3\,\text{sec}$

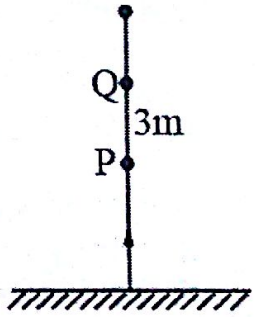
b. $v^2 = 2gy \Rightarrow v = \sqrt{2gy} = \sqrt{(2)(10)(45)} = 30\ \text{m/s}$

69. A man sitting on a tree limb wishes to drop vertically on to a horse racing under the tree at a speed of 10m/s. If the man is initially 3m above the saddle, what must be the horizontal distance between the man and the saddle when he jump?

Solution: $x = ut$ and $y = \frac{1}{2}gt^2$. Solving for t

$$\frac{x}{u} = \sqrt{\frac{2y}{g}} \Rightarrow x = (u)\left(\sqrt{\frac{2y}{2}}\right) = (10)\left(\sqrt{\frac{2(3)}{10}}\right) = 7.7\text{m}$$

70. A stone is thrown vertically upward. On its way up, it passes point P with speed V and point Q 3m above P with speed 0.5v. Calculate
a. The speed V
b. The height moved by the stone above Q



Solution: a. at P, $v^2 = u^2 - 2gh$

At Q, $(0.5v)^2 = u^2 - 2g(h+3)$

$$0.25\,v^2 = v^2 - 6g \Rightarrow v = 8.9\text{m/s}$$

b. $v^2 = u^2 - 2gh \Rightarrow h = \frac{(0.5\,v)^2}{2g} = \frac{(4.45)^2}{2(10)} = 1\text{m}$

71. Two objects begin a free fall from rest, from the same height one sec apart. How long after the first object begins to fall will the two objects be 10 apart?

Solution: $\Delta y = y_1 - y_2 = \frac{1}{2}gt^2 - \frac{1}{2}g(t-1)^2$

$10 = 5\,t^2 - 5\,(t^2 - 2t + 1)$, solve for t

$t = 1.5$ sec

72. Spider man steps from the top of a tall building. If he falls a distance if h/4 in the last 1 second of his fall, what is the height of the building?

Solution: $Y_n^{th} = u + \frac{g}{2}(2t-1)$

$$\frac{h}{4} = 5(2t-1) \text{ or } h = 20(2t-1) = 40t - 20$$

$$\text{and, } h = \frac{1}{2}gt^2 = 5t^2 \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (**)$$

From equation (*) and (**)

$t = 4$ sec

$\therefore h = 5t^2 = 5(4)^2 = 80m$

73. A rock is dropped from the top of 60m tall building. How far above the ground is the rock 1.2 sec before it reaches the ground?

Solution: $Y = \frac{1}{2}gt^2 \Rightarrow t_T = \sqrt{\frac{2y}{g}} = \sqrt{\frac{2(60)}{10}} = 3.5\text{sec}$, total time.

and 1.2 sec before it reach the ground,

$t = 3.5 - 1.2 = 2.3$ sec.

$$\text{therefore, } \Delta y = \frac{1}{2}gt_T^2 - \frac{1}{2}gt^2 = 60 - 5(2.3)^2 = 33.6m$$

74. A boy throws a ball upward and wants to hit it with a stone when it is 4m above the ground. If the stone is thrown 2 sec later with the same speed as the ball, what must be this speed?

Solution: For the ball, $y_b = ut_b - \frac{1}{2}gt_b^2 \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (*)$

For the stone, $y_s = ut_s - \frac{1}{2}gt_s^2 \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (**)$

But, $y_b = y_s$ and $t_s = t_b - 2$

$$\Rightarrow ut_b - \frac{1}{2}gt_b^2 = u(t_b - 2) - \frac{1}{2}g(t_b - 2)^2$$

$$ut_b - \frac{1}{2}gt_b^2 = ut_b - 2u - \frac{1}{2}gt_b^2 + 2gt_b - 2g$$

$$\Rightarrow t_b = \frac{u}{g} + 1 \text{ substitute it to equation (*)}$$

$$y_b = u\left(\frac{u}{g}+1\right)-\frac{1}{2}g\left(\frac{u}{g}+1\right)$$

$$4=\frac{u^2}{g}+u-\frac{1}{2}\frac{u^2}{g}-u-\frac{1}{2}g \Rightarrow u^2=180$$

$$u=\sqrt{180}=13.4\text{m/s}$$

75. Your sister who is in a 20m high building wants to hit you with an egg while you are walking beneath the building at a speed of 1.5m/s. How long before you pass beneath should she drop the egg? Assume you are 1.7 tall?

Solution: The time you moved horizontally is equal to the time the egg takes to move vertically.

$$x = ut \text{ and } y=\frac{1}{2}gt^2 \Rightarrow t=\frac{x}{u}=\sqrt{\frac{2y}{g}}$$

$$t=\sqrt{\frac{2(20-1.7)}{10}}=1.9\text{sec}$$

76. A stone is thrown up from the top of a tower, and reaches the ground in 5.6 sec. Another stone thrown down from the same position reaches the ground in 3.6 sec. If the initial velocities are equal. Determine

a. The initial velocities of the stones.

b. The height of the tower

Solution: a. Time taken by the first stone to reach the top of the tower on its way down is, t = 5.6 – 3.6 = 2sec and time to reach maximum height above the tower is 1 sec

$$t=\frac{u}{g}\Rightarrow u=(t)(g)=(1)(10)=10\text{ m/s}$$

b. $y=ut+\frac{1}{2}gt^2 = (10)(3.6) + 5(3.6)^2 =$ 101m

77. A body falling freely passes two points 10m a part in 0.2sec. From what height above the higher point did it start to fall?

Solution: For the motion from M to J,

$$y=ut+\frac{1}{2}gt^2 \Rightarrow 10=u\,(0.2)+5(0.2)^2$$

or $u = 49m/s$

For the motion A to M.

$u = 0$ and $v = 49m/s$

$$v^2 = u^2 + 2gh \Rightarrow h = \frac{v^2}{2g} = \frac{(49)^2}{(2)(10)} = 120m$$

78. An object falls 20m in a second. What time is required to cover the next 20m?

Solution: $y = ut + 0.5gt^2 \Rightarrow 20 = u(1) + 5(1)^2$ or $u = 15m/s$. The velocity after one second is then, $v = u + gt = 25m/s$ which is the initial velocity for the next 20m trip

$Y = ut + 0.5\,gt^2 \Rightarrow 20 = 25t + 5t^2$ or $t = 0.7$ sec

79. A body thrown vertically down from the top of a cliff covers 40m in the 4^{th} second of its fall. What is the initial velocity of the body?

Solution: $y_n^{th} = u + \frac{g}{2}(2t-1) \Rightarrow 40 = u + 5(2(4) - 1)$

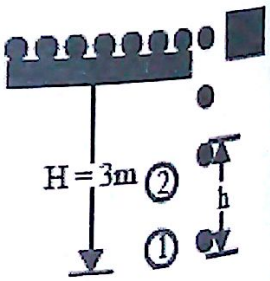
$u = 5m/s$

80. Small steel balls fall from rest through the opening at constant rate of two balls per second. Find the separation (h) between two consecutive balls when the lower one is dropped 3m.

Solution:

Ball 1, $H = \frac{1}{2}gt^2$ ..... (*)

Ball 2, $H - h = \frac{1}{2}g\left(t - \frac{1}{2}\right)^2$ ..... (**)



From equation (*) and (**)

$$\Rightarrow \frac{1}{2}gt^2 - h = \frac{1}{2}gt^2 - \frac{1}{2}gt + \frac{g}{8} = 2.6m$$

81. A ball is thrown straight upward with a speed u from a height h above the ground. Show that the time taken for the ball to reach the ground is, $t = \frac{u}{g}\left(1 + \sqrt{1 + \frac{2gh}{u^2}}\right)$

Solution: $-h = ut - \frac{1}{2}gt^2$, for the entire trip.

$t^2 - 2ut - \frac{2h}{g} = 0$. Using the quadratic formula

$t = \frac{u}{g}\left(1 + \sqrt{1 + \frac{2gh}{u^2}}\right)$, after rearrangement.

82. Two balls are released from the same height in one second difference. How long after the first ball begins to fall will they be 10m apart?

Solution: $\Delta y = y_1 - y_2 = \frac{1}{2}gt^2 - \frac{1}{2}g(t-1)^2$

$10 = 5t^2 - 5t^2 + 5t - 5$

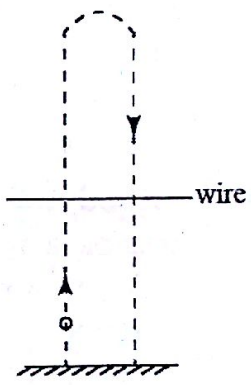
$\Rightarrow t = 1.5$ sec

83. A ball is thrown upward and it passes an overhead wire in 0.4 sec. on the way down, it passes the same wire 3.2 sec after being thrown. What is?

a. the time needed to reach h_{max}

b. the initial speed of the ball

c. the height of the wire from the ground



Solution: a. $0.4 + 2t = 3.2 \Rightarrow t = 1.4$ sec

$t_{max} = 0.4 + 1.4 = 1.8$ sec

b. $u = gt_{max} = (1.8)(10) = 18$m/s

c. $y = ut - 0.5g\,t^2 = 18(0.4) - 5\,(0.4)^2 = 6.4$m

84. A stone is dropped from a top of a building and passes through three equidistance marks. The time taken to travel from the first to the second and from the second to the third mark are 2 sec and 1 sec respectively. What is the separation between the successive marks?

Solution: $y_1 = y_2 = y = ut_1 + 0.5gt_1^2 = 2u + 20$ ..... (*)

$y_1 + y_2 = 2y = u(t_1 + t_2) + 0.5g\,(t_1 + t_2)^2 = 3u + 45$ .....(**)

From equation (*) and (**), u = 5m/s and y = 30m

85. A stone is dropped from a tower of 60m high. At the same time, another stone is thrown upward from the foot of the tower and meets the first stone at a height of 18m. What is the velocity of the second stone with which it is thrown?

Solution: $y = 0.5gt^2 \Rightarrow 60 - 18 = 5t^2$ or $t = 2.9$ sec

$18 = ut - 0.5\,gt^2 = 2.9\,(u) - 5\,(2.9)^2 \Rightarrow u = 20.7$ m/s

86. A stone is dropped from the top of a tower 50m high. At the same time, another stone is thrown upwards from the foot of the tower with a velocity of 25m/s. when and where the stones cross each other?

Solution: For stone one, $y = 5t^2$ ..... (*)

For stone two, $50 - y = 25t - 5t^2$ ..... (**)

From the two equation

$t = 2\text{sec}$ and $y = 20\text{m}$

87. A balloon is ascending at the rate of 12m/s. when it is at a height of 65m from the ground, a packet is dropped from it.

a. how long will it take to reach the ground?

b. with what velocity will the packet hit the ground?

Solution: a. The velocity of the packet is 12m/s upward due to inertia. As packet's motion is downward, initial velocity can be taken as negative.

$$y = ut + \frac{1}{2}gt^2 \Rightarrow -12t + 5t^2 = 65 \Rightarrow t = 5\text{ sec}$$

b. $v = -u + gt = -12 + (10)(5) = 38\text{m/s}$

88. A body of weight 200N is dropped from a certain height and it attains a maximum velocity before it hit the ground. With this velocity it moves at a constant of proportionality 5 kg/s. find

a. the terminal velocity

b. the height it moved

Solution: a. $kv = mg \Rightarrow v = \frac{mg}{k} = \frac{200}{5} = 40\text{m/s}$

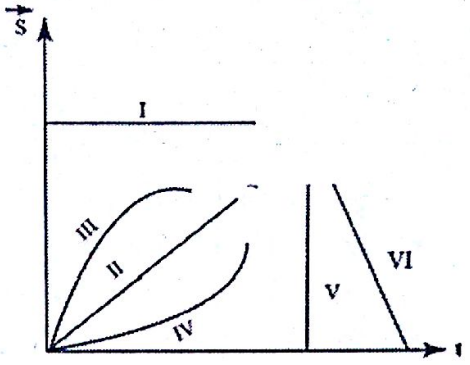
b. $v^2 = u^2 + 2gh \Rightarrow h = \frac{v^2}{2g} = \frac{(40)^2}{2(10)} = 80\text{m}$

3.6 Graphical Representation of Motion

A. Displacement – Time Graph, $\vec{s}$ - t graph

In a displacement – time graph, the displacement is plotted against y – axis and the time against x – axis. This is because, displacement is a dependent variable and time is an independent variable.

Keep In Mind the Following



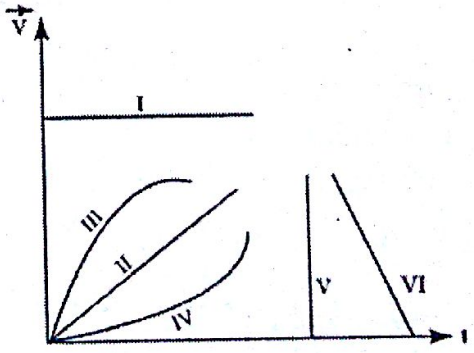
- The slope of $\vec{s}-t$ graph gives velocity.
- If the $\vec{s}-t$ graph is a straight line parallel to time – axis, the body has zero velocity, i.e, the body is at rest. This case is indicated by graph I.
- If the $\vec{s}-t$ graph is a straight line inclined to the time axis at an angle θ such that $0^{o} < \theta < 90^{o}$, then the body is moving with constant velocity. This case corresponds to graph II. If $\theta > 90^{o}$, then the velocity of the body is negative.
- In the case of graph III, the slope is decreasing with time. So, the velocity is decreasing with time. This is a case of retarded motion
- In the case of graph IV, the slope is increasing with time. So, velocity is increasing with time. This is a case of accelerated motion.
- If the $\vec{s}-t$ graph is perpendicular to time axis like graph V, then it represents infinite velocity. Practically speaking, this is an impossible situation.
- In the case of graph VI, velocity is constant but its direction is opposite to displacement.

B. Velocity – Time Graph, $\vec{v}$ - t graph

In a velocity – time graph, the velocity is plotted against y – axis and time against x – axis. This is because velocity is a dependent variable

Keep In Mind the Following

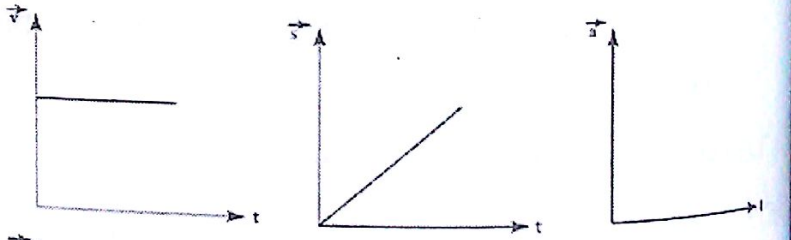
- The slope if $\vec{v}-t$ graph gives acceleration
- The area under $\vec{v}-t$ graph gives displacement
- If the $\vec{v}-t$ graph is a straight line parallel to time axis, then the body is moving with constant velocity. This is shown in graph **I**. In this case, the acceleration is zero.



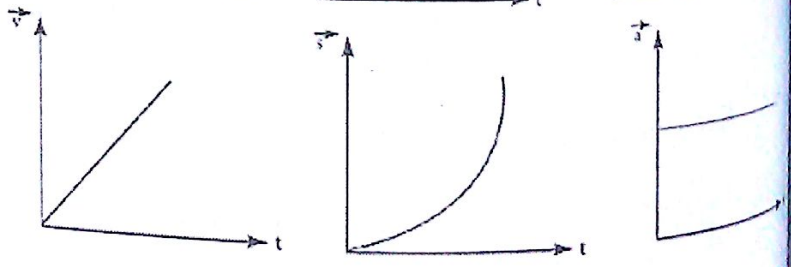
- If the $\vec{v}-t$ graph is a straight line inclined to the time – axis, at angle θ such that $0^o < \theta < 90^o$, the body is moving with constant acceleration. This is shown in graph II. If $\theta > 90^o$, then the body is moving with constant retardation.
- In the case of graph III, the slope is decreasing with time. So, acceleration is decreasing with time
- In the case of graph IV, the slope is increasing with time. So, acceleration is increasing with time.
- In the case of graph V, the body is moving with infinite acceleration. This is practically, impossible.
- If a body starts from rest, its $\vec{v}-t$ graph passes through the origin.
- If the initial velocity of a body is not zero, then its $\vec{v}-t$ graph does not pass through the origin.
- Graph VI represents constant retardation
- While calculating displacement from $\vec{v}-t$ graph, add up the areas with due regard for positive and negative signs
- While calculating distance from speed – time graph, use only positive sign.

A summary on most common Graphical Representations.

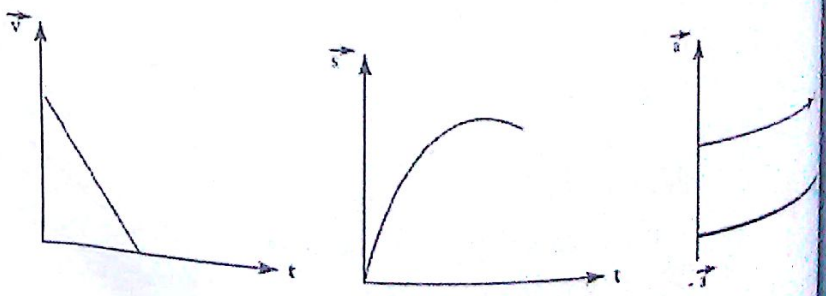
Uniform motion $\Delta\vec{v} = o, \vec{a} = 0$



Uniformly accelerated motion Motion, $\vec{a} > 0$



Uniformly decelerated motion, $\vec{a} < 0$



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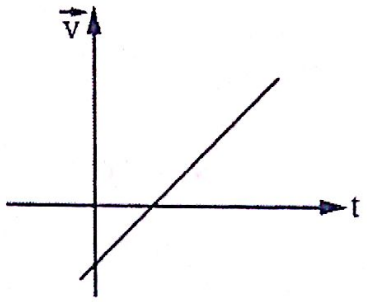
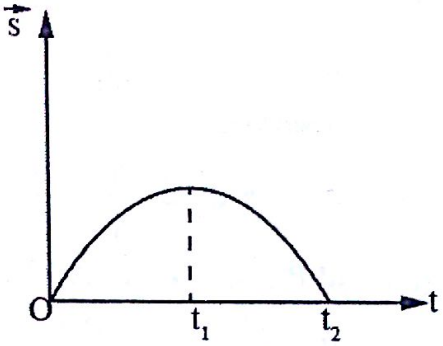
Conceptual Examples

89. For the $\vec{v}-t$ graph shown, is the acceleration
a. +ve or – ve
b. constant or varying



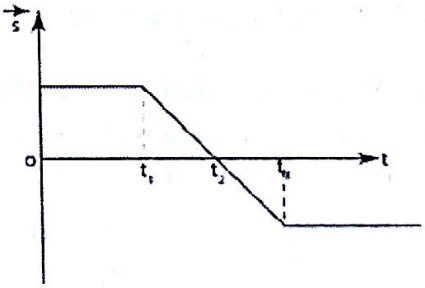
Explanation: a. +ve, increasing
b. constant, because it is a straight line

90. For the $\vec{s}-t$ graph shown, in which time interval is the velocity.
a. in the same direction as the displacement
b. opposite to the displacement
c. when does the object return to its initial position.



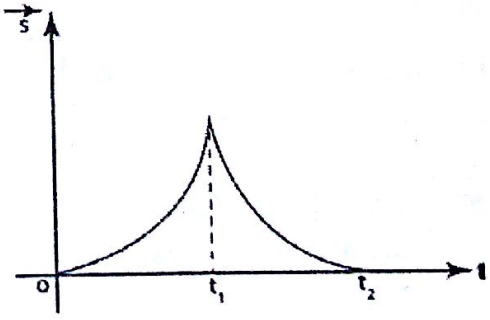
Explanation: a. From t_1 to t_2 b. From 0 to t_1 c. At t_2

91. Qualitatively describe the nature of the moving body whose $\vec{a}-t$ graph is shown



Explanation: The body is moving with constant acceleration from 0 to t_1. It is constantly retarding from t_1 to t_3. The body is then moving at a constant deceleration from t_3 onwards.

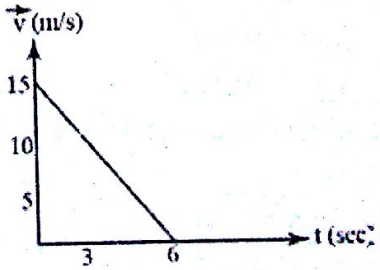
92. In which interval is the body
a. Accelerating
b. Retarding



Explanation:
a. in the interval 0 to t_1
b. in the interval t_1 to t_2

Conceptual Examples

93. The velocity time graph of a car that slows down at uninform rate is shown. Find
a. The acceleration
b. The distance travelled before it stops.



Solution:

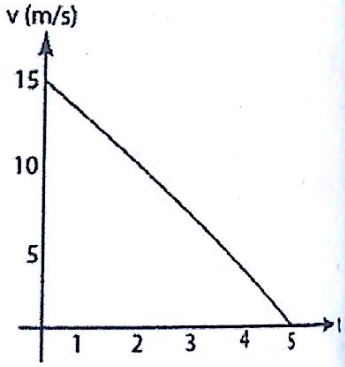
a. $\vec{a} = \text{slope} = \frac{\Delta v}{\Delta v} = \frac{0-15}{6-0} = -2.5 m/s^2$

b. $\vec{s} = \text{Area} = \frac{1}{2}vt = \frac{1}{2}(15)(6) = 45\,m$

94. The velocity – time graph of a moving object is shown below. How far does the object move in the first two seconds? (UEE: 2001)

A. 13m B. 26m C. –26m D. –13m

Solution:



$\vec{a} = \text{slope} = \frac{\Delta v}{\Delta v} = \frac{0-15}{5-10} = 3 m/s^2$

at t=2sec. v=u+at =15–(3)(2) = 9m/s

$S = \frac{v^2 - u^2}{2a} = \frac{(9)^2 - (15)^2}{-2(3)} = 26m$

3.7 Motion in a Plane

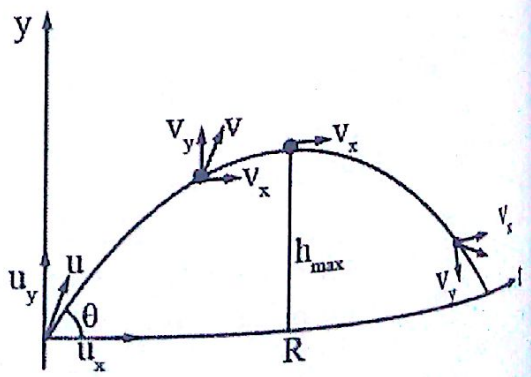
An important practical example of motion with uniform acceleration in two dimensions is the projectile

Projectile Motion

A missile shot from a gun, a rocket after its fuel is exhausted, a body dropped from air plane, a ball thrown into the air are all examples of projectile

Projectile: Is a body which is in flight through the air but is not drived on its way by any fuel.

☛ The path followed by a projectile is called *trajectory*. Now, consider a body that is thrown into the air with an initial velocity u at an angle θ from the horizontal.



The initial x and y components of the velocity u are:

$u_x = u\cos\theta$ and $u_y = u\sin\theta$ ..... (1)

The x and y components of the velocity at any time are:

$\boxed{v_x = u\cos\theta \text{ and } v_y = u\sin\theta - gt}$ .....(2)

The magnitude and direction of the velocity at any time is:

$$\boxed{v = \sqrt{v_x^2 + v_y^2} \text{ and } \tan\theta = \frac{v_y}{v_x}} \quad \ldots\ldots (3)$$

The x and y components of the displacement at any time is:

$$\boxed{x = (u\cos\theta)t \text{ and } y = (u\sin\theta)t - \frac{1}{2}gt^2} \quad \ldots\ldots (4)$$

The magnitude and direction of the displacement at any time is

$$\boxed{S = \sqrt{x^2 + y^2} \text{ and } \tan\theta = \frac{y}{x}} \quad \ldots\ldots (5)$$

The maximum height reached by the body is obtained from $v_y^2 = u_y^2 - 2g\,h_{max}$. But, $v_y = 0$ at h_{max}

$$\boxed{h_{max} = \frac{u_y^2}{2g} = \frac{u^2\sin^2\theta}{2g}} \quad \ldots\ldots (6)$$

Time to reach the maximum height is obtained from $v_y = u_y - gt$. But, $v_y = 0$ at hmax

$$\boxed{t = \frac{u_y}{g} = \frac{u\sin\theta}{g}} \quad \ldots\ldots (7)$$

Total time of flight is calculated from $y = (u\sin\theta)t - \frac{1}{2}gt^2$ but, $y = 0$

when the body return to the ground

$$\boxed{t = \frac{2u\sin\theta}{g}} \quad \ldots\ldots (8)$$

The maximum horizontal distance called range covered by the body is found from:

$x = (u\cos\theta)t$, here $t = \dfrac{2u\sin\theta}{g}$

$$\boxed{R = (u\cos\theta)\left(\frac{2u\sin\theta}{g}\right) = \frac{u^2 2\sin\theta\cos\theta}{g}} \quad \ldots\ldots (9)$$

but, since $\sin 2\theta = 2\sin\theta\cos\theta$

$$\boxed{R = \frac{u^2 2\sin\theta\cos\theta}{g} = \frac{u^2 \sin 2\theta}{g}}$$

Now, from $x = (u\cos\theta)t \Rightarrow t = \frac{x}{u\cos\theta}$ and we know

$y = (u\sin\theta)t - \frac{1}{2}gt^2$ substitute for t here

$$y = (u\sin\theta)\left(\frac{x}{u\cos\theta}\right) - \frac{1}{2}g\left(\frac{x}{u\cos\theta}\right)$$

$$= x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}, \frac{1}{\cos^2\theta} = \sec^2\theta$$

$$= x\tan\theta - \frac{gx^2}{2u^2}\sec^2\theta,\ \sec^2\theta = 1 + \tan^2\theta$$

$$\boxed{y = x\tan\theta - \frac{gx^2}{2u^2}(1+\tan^2\theta)} \longleftarrow \text{The equation of trajectory.}$$

☛ **Important points to remember**

- The horizontal and vertical motion components of a projectile are independent of each other. Thus, the motion of a projectile can be thought of as the result of two separate, simultaneous occurring components of motion.
- In the absence of air resistance, the horizontal motion of the projectile is uniform motion and has no acceleration. While, the vertical motion of the projectile is an accelerated motion controlled by the force of gravity.
- The horizontal range remains the same for angle θ or $90-\theta$ from the horizontal.
- The horizontal range remains the same for angle θ from the horizontal or from the vertical.

Conceptual Examples

95. A boy sitting in a moving car throws a ball upward into the air. Will the ball fail behind him? In front of him? In to his hand? Explain

Explanation: Into his hand if the car is moving at constant speed. Behind him if the car is accelerating. In front of him if the car is decelerating.

96. Suppose you run at constant speed and wish to throw an apple so that you can catch it as it comes back down. In which direction should you throw the apple? Explain

Explanation: Upward. Because, the apple is moving along with you, it will follow you with a horizontal velocity that is same as yours.

Illustrative Examples

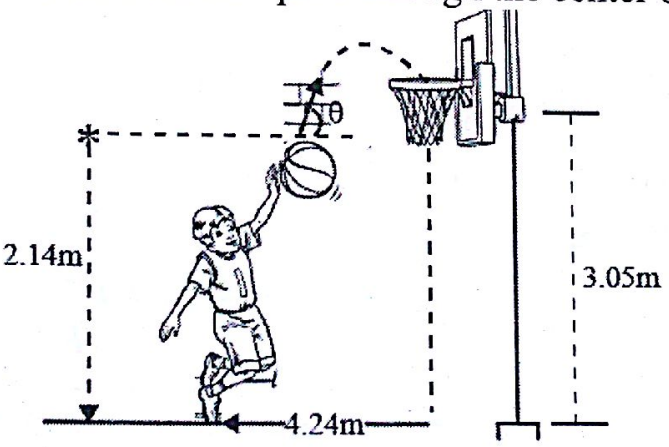
97. In his "I am a Disco Dancer" film, the Indian actor, Mitu throws his Guitar to his girlfriend who is as tall as he is and is 6.5m away from him. If he throws the guitar at an angle of 42°, find the guitar's

a. Initial speed b. Time of flight

Solution: a. $R = \frac{u^2 \sin 2\theta}{g} \Rightarrow 6.5 = \frac{u^2 \sin 2(42°)}{10} \Rightarrow u = 8\text{m/s}$

b. $t = \frac{2u \sin\theta}{g} = \frac{2(8)(\sin 42)}{10} = 1.1\text{sec}$

98. The number one basketball player, Michael Jordan, release his ball at 50° from the horizontal. What should be the initial speed to cause the ball pass through the center of the rim?



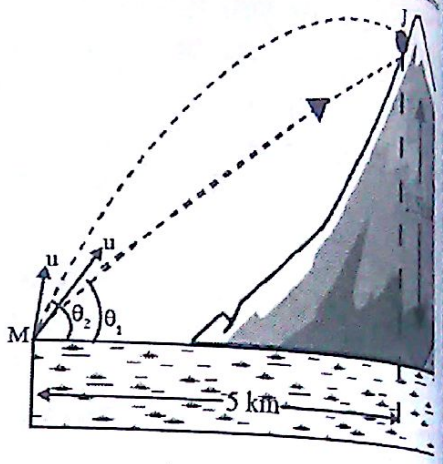
Solution: $x = (u \cos\theta)\, t \Rightarrow t = \frac{x}{(u)\cos\theta} = \frac{4.24}{(u)\cos(50)} = \frac{6.6}{(u)}$

and $y = u \sin\theta t - \frac{1}{2}gt^2$, $y = 3.05 - 2.14 = 0.91\text{m}$

$$0.91 = u \sin 50° \left(\frac{6.6}{u}\right) - \frac{1}{2}(10)\left(\frac{6.6}{u}\right)^2$$

$u = 7.25$ m/s

99. The muzzle velocity of a long range rifle at M is $u = 400$ m/s. Determine the two possible angles that allow the bullet to hit a target J on the mountain.



Solution:

Using the trajectory equation.

$$y = x\tan\theta - \frac{gx^2}{2u^2}(1+\tan^2\theta)$$

$$\Rightarrow 1500 = 5000\tan\theta - \frac{10(5000)^2}{2(400)^2}(1+\tan^2\theta)$$

$$\Rightarrow \tan^2\theta - 6.5\tan\theta + 2.96 = 0 \Rightarrow \theta_1 = 26^\circ \ \& \ \theta_2 = 80^\circ$$

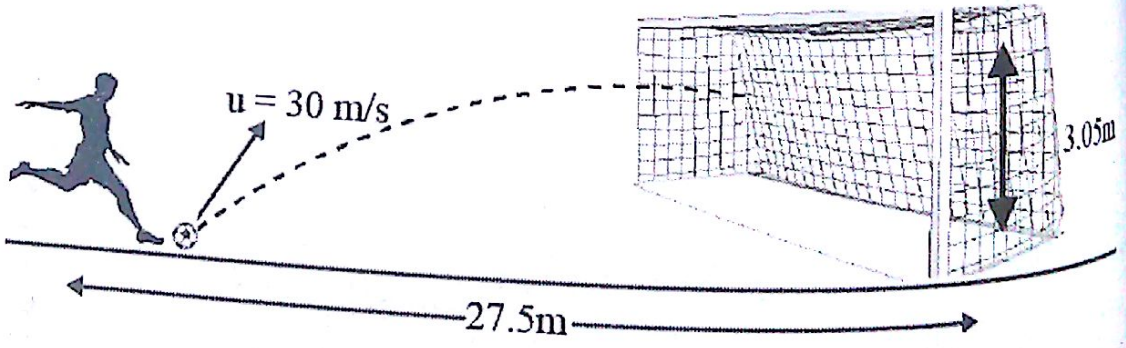
100. Saladin, the Ethiopian striker, shot a ball 27.5m from the goal. If he gives the ball an initial speed of 30 m/s, what minimum angle did he give the ball if it hit the cross bar of the goal?

Solution: $y = x\tan\theta - \frac{gx^2}{2u^2}(1+\tan^2\theta)$

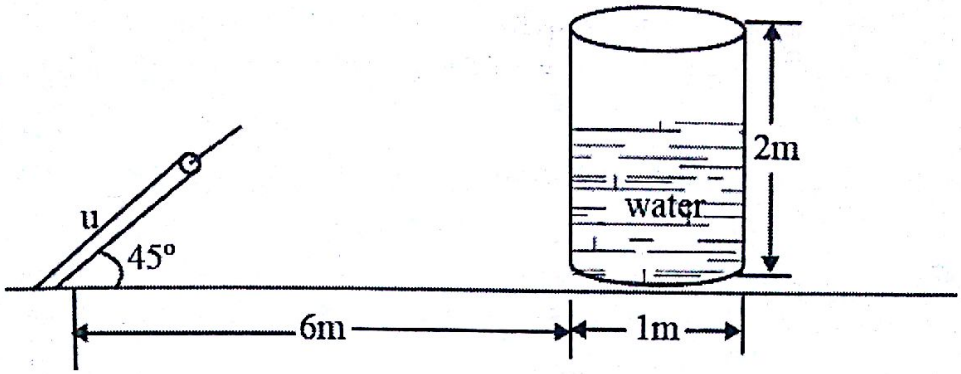
$$3.05 = 27.5\tan\theta - \frac{(10)(27.5)^2}{2(30)^2}(1+\tan^2\theta)$$

$$4.2\tan^2 - 27.5\tan\theta + 7.25 = 0$$

$$\Rightarrow \theta = 14.7^\circ \text{ or } \theta = 81^\circ$$



101. A water hose is used to fill a tanker of diameter 1m and height 2m. If the hose is 6m from the tanker and shoots water at 45° from the horizontal, for what range of launch speed will the water enter the tank?



Solution: From the trajectory equation;

$$y = x\tan\theta - \frac{gx^2}{2u^2}(1+\tan^2\theta)$$

$$y_1 = x_1\tan\theta - \frac{gx^2}{2u_1^2}(1+\tan^2\theta)$$

$$2 = 6\tan(45) - \frac{10(6)^2}{2u^2}(1+(\tan(45)^2)$$

$$\Rightarrow u_1 = 9.5 \text{ m/s}$$

$$\text{And } y_2 = x_2\tan\theta - \frac{gx^2}{2u_2^2}(1+\tan^2\theta)$$

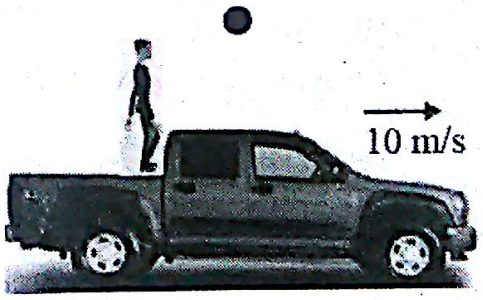
$$2 = 7\tan(45) - \frac{10(7)^2}{2u^2}(1+(\tan(45)^2))$$

$$\Rightarrow u_2 = 9.9 \text{ m/s}$$

Thus, the speed should be between u_1 and u_2 for water to enter into the tanker. That is;

$u_1 < u < u_2 \Rightarrow 9.5 \text{ m/s} < u < 9.9 \text{ m/s}$

102. A boy stands on a pick up that is moving at 10 m/s horizontally. If the boy throws a ball and catch it after the car has gone 20m. Find:



a) The angel at which the ball is thrown.

b) The initial speed of the ball.

Solution: a) R = (u cos θ) t, but u cos θ = speed of the car

$\Rightarrow t = \frac{R}{v} = \frac{20}{10} = 2\text{sec}$ and time of flight is also obtained by:

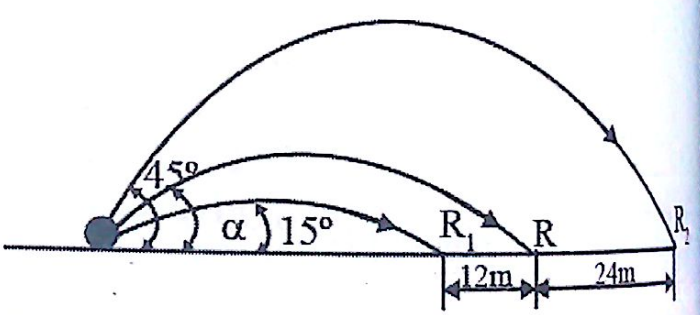
$$u = \frac{gt}{2\sin\theta} \Rightarrow t = \frac{2u\sin\theta}{g} \text{ and } R = \frac{2u^2\sin\theta\cos\theta}{g}$$

$$R = 2\left(\frac{gt}{2\sin\theta}\right)^2 \frac{\sin\theta\cos\theta}{g} = \frac{gt^2}{2\tan\theta} \Rightarrow \tan\theta = \frac{gt^2}{2R}$$

$$\theta = \frac{(10)(2)^2}{2(20)} \Rightarrow \theta = 45°$$

b) $u = \frac{gt}{2\sin\theta} = \frac{10(2)}{2\sin 45°} = 14\text{m/s}$

103. A bullet is aimed at a mark on the horizontal plane through the point of projection. It falls 12m short when the angle is 15°. And, it overshoots the mark by 24m when the angle is 45°. For what angle will the bullet hit the mark.



Solution:

$$R_1 = \frac{u^2\sin 2\alpha_1}{g} = (R-12) = \frac{u^2\sin 30}{g} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (*)$$

$$R_2 = \frac{u^2\sin 2\alpha_2}{g} = (R+24) = \frac{u^2\sin 90}{g} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (**)$$

Dividing eqn (*) by eqn (**)

$$\frac{R-12}{R+24} = \frac{0.5}{1} \Rightarrow R = 48\text{m}$$

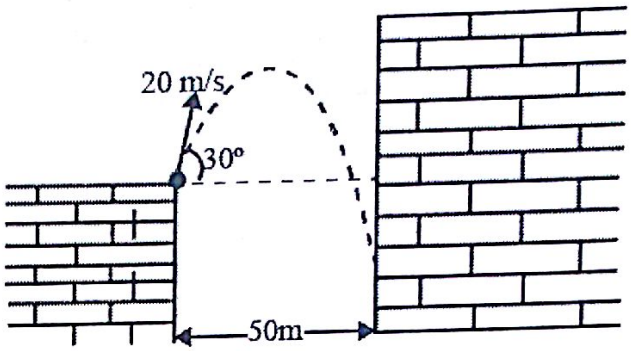
Substituting the value of R in to equation (**). We have,

$$48 + 24 = \frac{u^2}{10} \Rightarrow u = 26.8 \text{ m/s, and we know that;}$$

$$R = \frac{u^2\sin 2\alpha}{g} \Rightarrow \sin 2\alpha = \frac{Rg}{u^2} = \frac{(48)(10)}{720} = 0.667$$

$$\sin 2\alpha = 0.667 \Rightarrow 2\alpha = 4.8° \Rightarrow \alpha = 20.9°$$

104. A ball is thrown from the top of a building towards another building 50m away. If the initial velocity is 20 m/s at 30° above the horizontal. How far below its original level will the ball strike the opposite wall?



Solution:

$u_x = u\cos\theta = (20)(\cos 30) = 17.3$ m/s

$u_y = u\sin\theta = (20)(\sin 30) = 10$ m/s

We know, $x = u_x t \Rightarrow t = \dfrac{x}{u_x} = \dfrac{50}{17.8} = 2.9\,\text{sec}$

So we have, $y = u\sin\theta t - \dfrac{1}{2}gt^2$

$= (20)(\sin 30)(2.9) - \dfrac{1}{2}(10)(2.9)^2 = 13\text{m}\rightarrow$ below the level line.

105. A ball rolls off the top of a stairway with a horizontal velocity u. If the steps are h meters high and w meters wide, show that the ball will just hit the edge of n^{th} step if $n = \dfrac{2hu^2}{gw^2}$

Solution: In the n^{th} step, the horizontal and vertical distance travelled by the ball are; nw and hn respectively.

Therefore, from $x = ut$

$$nw = ut \ldots\ldots\ldots\ldots (*)$$

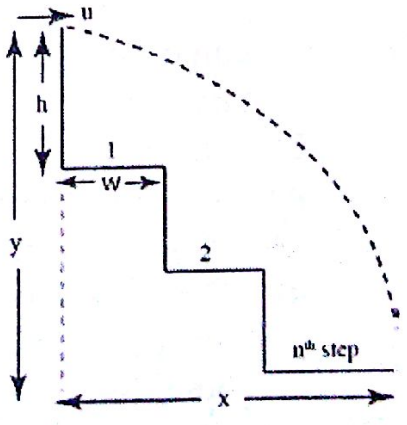
and, from $y = \dfrac{1}{2}gt^2$

$$nh = \frac{1}{2}gt^2 \ldots\ldots\ldots\ldots\ldots\ldots (**)$$

from equation (*) $t = \dfrac{nw}{u}$.

Substitute this to equation (**)

$$nh = \frac{1}{2}g\left(\frac{nw}{u}\right)^2 \quad \text{or} \quad n = \frac{2hu^2}{gw^2}$$



106. A ball is thrown with an initial velocity of $(3i + 4i)$ m/s from a 2.4m height above the ground.
   a. To what maximum height will the ball rise?
   b. What horizontal distance will the ball travel?

Solution: a. $h_{max} = 2.4 + \dfrac{v_y^2 - u_y^2}{2g} = 3.2m$

b. $-h = u_y t - \dfrac{1}{2}gt^2 \Rightarrow -2.4 = 4t - 5t^2$ or $t = 1.2sec$

107. A bird is sitting on the top of a tree 10m high with what velocity should a boy standing at a distance of 25m from the tree throw a stone at an angle of 30^o with the horizontal so as to hit the bird?

Solution: From the trajectory equation

$$y = x\tan\theta - \frac{gx^2}{2u^2}(1 + \tan^2\theta) \Rightarrow u = 28m/s$$

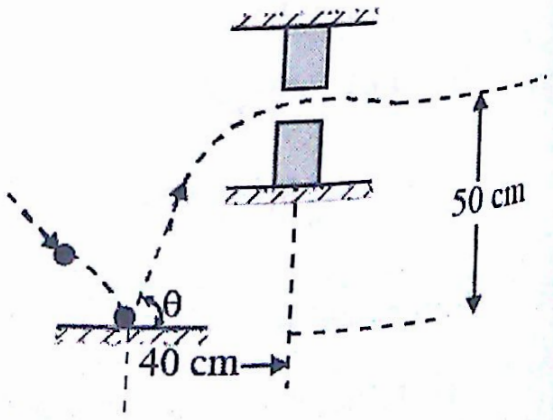
108. Jit Lee, the chinese actor jump from a bridge that is 45m above the water on to a boat that is moving at a constant speed. If he jumped when the boat is 12m from the point of impact, what was the speed of the boat?

Solution: $x = ut \Rightarrow t = \dfrac{x}{u}$ ..... (*)

$y = \dfrac{1}{2}gt^2$, substitute for t from equation (*)

$$y = \frac{1}{2}g\frac{x^2}{u^2} \text{ or } u = x\sqrt{\frac{g}{2y}} = 12\sqrt{\frac{10}{2(45)}} = 4m/s$$

109. A game in physics class requires the players to bounce a tennis ball and pass it through the opening shown. Calculate:
   a) The angle θ the ball makes while bounding.
   b) The velocity of the ball as it pass through the opening.



Solution: a) $h = \frac{u^2 \sin^2\theta}{2g}, R = \frac{u^2 \sin 2\theta}{g}$

$$\frac{h}{R} = \frac{1}{2}\tan\theta \Rightarrow \theta = 68°$$

b) $h = \frac{u^2 \sin^2\theta}{2g} \Rightarrow u = \sqrt{\frac{2gh}{\sin^2\theta}} = 3.4 m/s$

$u_x = u\cos\theta = 1.3$ m/s

110. A ball rolls off the edge of a horizontal table top 4m high. It strikes the floor at a point 5m horizontally away from the edge of the table. How long was the ball in the air in second? (UEE: 2002)

A. $\frac{3}{\sqrt{5}}$ B. $\frac{1}{\sqrt{5}}$ C. $\frac{2}{\sqrt{3}}$ D. $\frac{2}{\sqrt{5}}$

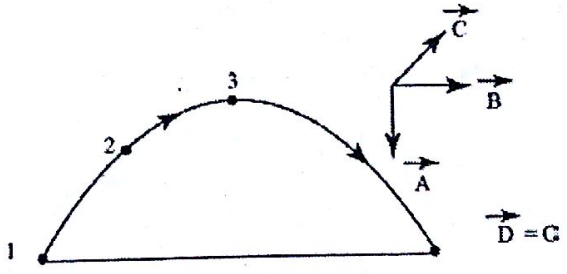
Solution: $h = \frac{1}{2}gt^2 \Rightarrow 4 = 5t^2 \Rightarrow t = \frac{2}{\sqrt{5}}$ sec **Answer: D**

111. A ball is thrown downward from the top of a tower. After being released, its downward (UEE: 2003)

A. velocity will be uniform
B. Acceleration will be equal to g
C. Acceleration will be greater than g
D. Acceleration will be smaller than g **Answer: B**

112. A bullet is fired from cannon at point 1 and follows the trajectory shown. Which vector best represents the bullets acceleration at point 2? (UEE: 2003)

A. $\vec{A}$
B. $\vec{B}$
C. $\vec{C}$
D. $\vec{D}$



Explanation: The acceleration of the bullet is acceleration due to gravity and it is always downward **Answer: A**

113. A projectile is fired at 60° with the horizontal. At the highest point of its path, the vertical component of its acceleration at this point? (UEE: 2004)

A. 20 m/s^2 B. 10 m/s^2 C. 5 m/s^2 D. 0 m/s^2 **Answer: B**

114. Two projectiles are fired from the ground level at equal speed but different angles. One is fired at an angle of 30° and the other at 60°. The projectile to hit the ground first will be the one fired at (UEE: 2006)

A. 60 degrees C. Both hit at same time
B. 30 degrees D. cannot be determined

Explanation: $t = \frac{2u\sin\theta}{g}$, smaller sin θ value smaller the time taken to land

Answer: B

115. If a long distance jumper athlete leaves the ground at an angle of 37° above the horizontal at a speed of 10m/s, how far does he jump in the horizontal direction (UEE: 2007)

A. 4.8 m B. 6 m C. 9.6 m D. 12 m

Solution: $R = \frac{u^2 2\cos\theta\sin\theta}{g} = \frac{(10)^2 2\cos 37°\sin 37°}{10} = 9.6\text{m}$

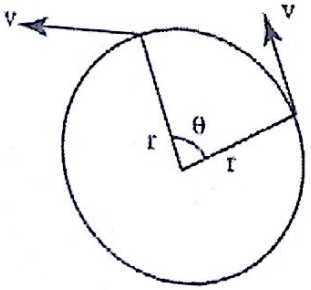
Answer: C

3.8 Uniform Circular Motion

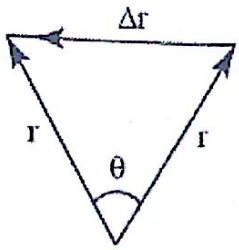
A car, a satellite, an ice Skater etc. moving around a circle with constant speed are all in uniform circular motion.

Uniform Circular Motion: Is the motion of a body around a circular path with constant speed.

Now, consider a body moving with constant speed v in a circle of radius r. to obtain an expression for the magnitude of position and velocity vectors. Notice that the triangles formed by position and velocity vectors are isosceles and similar.



Then,



Therefore, $\frac{\Delta v}{\Delta r} = \frac{v}{r}$

Rearranging and dividing by Δt, we get $\frac{\Delta v}{\Delta t} = \frac{v}{r}\frac{\Delta r}{\Delta t}$

In the limit as $\Delta t \to 0, \frac{\Delta v}{\Delta t}$ becomes acceleration and $\frac{\Delta r}{\Delta t}$ becomes velocity. And hence,

$$\boxed{a = \frac{v^2}{r}} \longrightarrow \text{centripetal acceleration}$$

From the fact that, for one complete revolution, distance moved by the body is equal to the circumference of the circle and the time taken is the period, we have:
The centripetal acceleration to have the form:

$$\boxed{a = \frac{4\pi^2 r}{T^2}}$$

☛ **Important points to remember**

- Even though the speed of a body moving in a circle is constant, it is accelerating because there is constant change in direction of the body.
- In uniform circular motion the velocity vector is always tangent to the path of the body and is perpendicular to the radius of the circle
- The acceleration vector is always perpendicular to the path and always point towards the center of the circle.

Conceptual Examples

116. An airplane travels from Addis Ababa to USA and its speed reads a constant value of 600 km/h. Is the velocity of the plane constant? Explain

Explanation: No. It is not. Due to the curvature of the earth, the plane is continuously changing direction even if its speed is constant.

Conceptual Examples

117. What is the centripetal acceleration of the earth as it moves in its own orbit around the sun?

Solution: $a = \frac{4\pi^2 r}{T}$, $r = 1.5\times10^8 km = 1.5\times10^{11} m$ $= 365$ days $= 3.15\times10^7$ sec

$$a = \frac{4(3.14)^2(1.5\times10^{11})}{(3.15\times107)^2} = 6\times10^{-3}\ m/s^2$$

118. A satellite is in circular orbit at an altitude of 200km above the earth's surface, where $g = 9.2\ m/s^2$. What is the orbital speed of the satellite?

Solution: $a\frac{v^2}{r}$, $r = R + h$ and $a = g$

$$\Rightarrow v = \sqrt{g(R+h)} = \sqrt{(9.2)(6.7\times10^6 + 200\times10^3)} = 7770m/s$$

119. The orbit of the moon about the earth has a radius of 3.84×10^8m. It takes the moon 27.3 days to complete one revolution about earth. Find the orbital speed of the moon.

Solution: $a = \frac{4\pi^2 r}{T^2} = \frac{(4\pi^2)(3.84\times10^8)}{(2.4\times106)^2} = 2.7\times10^{-3}m/s^2$

$$v = \sqrt{ar} = \sqrt{(2.7\times10^{-3})(3.84\times10^8)} = 1022m/s$$

120. Uniform circular motion is a (UEE: 2003)
   A. motion with constant velocity
   B. motion in one dimension
   C. motion with acceleration of constant magnitude
   D. motion in which the net force is constant **Answer: C**

3.9 Relative Velocity

In reality, everything is relative. There is no such thing absolute.

Relative velocity: Is the velocity of a body relative to a fixed body or a moving body.

Suppose you are travelling in car 1 and another car 2 moving in the same direction on the same road crosses your car. If you look at car 2, it seems to move slower than when someone standing on the ground watches both cars. If car 2 is moving with the same speed as your car, then, it will appear to you as though it were not moving at all. On the other hand, if car 2 is moving in the opposite direction and crosses your car, it will seem to you to move much faster. In such cases, we are speaking of the relative velocity of the bodies.

If body A and body B are moving with velocities V_A and V_B either in the same or opposite direction, then;

- The velocity of body A with respect to body B, V_{AB} is

$$\boxed{V_{AB} = V_A - V_B}$$

- The velocity of body B with respect to body A, V_{BA} is

$$\boxed{V_{BA} = V_B - V_A}$$

Conceptual Examples

121. A passenger in an elevator drop a coin while the elevator is descending at constant speed. What can you say about the acceleration of the coin compared to g, when observed by
   a. a passenger in the elevator
   b. someone standing on the ground

Explanation: a. a = g b. a = g, it is the same

122. A ball is thrown up in the air by a passenger on a train that is moving with constant velocity. Describe the path of the ball as seen by
   a. The passenger
   b. Stationary observer on the ground

Explanation: a. vertically up and then down
   b. parabolic path

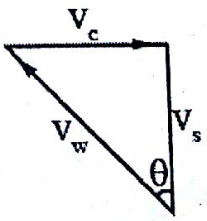
Illustrative Examples

123. A swimmer in Bahir Dar can swim at a speed of 0.9m/s with respect to the water. He wants to cross a river which is 70 m wide and has a current of 0.7m/s
   A. In what direction must he swim to cross the river directly opposite from his starting point?
   B. How long will it take him to cross
   C. How long will it take him to cross in the shortest possible time?
   D. How far from the intended point will he land in the other side?

Solution: A. $V_s = V_w + V_c$ and $\sin\theta = \frac{V_c}{V_w} = \frac{0.7}{0.9} \Rightarrow \theta = 51^\circ$

B. $V_s = V_w \cos\theta = 0.9 \cos 51 = 0.89 m/s$

$$t = \frac{s}{v_s} = \frac{70}{0.89} = 78 sec$$



C. $t = \frac{s}{v_w} = \frac{70}{0.9} = 77.8 sec$

D. $d = v_c t = (0.7)(78) = 54m$

124. A man in an elevator that is descending at a constant speed of 5m/s drops his pen. 3 sec after release of the pen, what is its position, velocity and acceleration as seen by
   a. The man in the elevator

b. Someone on the ground

Solution: a. $S_o = \frac{1}{2}gt^2 = 45m$, $V = gt = 30m/s$, $a = g = 10\,m/s^2$

b. $s = s_o - vt = 45 - 5\,(3) = 30m$, $v_1 = v - v_o = 30 - 5 = 25m/s$, $a = g = 10m/s^2$

125. A river is flowing east at 30m/s and a boat is traveling south at 40 m/s w.r.t the river. What is the speed and direction of the boat w.r.t the earth?

Solution: $V_R = (30i, 0j)$ m/s $V_B = (0i, 40j)$ m/s

$V_{BE} = (0 - 30)\,i + (-40 - 0)\,j$ m/s $= (-30i - 40j)$ m/s

$V_{BE} = \sqrt{(30)^2 + (-40)^2} = 50m/s$ south of east.

126. Dire Dawa is about 300 km east of Addis Ababa. A plane leaves Addis Ababa due east at 500 km/hr while the wind is blowing to the south at 100km/hr. what is the velocity of the plane relative to the ground? (UEE: 2001)

A. 510 km/hr north of east C. 490 km/hr north of west

B. 510 km/hr south of east D. 490 km/hr south of west

Solution: $V = \sqrt{(500)^2 + (100)^2} = 510km/h$, south east.

Answer: B

127. A boat is traveling at 15km/hr w. r. t the water. The water flows at 6km/hr w. r. t the ground. A child on the boat walks from rear seat to front at 5km/hr. w. r. t. the boat. What are the velocities of the boat and the child in km/hr w. r. t the ground respectively? (UEE: 2002)

A. 6, 12 B. 5, 9 C. 21, 12 D. 9, 14

Solution: $V_{bw} = V_{bg} + V_{wg} \Rightarrow 15 = V_{bg} + 6$ or $V_{bg} = 9km/h$

$V_{cb} = V_{cg} - V_{bg} \Rightarrow 5 = V_{cg} - 9$ or $V_{cg} = 14$ km/h

Answer: D

128. A small car and a bus move with velocities given by $\vec{V}_c = (3i - 2j)m/s$ and $\vec{V}_b = (-5i - 4j)$ m/s respectively, according to an observer on the ground. What is the velocity of the bus according to the car driver? (UEE: 2003)

A. $(-2i + 6j)m/s$ B. $(-2i - 6j)$ m/s

C. $(-8i + 2j)$ m/s D. $(-8i - 2j)$ m/s

Solution: $V_{bc} = V_{bg} - V_{cg} = (-si - 4j) - (3i - 2j) = (-8i - 2j)$ m/s

Answer: D

Additional Examples

129. If a body starts from a point and returns back to the same point, then its
A. average velocity is zero, but not average speed
B. average speed is zero, but not average velocity
C. both average speed and velocity are zero
D. average speed and velocity depend upon the path
Explanation: when a body returns to the starting point the displacement and hence the average velocity is zero. But both distance and average speed are not zero. **Answer: A**

130. Which of the following changes when a particle is moving with uniform velocity?
A. position vector B. speed
C. velocity D. acceleration
Explanation: Both speed and velocity are constant for a body moving with uniform velocity. A body moving with uniform velocity has zero acceleration. **Answer: A**

131. Two cars are moving in opposite directions with speed v. What is the magnitude of their relative velocity?
A. o B. v C. v/2 D. 2v
Explanation: $V_{AB} = V_A + V_B = V + V = 2\,v$ **Answer: D**

132. A train is moving east at a speed of 5m/s. A bullet fired west wards with a velocity of 10m/s crosses the train in 8 sec. The length of the train is?
A. 120 m B. 60 m C. 30 m D. 15 m
Explanation: Relative velocity is 5 + 10 = 15 m/s length of train = vt = (15) (8) = 120m. **Answer: A**

133. A car's odometer reads 22687 km at the start of a trip and 22791 km at the end of the trip. If the trip takes 4 hours, then what is the average speed of the car?
A. 0 B. 26km/hr C. 7.2 km/hr D. 4km/hr
Solution: $\frac{S}{t_T} = \frac{22791 - 22687}{4} = 26\text{km/hr}$ **Answer: B**

134. The relative velocity of a car moving with 40 km/h due east with respect to itself is?
A. 0 B. 10km/h C. 50km/h D. 60km/hr
Solution: $V_{cc} = 40 + (-40) = 0$ **Answer: A**

135. A girl moves with a velocity of 18 km/h along a straight line joining two stationary objects. She will observe that both the objects

A. move in the same direction with the same speed of 18 km/h
B. move in different directions with the same speed of 18 m/h
C. move towards her
D. remains stationary

Explanation: Both the objects move at 18 km/h in opposite direction to the girl. **Answer: A**

136. A toy car travels from A to B at a constant speed of 30 km/h and without stopping at B returns to A at a constant speed v. If the average speed of the car is 24 km/h, then v is

A. 24 km/h B. 28 km/h
C. 30 km/h D. 32 km/h

Solution: $V_{av} = \frac{S_T}{t_T} = \frac{S_1 + S_2}{t_1 + t_2} = \frac{u_1 t_1 + u_2 t_2}{t_1 + t_2}$

Since, $s_1 = s_2 \Rightarrow v_1 t_1 = u_2 t_2 \Rightarrow \frac{v_1}{v_2} = \frac{t_2}{t_1}$

$\frac{20}{v} = \frac{t_2}{t_1}$ but, $24 = \frac{20t_1 + vt_2}{t_1 + t_2}$

$24\, t_1 + 24t_2 = 20t_1 + v\, t_2$, divide by t_1

$24 + 24\frac{t_2}{t_1} = 20 + \frac{vt_2}{t_1}, \frac{t_2}{t_1} = \frac{20}{v}$

$24 + 24\left(\frac{20}{v}\right) = 20 + (v)\left(\frac{20}{v}\right) = 30$ km/hr **Answer: C**

137. Which of the following changes in the case of uniform motion?

A. speed B. velocity
C. acceleration D. position

Explanation: When a particle is in motion, its position changes continuously with time. **Answer: D**

138. A boy walks to his school with constant speed of 2.5 km/h and walks back with a constant speed of 4 km/h. His average speed for round trip, is?

A. $\frac{1}{2}$km/h C. $\frac{24}{13}$km/hr

B. 3 km/h D. $\frac{40}{13}$km/hr

Solution: $V_{av} = \frac{S_1 + S_2}{t_1 + t_2} = \frac{u_1 t_1 + u_2 t_2}{t_1 + t_2} = \frac{v_1\left(\frac{s}{v_1}\right) + v_2\left(\frac{s}{v_2}\right)}{\left(\frac{s}{v_1}\right) + \left(\frac{s}{v_2}\right)}$

$= \frac{s+s}{\frac{v_2 s + v_1 s}{(v_1)(v_2)}} = \frac{2s(v_1)(v_2)}{s(v_1+v_2)} = \frac{2v_1 v_2}{v_1 + v_2}$

$= \frac{2(2.5)(4)}{(2.5)+4} = \frac{20}{6.5}\frac{\text{km}}{\text{h}} = \frac{40}{13}$km/h **Answer: D**

139. The numerical ratio of displacement to distance is
A. always = 1 C. always > 1
B. always < 1 D. may be ≤ 1

Answer: D

140. If the displacement of a particle is zero, the distance covered
A. must be zero C. is negative
B. cannot be zero D. may or may not be zero

Answer: D

141. A man leaves his house for a cycle ride. He comes back to his house after half an hour after covering a distance of one km. what is his average velocity for the ride?

A. 0 km/h B. 2 km/s C. Km/s D. $\frac{1}{2}$km / h

Explanation: Displacement is zero. So does velocity.

Answer: A

142. A car travels from A to B at a speed of 30km/h. the average speed of the car for the whole journey is?
A. 5 km/h B. 24 km/h
C. 25 km/h D. 50 km/h

Solution: $V_{av} = \frac{2v_1 v_2}{v_1 + v_2} = 2\frac{(20)(30)}{20+30} = \frac{1200}{50} = 24\text{km/h}$

Answer: B

143. The numerical ratio of velocity and speed of a particle is always
A. equal to or less than one C. less than one
B. equal to one D. zero **Answer: A**

144. When two bodies move uniformly towards each other, the distance decreases by 6m each second. If both bodies move in the same direction with the same speeds as before the distance between them increases by 4m each second. What are she speed of the bodies:

A. 3m/s and 3m/s
B. 4 m/s and 2 m/s
C. 5 m/s and 1m/s
D. 7 m/s and 3 m/s

Solution: When two bodies move toward each other

$$V_{AB} = V_A + V_B \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (*)$$

When two bodies move in the same direction

$$V_{AB} = V_A - V_B \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (**)$$

From the two equations

$$6 = V_A + V_B$$
$$4 = V_A - V_B \qquad \text{solving at a time}$$
$$\Rightarrow V_A = 5\text{m/s and } V_B = 1\text{m/s}$$

Answer: C

145. A drunkard is walking along a straight road. He takes 5 steps forward and 3 steps backward, followed by 5 steps forward and 3 steps backward and so on. Each step is one meter long and takes one second. There is a pit on the road 11m away from the starting point. The drunkard will fall into the pit after?

A. 21 sec B. 29 sec C. 31sec D. 37 sec

Explanation: His displacement is 2m in 8 sec. For (11 - 5)m = 6m it takes him $3 \times 8 = 24$ sec. Last 5m will take 5 sec. so, total time is $t = 24 + 5 = 29$ sec

Answer: B

146. Wind is blowing west to east along two parallel tracks. Two trains moving with same speed in opposite direction have the steam track of one double than other. The speed of each train is

A. half that of wind
B. equal to that of wind
C. double that of wind
D. three times that of wind

Solution: $v + w = 2(v - w) \Rightarrow v = 3w$

Answer: D

147. A 120 m long train is moving west at a speed of 10m/s. A small bird flying east at a speed of 5m/s crosses the train. What is the time taken by the bird to cross the train?

A. 4 sec
B. 8 sec
C. 12 sec
D. 24 sec

Solution: $V_{bt} = V_b + V_t = 5 + 10 = 15\text{m/s}$

$$t = \frac{s}{t} = \frac{120}{15} = 8\text{sec}$$

Answer: B

148. A train of length 200m travelling at 30 m/s over takes another train of length 300m travelling at 20m/s. the time taken by the first train to pass the second is

A. 10 sec B. 30 sec
C. 40 sec D. 50 sec

Solution: Time $= \frac{S_1 + S_2}{V_1 + V_2} = \frac{500}{50} = 10\,\text{sec}$ **Answer: A**

149. The displacement – time graphs of two particles A and B are straight lines making angles of 30° and 60° with the time axis respectively. If the velocity of A is V_A and that of B is VB, then the ratio $\frac{V_A}{V_B}$ is

A. $\frac{1}{2}$ B. $\frac{1}{\sqrt{3}}$ C. $\sqrt{3}$ D. $\frac{1}{3}$

Explanation: $\frac{V_A}{V_B} = \frac{\tan 30^\circ}{\tan 60^\circ} = \frac{1}{\sqrt{3}} \times \frac{1}{\sqrt{3}} = \frac{1}{3}$ **Answer: D**

150. A particle travels at a constant velocity of 3m/s for 20 sec. It then stops for 10 sec before completing its journey in a further 30 sec at a constant velocity of 2m/s. Its average velocity for the whole journey is

A. 2m/s B. 2.4m/s C. 2.5m/s D. 3m/s

Solution: $S_T = u_1t_1 + v_2t_2 + v_3\,t_3$

$S_T = (3)\,(20) + 0 + (2)\,(30) = 120\text{m}$

$t_T = t_1 + t_2 + t_3 = 20 + 10 + 30 = 60\ \text{sec}$

$V_{av} = \frac{S_T}{t_T} = \frac{120}{60} = 2\text{m/s}$ **Answer: A**

151. A railway train 200m long passes over a bridge 600m long. The time taken by the train to cross the bridge with uniform speed of 36km/h is

A. 60 sec B. 80 sec C. 800 sec D. 20 sec

Solution: $V = \frac{360\text{km}}{\text{h}} = 36\left(\frac{1000}{3600}\right) = 10\text{m/sec}$

$t = \frac{S_1 + S_2}{v} = \frac{600 + 200}{10} = 80\,\text{sec}$ **Answer: B**

Uniformly Accelerated Motion

152. If a ball is thrown vertically upward with speed u, the distance covered during the last t seconds of its ascent is

A. ut B. $\frac{1}{2}gt^2$ C. $ut-\frac{1}{2}gt^2$ D. $ut-\frac{1}{2}gt^2$

Solution: $h=\left(ut-\frac{1}{2}gt^2\right)-\left[u(t-1)-\frac{1}{2}g(t-1)^2\right]$

$\Rightarrow h=\frac{1}{2}gt^2$ **Answer: B**

153. The velocity of a body depends on time according to equation, $v = 20 + 0.1\ t^2$. The body is undergoing

A. uniform retardation C. non-uniform acceleration
B. uniform acceleration D. zero acceleration

Explanation: Differentiating, $v = 20 + 0.1\ t^2 \Rightarrow a = 0.2t$ since the acceleration is time dependent, it is non-uniform.

Answer: C

154. Ball A is dropped to the ground from the top of a tower while another ball B is thrown towards the earth with some initial velocity from same tower. Neglecting air resistance, the acceleration is.

A. same for both the ball
B. greatest for the dropped ball A
C. greatest for ball b
D. depends up on the size of A and B

Hint: the acceleration is due to gravity for both balls.

Answer: A

155. Two stones of different masses are dropped simultaneously from the top of a building. Which of the following is correct?

A. Larger stone hits the ground earlier
B. Smaller stone hits the ground earlier
C. Their speed depends on their composition
D. Both the stones reach the ground simultaneously

Explanation: The stones reach the ground simultaneously does not depend upon the mass of the body, in vacuum.

Answer: D

156. A body starts from rest and moves with uniform acceleration a. the distance covered during n^{th} second will be?

A. $u+\frac{a}{2}(2n-1)$ B. $\frac{a}{2}\left(2n-\frac{1}{2}\right)$

C. $a\left(2n-\frac{1}{2}\right)$ D. $a\left(n-\frac{1}{2}\right)$

Explanation: $S_n^{th} = S_n - S_{n-1} = \frac{1}{2}an^2 - \frac{1}{2}a(n-1)^2$

$$S_n^{th} = a\left(n-\frac{1}{2}\right)$$ **Answer: D**

157. A body having uniform acceleration of $10m/s^2$ has a velocity of 100m/s. In what time, the velocity will be doubled?

A. 8 sec B. 10 sec

C. 12 sec D. 14 sec

Solution: $v = u + at \Rightarrow 200 = 100 + 10t \Rightarrow t = 10$ sec **Answer: B**

158. The equation $\sqrt{x} = t+9$ gives the variation of displacement with time. Which of the following is correct?

A. velocity is proportional to time

B. velocity is inversely proportional to time

C. acceleration depends upon time

D. acceleration is constant

Hint: squaring both sides of $\sqrt{x} = t+9 \Rightarrow x = t^2 + 18t + 81$

which resembles, $s = ut + \frac{1}{2}at^2$ **Answer: D**

159. Which one of the following equations represents the motion of a body with constant acceleration?

A. $y = at$ B. $y = at + bt^2$

C. $y = at + bt^2 + ct^3$ D. $y = \frac{a}{t} + bt$

Hint: $y = at + bt^2$, resembles, $S = ut + \frac{1}{2}at^2$ **Answer: B**

160. One body is dropped, while a second body is thrown downward with an initial velocity of 1m/s simultaneously. The separation between them is 18m after a time of

A. 4.5 sec B. 9 sec C. 18 sec D. 36 sec

Solution: For the dropped, $y_1 = \frac{1}{2}gt^2$

For the thrown, $y_2 = ut + \frac{1}{2}gt^2$

$\Delta y = y_2 - y_1 = ut \Rightarrow t = \frac{y_2 - y_1}{u} = \frac{18m}{1m/sec} = 18$ **Answer: C**

161. Given $v = 0.6 - 0.3t$. The acceleration is

A. $-0.3\ m/s^2$ B. $0.6\ m/s^2$ C. $2.4\ m/s^2$ D. $3.6\ m/s^2$

Solution: $a = \frac{dv}{dt} = -0.3m/s^2$ **Answer: A**

162. Two bodies of masses m_1 and m_2 fall from heights h_1 and h_2 respectively. The ratio of their velocities when they hit the ground is?

A. $\frac{h_1}{h_2}$ B. $\sqrt{\frac{h_1}{h_2}}$ C. $\frac{m_1h_1}{m_2h_2}$ D. $\frac{h_1^2}{h_2^2}$

Solution: $v_1^2 = 2gh_1$ and $v_2^2 = 2gh_2$

$\Rightarrow \frac{v_1^2}{v_2^2} = \frac{2gh_1}{2gh_2} \Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{h_1}{h_2}}$ **Answer: B**

163. Two stones are thrown from the top of a tower, one straight down with an initial speed u and the second straight up with the same speed u. when the two stones hit the ground, they will have speeds in the ratio.

a. 2:3 B. 2:1 C. 1:2 D. 1:1

Solution: For downward throw, $v_1^2 = u^2 + 2gh$

For upward throw, $v_2^2 = u^2 - 2g(-h)$

taking ratio, $\frac{v_1^2}{v_2^2} = \frac{1}{1} \Rightarrow \frac{v_1}{v_2} = \frac{1}{1} = 1:1$ **Answer: D**

164. A car starts from rest and moves with constant acceleration. The ratio of distances covered by the car in n^{th} second to that covered in n seconds is

A. $\frac{2n-1}{2n^2}$ B. $\frac{n^2}{2n-1}$

C. $\frac{2n-1}{n^2}$ D. $1:n$

Solution: $S_n^{th} = \frac{a}{2}(2n-1)$ and $S_n = \frac{1}{2}an^2$

taking ratio: $\frac{S_n^{th}}{S_n} = \frac{a(2n-1)}{2\left(\frac{1}{2}an^2\right)} = \frac{2n-1}{n^2}$ **Answer: C**

165. A particle is dropped from a height of 3.48m. In what time the particle would travel the first meter distance

A. $\sqrt{\frac{1}{2}}$ sec B. $\sqrt{\frac{1}{3}}$ sec C. $\frac{1}{2}$ sec D. $\sqrt{\frac{1}{5}}$ sec

Solution: $h = \frac{1}{2}gt \Rightarrow t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{(2)(1)}{10}} = \sqrt{\frac{1}{5}}$ sec

Answer: D

166. A particle starts with initial velocity u and retardation a. It will reach the point of start in time

A. $\frac{u}{a}$ B. $\frac{u}{a^2}$ C. $\frac{u^2}{a}$ D. $\frac{2u}{a}$

Solution: when it return, s = 0 and,

$$S = ut - \frac{1}{2}at^2 \Rightarrow ut = \frac{1}{2}at^2 \Rightarrow t = \frac{2u}{a}$$ **Answer: D**

167. The initial velocity, of a particle is 10m/s and its retardation is $2m/s^2$. The distance moved by the particle in 5^{th} second of its motion is

A. 75m B. 50m C. 19m D. 1m

Solution: $S_n^{th} = u + \frac{a}{2}(2n-1) = 10 - \frac{2}{2}(2(5)-1) = 1m$

Answer: D

168. A car at rest accelerates uniformly to a speed of 144 km/h over a distance of 400m. The time taken is

A. 5sec B. 10 sec C. 15 sec D. 20 sec

Solution: $S = \left(\frac{v+u}{2}\right)t \Rightarrow 400 = \left(\frac{\frac{144\times1000}{3600}+0}{2}\right)t$

Answer: D

$\Rightarrow t = 20$ sec

169. A trolley runs down a slope from rest with uniform acceleration. in the first second, it travels 1.6. Its acceleration is

A. $1.6\ m/s^2$ B. $3.2\ m/s^2$

C. $4.8\ m/s^2$ D. $6.4\ m/s^2$

Solution: $S = ut + \frac{1}{2}at^2 \Rightarrow 1.6 = (0)(1) + \frac{1}{2}(a)(1)^2$

$\Rightarrow a = 3.2\ m/s^2$. **Answer: B**

170. A ball thrown vertically up with a speed v comes back to the starting point with speed v. The ratio of distance covered and displacement is

A. zero B. 1 C. –1 D. infinite

Hint: $S = 2h$ and $\vec{S} = 0 \Rightarrow \frac{s}{\vec{s}} = \frac{2h}{0}$, infinite **Answer: D**

171. A ball thrown upward with a speed of v attains height h_1. Another ball thrown upwards from the same point with a speed of 2v attains a height h_2. Then, h_2 is equal to

A. h_1 B. $2h_1$ C. $3h_1$ D. $4h_1$

Solution: $v_2 = u_2 - 2gh$, at h_{max}, $v = 0$

$$\Rightarrow h_1 = \frac{v_1^2}{2g} \text{ and } h_2 = \frac{v_2^2}{2g}$$

Taking ratio, $\frac{h_2}{h_1} = \frac{v_2^2}{v_1^2} = \frac{(2v_1)^2}{v_1^2} = 4 \Rightarrow h_2 = 4h_1$ **Answer: D**

172. The displacement of a body is given by $y = 8t - 3t^2$. The average velocity of the body in the time interval of t = 0 and t = 1 sec is?

A.–4m/s B. zero C. 5m/s D. 6m/s

Solution: $V_{av} = \frac{\Delta s}{\Delta s} = \frac{s_2 - s_1}{t_2 + t_i} = \frac{(8(1) - 3(1)^2) - (8(0) - 3(0))}{1 - 0}$

$V_{av} = 5m/s$ **Answer: C**

173. If two balls of same density but different masses are dropped from a height of 100m, then

A. both will come together on earth

B. both will come late on the earth

C. first will come first and second after that

D. second will come first and first after that **Answer: A**

Hint: the acceleration is same for both

174. A body is thrown upward with a speed of 100 m/s on the return journey, the speed at the starting point is?
A. 9.8 m/s B. 10 m/s C. 100 m/s D. 200 m/s
Solution: $v^2 = = u^2 - 2gh$, but $h = 0$
$v^2 = u^2 \Rightarrow v = u = 100$ m/s **Answer: C**

175. You observe an object covering distances in the direct proportion to the time elapsed. What conclusion might you draw about its acceleration?
A. It is zero B. It is constant
C. It is decreasing D. It is increasing
Hint: $s \sim t \Rightarrow s = kt$, $k = v$, acceleration is zero **Answer: A**

176. A body is thrown upwards and reaches its maximum height. At that position
A. it acceleration is minimum
B. its velocity is zero and its acceleration is also zero.
C. its velocity is zero but its acceleration is maximum
D. its velocity is zero and its acceleration is the acceleration due to gravity **Answer: C**

177. The correct statement from the following is
A. A body having zero velocity will not necessarily have zero acceleration.
B. A body having zero velocity will necessarily have zero acceleration
C. A body having uniform speed can have only uniform acceleration
D. A body having non – zero velocity will have zero acceleration.

Explanation: i.
i. at maximum height velocity is zero but $g = 9.8 m/s^2$
ii. at constant velocity, acceleration is zero
Answer: A and D

178. A ball released from a certain height falls 5m in one sec. In 4 sec, it falls through
A. 80m B. 40 m C. 1.25m D. 20m
Solution: $y = \frac{1}{2}gt^2 = \frac{1}{2}(10)(4)^2 = 80m$ **Answer: A**

179. Which of the following four statements is false?
A. A body can have a constant velocity and still have a varying speed

B. A body can have zero velocity and still be accelerated
C. The direction of the velocity of a body can change when its acceleration is constant in magnitude
D. A body can have a constant speed and still have a varying velocity.

Answer: A

180. A ball is thrown upward, after it has left the hand, its acceleration
A. remains constant B. increases
C. decreases D. is zero

Answer: A

181. P, Q and R are three balloons ascending with velocities u, 4u and 8u respectively. If stones of the same mass are dropped from each, when they are at the same height, then
A. stone from P reaches the ground first
B. stone from Q reaches the ground first
C. stone from R reaches the ground first
D. All reach at the same time

Explanation: $h = -vt + \frac{1}{2}gt^2$ or $\frac{1}{2}gt^2 - vt - h = 0$

if u is minimum, t is minimum

Answer: A

182. The velocity of a body travelling with a uniform acceleration of $2m/s^2$ is 10m/s. What is its velocity after a time interval of 4sec?
A. 18m/s B. 16m/s C. 14 m/s D. 12 m/s

Solution: $v = u + at = 10 + (2)(4) = 18$ m/s

Answer: A

183. The displacement of a freely falling body is proportional to the
A. time of fall
B. square of time of fall
C. mass of the body
D. square of mass of the body

Hint: $y = \frac{1}{2}gt^2$

Answer: B

184. A body starts from rest and falls vertically from a height of 19.6m. If $g = 9.8m/s^2$, then the time taken by the body to fall through the last meter of its fall is
A. 2 sec B. 0.05 sec
C. 0.45 sec D. 1.95 sec

Solution: $y = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{\frac{2y}{g}}$

$$\Delta t = \frac{(2)(19.6)}{\sqrt{9.8}} - \frac{(2)(18.6)}{\sqrt{9.8}} = 0.05 \text{sec}$$

Answer: B

185. A rubber ball is dropped from a height of 5m on a plane. On bouncing, it rises to a height of 1.8m. On bouncing, the ball loses its velocity by a factor of

A. $\frac{3}{5}$ B. $\frac{9}{25}$ C. $\frac{2}{5}$ D. $\frac{16}{25}$

Solution: Fractional loss in velocity $= \frac{v_1 - v_2}{v_1}$

$v^2 = 2gh \Rightarrow v = \sqrt{2gh}$

Fraction loss, $\frac{\sqrt{2gh_1} - \sqrt{2gh_2}}{\sqrt{2gh_1}} = 1 - \frac{3}{5} = \frac{2}{5}$ **Answer: C**

186. On the moon, the acceleration due to gravity is $\frac{1}{6}$g. How long would it take an object to fall 100m at the surface of the moon?

A. 4.5 sec B. 8 sec C. 9 sec D. 11 sec

Solution: $y = \frac{1}{2} g_m t^2 \Rightarrow 100 = \frac{1}{2}\left(\frac{1}{6}\right)(10)t^2 \Rightarrow t \simeq 11 \text{sec}$

Answer: D

187. For a particle moving in a straight line, the displacement of the particle at time t is given by

$S = t^3 - 6t^2 + 3t + 7$

What is the velocity of the particle when its acceleration is zero?

A. −9m/s B. −12m/s C. 3m/s D. 42m/s

Solution: $S = t^3 - 6t^2 + 3t + 7$

$$v = \frac{ds}{dt} = 3t^2 - 12t + 3$$

$$a = \frac{du}{dt} = 6t - 12$$

When $a = 0 \Rightarrow t = 2\text{sec}$

V at $t = 2\text{sec} \Rightarrow v = 3(2)^2 - 12(2) + 3 = -9$ m/s **Answer: A**

188. A stone is thrown upwards with velocity u. At half the height, its velocity is

A. zero B. $\frac{u}{4}$ C. $\frac{u}{2}$ D. $\frac{u}{\sqrt{2}}$

Solution: $v^2 = u^2 - 2gh$, $v \sim \sqrt{h}$ and $v^1 \sim \sqrt{\frac{h}{2}}$

$$\frac{v_1}{v} = \frac{1}{\sqrt{2}} \text{ or } v_1 = \frac{u}{\sqrt{2}}$$ **Answer: D**

189. The distance through which a body falls in the n^{th} second is h. The distance through which it falls in the next second is

A. h B. $h + \frac{g}{h}$ C. $h - g$ D. $h+g$

Solution: $y = \frac{1}{2}g(n+1)^2 - \frac{1}{2}gn^2 = \frac{g}{2}(2n+1)$

Also, $h = \frac{g}{2}(2n-1) \therefore y = \frac{g}{2}\left(\frac{2h}{g} + 1 + 1\right) = h + g$ **Answer: D**

190. The velocity of a partile with displacement varies with $V = \sqrt{3\times + 16}$. What are the acceleration and initial velocity of the particle respectively?

A. 1.5 units and 4 units C. 16 units and 1.6 units
B. 3 units and 4 units D. 16 units and 3 units

Solution: $v = \sqrt{3\times 16} \Rightarrow v^2 = 3\times + 16$ comparing **Answer: B**
$v^2 - 16 = 2\times$ with. $v^2 - u^2 = 2a$, $u = 4$ and $a = 3$

191. A stone is allowed to fall from the top of a tower 100m high and at the same time another stone is projected vertically upwards from the ground with a velocity of 25m/s. The two stones will meet after

A. 4 sec B. 0.4 sec C. 0.04sec D. 4 sec

Solution: $y = \frac{1}{2}gt^2$ and $100 - y = 25t - \frac{1}{2}gt^2$ adding these two, **Answer: A**

$25t = 100$ or $t = 4$ sec

192. A sprinter can reach his maximum speed of 10m/s by accelerating at 2.5 m/s^2. In order to cover a distance of 100m, he will take

A. 9sec B. 10 sec C. 11sec D. 12sec

Solution: time to reach maximum speed

$$v = 0 + at \Rightarrow t_1 = \frac{v}{a} = \frac{10}{2.5} = 4\text{sec}$$

Distance covered in 4 seconds is

$$s_1 = \frac{1}{2}at^2 = \frac{1}{2}(2.5)(4)^2 = 20\text{m}$$

That means, the 80m is covered by constant speed.

$$t_2 = \frac{s_2}{v} = \frac{80}{10} = 8\text{sec}$$

$t_T = t_1 + t_2 = 4 + 8 = 12$ sec **Answer: D**

193. Two bodies begin a free fall from rest, from the same height 2 sec apart. How long after the first body begins to fall, the two bodies will be 40m apart

A. 1 sec B. 2 sec C. 3 sec D. 4 sec

Solution: $y_1 = \frac{1}{2}gt^2$ and $y_2 = \frac{1}{2}g(t-2)^2$

$$y_1 - y_2 = 40\text{m} \Rightarrow \frac{1}{2}gt^2 - \frac{1}{2}g(t-2)^2 = 40$$

$\Rightarrow st^2 - 5\,(t-2)^2 = 40$ or $t = 3$ sec **Answer: C**

194. Two balls are projected simultaneously with the same speed from the top of a tower, one upward and the other downward. If they reach the ground in 9 sec and 4 sec respectively. The height of the tower is

A. 100m B. 120 m C. 180 m D. 200 m

Solution: upward, $y = -ut + \frac{1}{2}gt^2 = -9u + (5)(81)$

Down ward, $y = ut + \frac{1}{2}gt^2 = 4u + (5)(16)$

So we have, u=25m/s and y = 180m

Answer: C

195. A lift in which a man is standing is moving up with a speed of 10m/s. The man drops a coin from a height of 4.9m, then the coin reaches the floor of the lift after a time

A. $\sqrt{2}$ sec B. 1 sec C. $\frac{1}{2}$sec D. $\frac{1}{\sqrt{2}}$sec

Solution: Consider relative velocity, acceleration and displacement of the coin with respect to the lift floor

$u = 10 - 10 = 0$, $a = 9.8 m/s^2$, $s = 4.9m$

$$S = ut + \frac{1}{2}gt^2 = 0 + \frac{1}{2}(9.8)\left(t^2\right) = 4.9 \Rightarrow t = 1 sec$$

Answer: C

196. A particle is projected upward with a speed of 50m/s from point A and another is dropped simultaneously from B which is 200m above A. They cross each other after?

A. 8 sec B. 6 sec C. 4 sec D. 5 sec

Solution: $y = ut - \frac{1}{2}gt^2$ and $-(200 - y) = \frac{1}{2}gt^2$

Subtracting, $200 = 50t$ or $t = 4$ sec

Answer: C

197. A bus starts from rest with an acceleration of $1 m/s^2$. A man who is 48m behind the bus starts running with a uniform velocity of 10m/s. The minimum time after which the man will catch the bus is?

A. 4.8 sec B. 8 sec C. 10 sec D. 12 sec

Solution: $vt = s_o + \frac{1}{2}at^2$

$$\Rightarrow 10t = 48 + \frac{1}{2}(1)\left(t^2\right) \Rightarrow t = 8 \text{ sec or } 12 \text{ sec}$$

Answer: B

198. A body travelling with uniform acceleration crosses two points A and B with velocities 20m/s and 30m/s respectively. The speed of the body at the midpoint of A and B is?

A. 24 m/s B. 25 m/s C. 25.5 m/s D. $10\sqrt{6} m/s$

Solution: $V_B^2 - V_A^2 = 2as$ and $V_B^2 - V^2 = 2a\left(\frac{s}{2}\right)$

$$\Rightarrow V_B^2 - V^2 = \left(\frac{V_B^2 - V_A^2}{2}\right) = \frac{30^2 - 20^2}{2} = \frac{900 - 400}{2} = 250$$

$$V^2 = V_B^2 - 250 = 900 - 250 = 650 \Rightarrow V = \sqrt{650} = 25.5 m/s$$

Answer: C

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199. A steel ball is bouncing up and down on a steel plate with a period of oscillation of one second. The ball bounces up to a height of?

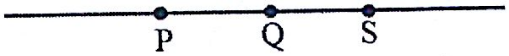
A. 5M B. 10m C. 2.5m D. 1.25m

Solution: Time of fall is t/2 second.

$$\Rightarrow y = \frac{1}{2}g\left(\frac{t}{2}\right)^2 = \frac{1}{2}10\left(\frac{1}{2}\right)^2 = 1.25m$$ **Answer: D**

200. A particle is moving with uniform acceleration along a straight line. The average velocity of the particle from P to Q is 8m/s and that from Q to S is 12m/s. If QS = PQ, then the average velocity from P to S is

A. 9.6 m/s
B. 12.87 m/s
C. 64 m/s
D. 327 m/s



Solution: $V_{av} = \frac{2V_1V_2}{V_1 + V_2} = \frac{(2)(8)(12)}{8+12} = 9.6m/s$ **Answer: A**

201. Drop of water fall from the roof of a building 9m high at regular intervals of time, the first drop reaching the ground at the same instant the fourth drop starts its fall. What are the distance of the second and third drop from the roof?

A. 4m and 1m B. 4m and 2m
C. 6m and 2m D. 6m and 3m

Solution: For the fourth drop, t = 0 for the third drop time = t for the second drop, time = 2t, for the first drop, time = 3t

Now, for the first drop,

$$y = \frac{1}{2}g(3t)^2 \Rightarrow \frac{1}{2}gt^2 = 1m$$

For the second drop

$$y = \frac{1}{2}g(2t)^2 = 4\left(\frac{1}{2}gt^2\right) = 4(1) = 4m$$

For the third drop

$$y = \frac{1}{2}g(t)^2 = 1m$$

Hence, the distances of the second and third drops are 4m and 1m respectively. **Answer: A**

202. A train is moving with a velocity of 30 m/s. when brakes are applied, it is found that the velocity reduces to 10m/s in 240 m when the velocity of the train is zero, the distance travelled is
A. 220 m B. 240 m C. 250 m D. 270 m

Solution: $V_2^2 - V_1^2 = 2a\,S_1 \Rightarrow a = \dfrac{-5}{3}$

$0 - V_2^2 = 2as_2 \Rightarrow s_2 = 30\text{m}$

$S = s_1 + s_2 = 240 + 30 = 270$ m **Answer: D**

203. A ball is thrown vertically upward with a speed of 18km/h. Then the maximum height attained by the ball is?
A. 1.25 m B. 3 m C. 10 m D. 180 m

Solution: $u = \dfrac{18\text{km}}{\text{h}} = \dfrac{18(1000)\text{m}}{3600\text{sec}} = 5\text{m/s}$

$h_{max} = \dfrac{u^2}{2g} = \dfrac{(5)^2}{2(10)} = 1.25\text{m}$ **Answer: A**

204. Two bodies begin a free fall from rest, from the same height, 1 second apart. How long after the first body begins to fall will the two bodies be 10m apart?
A. 2 sec B. 3.5 sec C. 0.5 sec D. 1.5 sec

Solution: $y_1 = \dfrac{1}{2}gt^2$ and $y_2 = \dfrac{1}{2}g(t-1)^2$

$\Delta y = y_1 - y_2 = 10$ solving $t = 1.5$ sec **Answer: D**

205. A body travels 2m in the first two second and 2.2 m in the next 4 second with uniform deceleration. The velocity of the body at the end of 9 second is
A. -10m/s B. -0.2m/s C. -0.4m/s D. -0.8m/s

Solution: $S = ut + \dfrac{1}{2}at^2 \Rightarrow 2 = 2u + \dfrac{1}{2}a(2)^2 \Rightarrow u + a = 1$

In the next 4 second

$4.2 = 6u + \dfrac{1}{2}a(6)^2 \Rightarrow a = -0.15\text{m/s}^2$ and $u = 1.15\text{m/s}$

$v = u + at = 1.15 - (0.15)(9) = -0.2\ \text{m/s}^2$ **Answer: B**

206. A particle starting from rest travels a distance x in first 2 second and a distance y next 2 second, then
A. $y = 2x$ B. $y = 3x$
C. $y = 4x$ D. $y = x$

Solution: $x = \frac{1}{2}a(2)^2 = 2a$ and $x + y = \frac{1}{2}a(4)^2 = 8a$

$\Rightarrow x + y = 8a \Rightarrow y = 3x$ **Answer: B**

207. Two bodies are thrown vertically upwards with their initial velocities in the ratio 2:3. Then the ratio of the maximum heights attained by them is

A. 1: 1 B. 4: 9 C. 2: 3 D. $\sqrt{2}:\sqrt{3}$

Solution: $h_1 = \frac{u_1^2}{2g}$ and $h_2 = \frac{u_2^2}{2g}$

$$\frac{h_1}{h_2} = \frac{u_1^2}{u_2^2} = \left(\frac{2}{3}\right)^2 = 4:9$$ **Answer: B**

208. A car travels equal distance in the same direction with velocities 60 km/h, 20km/h and 10km/h respectively. The average velocity of the car over the whole journey of motion is

A. 8 m/s B. 7 m/s C. 6 m/s D. 5 m/s

Solution: $V_{av} = \frac{s_T}{t_T} = \frac{3x}{\frac{x}{60} + \frac{x}{20} + \frac{x}{10}} = 18\text{km/h} = 5\text{m/s}$

Answer: D

209. A body starts from rest and is uniformly accelerated for 30 sec. The distance travelled in the first 10 sec is x_1, next 10 sec is x_2 and last 10 sec is x_3. Then, $x_1: x_2: x_3$ is the same as

A. 1: 2: 4 B. 1: 2: 5 C. 1: 3: 5 D. 1: 3: 9

Solution: $x_1 = \frac{1}{2}a\,(10)^2 = 50a$

$$x_1 + x_2 = \frac{1}{2}a\,(20)^2 = 200a$$

$$x_1 + x_2 + x_3 = \frac{1}{3}a\,(30)^2 = 450a$$

$x_2 = 200a - 50a = 150a$

$x_3 = 450a - 200a = 250\,a$

$x_1 : x_2 : x_3 = 50a: 150a: 250a = 1: 3: 5$ **Answer: C**

210. A stone thrown upward attains a maximum height of 45m. In what time will its velocity become half of its initial velocity

A. 2 sec B. 1.5 sec C. 1 sec D. 0.5 sec

Solution: $u = \sqrt{2gh} = \sqrt{2(10)(45)} = 30\text{m/s}$

$$v = u - gt \Rightarrow \frac{u}{2} = u - gt \text{ or } t = 1.5\text{sec}$$

Answer: B

211. Two trains A and B 100km apart are moving towards each other with same initial speed of 50 km/h. If train A is accelerating at 18km/h^2 and train B is decelerating at 18km/h^2. Where will they meet as measured from train A?

A. 32 km B. 68 km C. 59 km D. 50 km

Solution: $x = 50t + \frac{1}{2}(18)(t)^2$ and $100 - x = 50t - \frac{1}{2}(18)t^2$

Adding, $t = 1$ hr and $x = 59$ km

Answer: C

212. The velocity of a particle at an instant is 10m/s. After 5 sec, the velocity of the particle is 20m/s. the velocity at 3sec before the given instant is

A. 4 m/s B. 6 m/s C. 7 m/s D. 8 m/s

Solution: $20 = 10 + 5a \Rightarrow a = 2\text{m/s}^2$, and

$10 = u + 2(3) \Rightarrow u = 4\text{m/s}$

Answer: A

213. A ball is dropped from a height of 20m and rebounds with a velocity which is $\frac{3}{4}$ th of the velocity between the first and second bounces.

A. 3 sec B. 4 sec C. 5 sec D. 6 sec

Solution: $v = \sqrt{2gh} = 20\text{m/s}$

$$v^1 = \frac{3}{4}(20) = 15\text{m/s}$$

$$t = \frac{2(15)}{10} = 3\text{sec}$$

Answer: A

214. With what speed must a ball be thrown down for it to bounce 10m higher than its original level?

A. 20 m/s B. 14 m/s C. 5 m/s D. 1 m/s

Solution: $\frac{u^2}{2g} = 10 \Rightarrow v = 14\text{m/s}$

Answer: B

215. A balloon going upwards with a velocity of 12m/s releases a packet when it is at a height of 65m from the ground. How much time the packet will take to reach the ground?

A. 5sec B. 6 sec C. 7 sec D. 8 sec

Solution: $65 = -12t + \frac{1}{2}(10)(t^2) \Rightarrow t = 5\,\text{sec}$ **Answer: A**

216. A body moving with uniform retardation covers 3km before its speed is reduced to half its initial value. It comes to rest in another distance of

A. 1km B. 2km C. 3km D. 0.5 km

Solution: $\frac{u}{2}^2 - v^2 = 2(a)(3) \text{and} (0)^2 - \left(\frac{u}{2}\right)^2 = 2a(s) \Rightarrow s = 1\text{km}$

Answer: A

217. The displacement of a body is given by $x = \frac{-2}{3}t^2 + 16t$. In what time with the body comes to rest?

A. 6 sec B. 12 sec C. 18 sec D. 20 sec

Solution: $v = \frac{\Delta x}{\Delta t} = \frac{dx}{dt} = -\frac{4}{3}t + 16 = 0 \Rightarrow t = 12\,\text{sec}$ **Answer: B**

218. A rifle bullet loses $\frac{1}{20}^{\text{th}}$ of its velocity in passing through a plank. The least number of suck planks required to just stop the bullet is?

A. 5 B. 10 C. 11 D. 20

Solution: using $v^2 - u^2 = 2as$ $\left(\frac{19v}{20}\right)^2 - v^2 = 2ax$

$0 - v^2 = 2an\,x$

$n = \frac{-v^2}{\left(\frac{1g}{20}v\right)^2 - v^2} = 10.3 \simeq 11\text{plank}$ **Answer: C**

219. A stone is dropped into water from a bridge 44.1m. Another stone is thrown down 1sec later. Both strike the water simultaneously. What was the initial speed of the second stone?

A. 12.25 m/s B. 14.75 m/s C. 16.23 m/s D. 17.15 m/s

Solution: $t_1 = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2(44.1)}{9.8}} = 3\,\text{sec} \Rightarrow t^2 = 3 - 1 = 2\,\text{sec}$